

Price Policy for Sugarcane

2026-27 Sugar Season



Commission for Agricultural Costs & Prices

Department of Agriculture & Farmers Welfare

Ministry of Agriculture & Farmers Welfare

Government of India, New Delhi

November 2025

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Preface & Acknowledgments

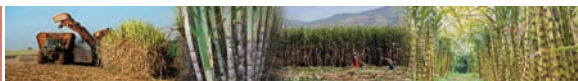
It is a great honour and privilege to submit the report of “**Price Policy for Sugarcane: 2026-27 Sugar Season**”. The report contains recommendation of Fair and Remunerative Price (FRP) of sugarcane and non-price measures. I am sure the recommendations will serve interest of all stakeholders including farmers, sugar mills, and consumers and incentivise sugarcane farmers and sugar factories to adopt modern technologies to promote productivity, efficiency and global competitiveness.

I am grateful to Economics, Statistics and Evaluation (ES&E) Division, Ministry of Agriculture & Farmers Welfare, Government of India for providing data on cost estimates for sugarcane production. My sincere thanks to the Central Ministries/Departments, State Governments, sugar industry associations, research institutions, and other stakeholders for providing valuable inputs and detailed information for preparation of this report.

My sincere appreciation to Dr. Prem Chand, Member (Official) and Sh. Ratan Lal Daga, Member (Non-official) for their outstanding contribution in preparation of the report. I would like to acknowledge contribution and coordination support of Sh. Cyriac George, Member Secretary. I express my great appreciation to Advisors, Dr. R. Sathish, Dr. Seema and Sh. Vivek Shukla for their excellent work and contributions. Special thanks to Mrs. Divya Sharma, Mr. Baidyanath Guru, Mrs. Shivani, Mrs. Shraddha Pal, Mr. Deepak Peeceeyen S, Mr. Vipin Singh and Ms. Parul Singh for their invaluable support in analysis and report writing. The report would not have been possible without active support of Mr. Md. Abdul Aleem, Mr. Mintu Kumar, Mr. Chandra Kumar, Mr. B.K. Barnawal, Mr. Ravi Kumar Sharma, Mr. Ashok Kumar, Mrs. Savitri Ramani, Mr. Mahender Kumar Gupta, Mr. Md. Shoeb Malik, Mrs. Jagrity Anand, Mrs. Anindita Das and I gratefully acknowledge the support received from them. I express my gratitude to the officers and staff members of the Commission for administrative support.

28th November 2025

(Vijay Paul Sharma)





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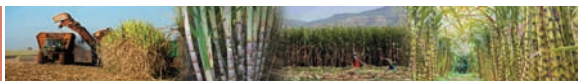


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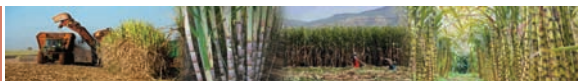


Acronyms

Acronym	Description
A_2	Paid out cost
A_2 +FL	Paid out cost plus imputed value of family labour
ABARES	Australian Bureau of Agricultural and Resource Economics and Science
AI	Artificial Intelligence
BHM	B-Heavy Molasses
C_2	Paid out cost plus imputed value of family labour plus rental value of own land and interest on value of own fixed capital assets (excluding land)
CACP	Commission for Agricultural Costs & Prices
CAGR	Compound Annual Growth Rate
CHCs	Custom Hiring Centres
CHM	C-Heavy Molasses
CIAE	Central Institute of Agricultural Engineering
CIPI	Composite Input Price Index
CoC	Cost of Cultivation
CoP	Cost of Production
CS	Comprehensive Scheme
CV	Coefficient of Variation
DBT	Direct Benefit Transfer
DFG	Damaged Food Grains
DGCIS	Directorate General of Commercial Intelligence and Statistics
DGFT	Directorate General of Foreign Trade
EBP	Ethanol Blended Petrol
ECA	Essential Commodities Act
ES&E	Economics, Statistics & Evaluation
ESY	Ethanol Supply Year
EU	European Union
FAO	Food and Agriculture Organization
FCI	Food Corporation of India
FRP	Fair and Remunerative Price
GJ	Gujarat
GST	Goods and Services Tax
GVO	Gross Value of Output
HH	High area-High yield
HL	High area-Low yield
HR	Haryana
HSD	High Speed Diesel
ICAR	Indian Council for Agricultural Research



Acronym	Description
ICE	Intercontinental Exchange
IISR	Indian Institute of Sugarcane Research
IND	India
ISMA	Indian Sugar & Bio-energy Manufacturers Association
ISO	International Sugar Organization
KA	Karnataka
LH	Low area-High yield
LL	Low area-Low yield
LMT	Lakh Metric Tonne
LPA	Long Period Average
MAEQ	Maximum Admissible Export Quantity
MH	Maharashtra
MIEQ	Minimum Indicative Export Quota
MPS	Market Price Support
MSP	Minimum Selling Price
MT	Metric Tonne
NFCSF	National Federation of Cooperative Sugar Factories Ltd.
NFSNM-CC	National Food Security and Nutrition Mission-Commercial Crops
OECD	Organisation for Economic Cooperation and Development
OMCs	Oil Marketing Companies
PMKSY-PDMC	<i>Pradhan Mantri Krishi Sinchayee Yojana- Per Drop More Crop</i>
PSF	Price Stabilisation Fund
R&D	Research and Development
RSF	Revenue Sharing Formula
SAP	State Advised Price
SBI	Sugarcane Breeding Institute
SMAM	Sub-Mission on Agricultural Mechanization
SMP	Statutory Minimum Price
SRR	Seed Replacement Rate
SUR	Stocks-to-Use Ratio
TE	Triennium Ending
TN	Tamil Nadu
TRQ	Tariff Rate Quota
UP	Uttar Pradesh
USA	United States of America
USDA	United States Department of Agriculture
UTs	Union Territories
VRR	Varietal Replacement Rate
WPI	Wholesale Price Index



Summary of Recommendations

Fair and Remunerative Price (FRP) Recommendations

- S.1 The Commission has considered various factors including the cost of production of sugarcane, recovery of sugar from sugarcane, demand-supply situation and prices of sugar in domestic and global market, realization from primary by-products, reasonable margins for farmers on account of risk and profits, returns from alternative crops, availability of sugar to consumers at a fair price and general price trend for recommending the Fair and Remunerative Price (FRP) for sugarcane. The Commission also held extensive consultations with various stakeholders, including the Central and State Governments, sugar industry representatives and industry associations and research institutions, before finalizing the price and non-price recommendations.
- S.2 For Sugar Season 2026-27, the Commission recommends FRP of sugarcane at ₹365 per quintal at basic sugar recovery rate of 10.25 percent and a premium of ₹3.56 for every 0.1 percentage point increase in sugar recovery over and above the basic recovery, and discount of ₹3.56 for every 0.1 percentage point reduction in sugar recovery rate below the basic recovery. The recommended FRP gives a margin of 100.5 percent over $A_2 + FL$ cost of production.

Non-Price Recommendations

Improve Varietal and Seed Replacement Rate

- S.3 The varietal replacement rate (VRR) and seed replacement rate (SRR) in sugarcane remain extremely low, mainly due to high cost of seed cane and non-availability of quality seed/seedlings and suitable varieties, which make crop more susceptible to pests and diseases and adversely affect cane and sugar yield. Therefore, the Commission recommends that State Governments should formulate an action plan in collaboration with sugar mills and research institutions for developing climate resilient high-yielding varieties and ensuring availability of quality seed cane through participatory seed production at farmers' fields under strict monitoring. Special training and awareness campaigns should be launched to educate farmers about importance of seed treatment and ensure availability of seed treatment devices to promote seed treatment and minimize seed-borne diseases.

Encourage Intercropping in Sugarcane

- S.4 Various sugarcane-based intercropping models developed by the Indian Council of Agricultural Research (ICAR) and State Agricultural Universities (SAUs) have shown promising results in improving soil



health and resource use efficiency but adoption is quite low. Therefore, there is a need to promote intercropping in sugarcane cultivation and crop rotation by organizing large-scale field demonstrations and awareness programmes.

Promote Efficient Water Management

- S.5 Sugarcane is a water-intensive crop and groundwater is a major source of irrigation. High groundwater extraction and low coverage of micro-irrigation in major sugarcane growing regions calls for efficient groundwater management and conservation by promoting efficient water management technologies and practices such as micro-irrigation, skip-furrow system, trash mulching, etc. The Commission, therefore, recommends that a separate allocation for promoting micro-irrigation in sugarcane cultivation should be made under the *Pradhan Mantri Krishi Sinchayee Yojana- Per Drop More Crop* (PMKSY-PDMC). The State Governments and sugar mills should be encouraged to provide additional subsidy on micro-irrigation and actively involved in promoting efficient water management and conservation practices and technologies.

Special Focus on Mechanisation

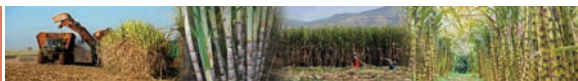
- S.6 Mechanization in sugarcane cultivation is crucial for minimizing cost and improving profitability, given the high share of human labour cost in cost of cultivation. The Government has taken several steps and the recent reduction in GST on tractors and farm machinery will make farm machinery more affordable and accessible to small farmers. However, high cost of machinery & equipment, particularly sugarcane harvesters, and small operational holding are the major constraints in adoption of mechanization. The Commission reiterates its recommendation that special provision should be made for sugarcane under Sub-Mission on Agricultural Mechanization (SMAM). There is a need for strengthening R&D for developing improved, low-cost tools and machinery and promote large machinery including harvesters through Custom Hiring Centres (CHCs), Farm Machinery Banks, Farmer Producer Organizations (FPOs), cooperatives and farmer groups.

Address Distortions in Sugarcane Pricing

- S.7 The State Advised Price (SAP) or additional incentive over and above FRP by some State Governments distort sugar sector and does not promote efficiency. The Commission reiterates its earlier recommendation of discontinuation of SAP and adopting Revenue Sharing Formula (RSF) or FRP. The Commission is of the view that those States that want to continue with SAP/incentive should pay the difference between FRP/RSF and SAP directly to farmers through Direct Benefit Transfer (DBT) as adopted by Government of Punjab, Haryana and Tamil Nadu to reduce financial burden on sugar mills.

High Powered Expert Committee

- S.8 The Indian sugar industry has undergone significant transformation by diversifying into ethanol and substantial capacity addition and expansion during the last decade. Rapid expansion in capacity not accompanied by growth in cane availability and market demand for sugar has contributed to lower capacity utilization and closure of over 30 percent of sugar factories in 2024-25 season. The Commission



recommends that a High-Powered Expert Committee comprising key stakeholders, including Central and State Government, sugar industry, academia, and farmers, should be constituted to review issues related to cane area reservation, minimum distance criteria, Revenue Sharing Formula, dual pricing, etc.

Review Cane Area Reservation and Minimum Distance Criteria

- S.9 Cane area reservation and minimum distance criteria are intended to ensure adequate cane supply to mills, prevent unhealthy competition, and ensure the sugar mill procures sugarcane from farmers at FRP/SAP. However, sugar industry has witnessed capacity addition and expansion of existing factories over the last two decades and resulted in low capacity utilization and closure of 240 factories in 2024-25 season. Therefore, the Commission recommends that cane area reservation and minimum distance criteria should be reviewed to avoid unhealthy competition amongst sugar factories and ensure adequate supply of sugarcane for running sugar factories.

Link Ethanol Price to FRP of Sugarcane

- S.10 The administered price mechanism for ethanol produced from sugarcane-based feedstock and guaranteed procurement by the public sector Oil Marketing Companies (OMCs) under the Ethanol Blended Petrol (EBP) Programme have helped in stabilizing Indian sugar sector and ensured timely cane payment to farmers. However, there has been a marginal increase in prices of ethanol produced from sugarcane juice/sugar/sugar syrup and B-Heavy molasses during last five years compared with increase in FRP of sugarcane and price of grain-based ethanol. The Commission, therefore, recommends that the Government should revise the price of ethanol from sugarcane-based feedstocks in view of increase in FRP of sugarcane.

Dual Pricing of Sugar

- S.11 The Commission recommends that feasibility of a dual pricing mechanism of sugar for domestic and commercial users should be explored as industrial/commercial sector uses about 60-65 percent of total sugar production in the country. Various State Governments, sugar industry and industry associations have also suggested dual price policy for sugar.



CHAPTER 1



Overview

- 1.1 India is the world's second largest sugarcane producer, with about 5.7 million hectare cultivated area and 466 million tonnes of production at an average yield of about 82 tonnes per hectare in Triennium Ending (TE) 2024-25. Sugarcane ranks third in terms of contribution to total value of output from crops, after rice and wheat, with 6.2 percent contribution of sugar and *gur* (at 2011-12 prices) in TE2023-24. However, contribution is significantly high in some States, like Uttar Pradesh (23.5%), Uttarakhand (17.4%), Maharashtra (11.8%) and Karnataka (9.2%). The value of sugarcane output grew at a Compound Annual Growth Rate (CAGR) of 3.9 percent during last 10 years, higher than crop sector output (3.2%). The crop supports livelihood of about 50 million farmer households and provides raw material for various sugarcane-based industries like sugar, ethanol, etc., which provide direct employment to about five lakh workers.
- 1.2 Over the last decade, Indian sugarcane economy has transformed beyond sugar production by diversifying into bio-ethanol, cogeneration, compressed biogas, etc. However, sugarcane production in the country is still subjected to cyclical fluctuations, primarily due to agro-climatic factors and fluctuations in prices, but volatility has reduced over the years. Global sugar market remains one of the most protected and policy distorted agricultural commodities markets through various trade restrictions and price support mechanisms. For example, United States of America (USA) protects domestic sugar industry through a blend of import tariffs, Tariff Rate Quotas (TRQs), price guarantees, domestic marketing allotments, etc. In addition, Market Price Support (MPS), which supports domestic prices, is also quite high at 36.8 percent of gross receipts in 2022-24, as per the OECD. This chapter presents an overview of the global and domestic sugarcane and sugar sectors and discusses major issues facing the sector.

Sugarcane and Sugar Production

Sugarcane

Global Scenario

- 1.3 According to Food and Agriculture Organization (FAO), global sugarcane production crossed 2,025 million tonnes in 2023, an increase of 5.2 percent over previous year. Brazil is a global leader in sugarcane production with a 38.3 percent share in TE2023, followed by India (23%) and China



(5.4%). World production is strongly concentrated in a few countries with top-five producers accounting for more than three-fourth of global area and production. Over the last two decades, Brazil has increased its share in world production from 32.4 percent in TE2006 to 38.3 percent in TE2023 and the country is expected to maintain its leading position until 2034. The share of India has increased from 18.5 percent to 23 percent during past two decades and is expected to remain at the same level over the next ten years. In 2025-26, sugarcane production in Brazil is forecast to fall by about 2.6 percent due to lower yield owing to water stress, while sugarcane production in India is expected to increase driven by favourable weather conditions and area expansion.

- 1.4 Sugarcane productivity is significantly higher in India among major producers, at 83.9 tonnes per hectare in TE2023, compared with Brazil (74.3 t/ha), China (80.4 t/ha), Thailand (56.9 t/ha) and world average of 73.5 tonnes per hectare. Over the last decade, world sugarcane production expanded at a moderate pace (CAGR 0.5%) driven by yield gains (0.6%), while CAGR of area was negative (-0.1%). However, India recorded much higher CAGR in production (3.5%) and yield (2.5%) during the period.

Domestic Scenario

- 1.5 Over the past decade, sugarcane production in the country has expanded significantly with Government's strong push for Ethanol Blended Petrol (EBP) programme and remunerative prices to farmers. Sugarcane acreage increased from 47 lakh hectare in TE2017-18 to 56.9 lakh hectare in TE2024-25, an increase of 21.1 percent, while average yield rose from 73.3 tonnes per hectare to 81.9 tonnes per hectare during the period, resulting in 35.2 percent increase in production, from 344.8 million tonnes to 466.1 million tonnes. Sugarcane area has declined during the last two years but yield, which declined by 5.3 percent in 2023-24, rebounded with 5.7 percent increase in 2024-25 leading to marginal rise in production. The country is estimated to have a larger sugarcane crop at 475.6 million tonnes in 2025-26 due to increase (4.5%) in area. The area, production and yield of sugarcane in major producing States from 2011-12 to 2025-26 are given in Annex Tables 1.1 to 1.3.
- 1.6 The release of Co 0238, a high-yielding early-maturing and high sugar content variety, for commercial cultivation in North-West zone in 2009, increased the average yield by about 40 percent during the last one and half decades and helped in doubling of production in Uttar Pradesh and its share in total cane production from 36.8 percent in TE2011-12 to 47.3 percent in TE2024-25. However, Co 0238 variety has become susceptible to red rot and wilt diseases, which adversely affected cane yield and sugar recovery in the region. Therefore, there is a need to phase out Co 0238 variety and replace with other suitable varieties.
- 1.7 The share of Maharashtra, the second largest producer of sugarcane, remained almost stable at 23-25 percent, while Tamil Nadu registered a significant decline in both area (from 6.8% to 2.6%) and production share (10.3% to 3.4%) between TE2011-12 and TE2024-25, primarily driven by low sugar recovery rates due to erratic monsoons, shortage of agricultural labour and predominance of old varieties. For example, Co 86032, more than 25 years old variety, is still being cultivated on about 50 percent of total sugarcane area in the State of Tamil Nadu. The share of Karnataka has remained almost stable during the last one and a half decades.



Sugar

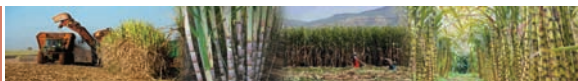
World Economy

- 1.8 Nearly 85 percent of global sugar production comes from sugarcane while the remaining share is derived from other sources, mainly sugar beet. Brazil and India account for approximately 40 percent of global sugar production, as shown in Table 1.1, which presents major sugar producers, consumers, exporters and importers for TE2024. Brazil is also the largest exporter of sugar with about 34 million tonnes of sugar accounting for 51.4 percent of the world exports. Brazil's export volume was nearly five times more than that of Thailand, the second largest exporter. India is the second largest producer (17.1%) of sugar globally after Brazil (24.9%), with a production of 30.4 million tonnes. However, India is the largest consumer of sugar, accounting for 15.7 percent of global consumption at 28.2 million tonnes, mainly due to large population and high demand from food and beverage industry. Despite a high internal demand that uses up most of the domestic output, India accounted for 7.3 percent of global sugar exports, supported by surplus production and favourable prices. China and Indonesia are top importers with a combined share of 11 percent. Other importers are USA, Bangladesh, India, Algeria, Malaysia and South Korea.
- 1.9 World sugar production and exports are highly concentrated with top-5 countries/country groups accounting for 61.4 percent of production and 75.2 percent of global exports. On the import side, sugar markets are less concentrated with share of top-5 importers less than 30 percent.

Table 1.1: Major Sugar Producers, Consumers, Exporters and Importers: TE2024

Producers	Million	Consumers	Million	Exporters	Million	Importers	Million
	Tonnes		Tonnes		Tonnes		Tonnes
Brazil	44.3	India	28.2	Brazil	34.1	China	5.6
India	30.4	EU	16.0	Thailand	6.8	Indonesia	5.4
EU	14.8	China	15.8	India	4.9	USA	3.0
China	10.0	Brazil	11.2	Australia	2.7	Bangladesh	2.5
Thailand	9.8	USA	10.5	Guatemala	1.4	India	2.3
USA	7.8	Indonesia	7.9			Algeria	2.2
Russia	6.5	Pakistan	6.1			Malaysia	2.1
Pakistan	6.4	Russia	5.9			S. Korea	2.0
						Nigeria	1.8
						Saudi Arabia	1.8
						UAE	1.7
						EU	1.7
						Morocco	1.6
						Vietnam	1.5
						Egypt	1.4
World	178.1	World	180.2	World	66.2	World	67
Top-5 (%)	61.4	Top-5 (%)	45.4	Top-5 (%)	75.2	Top-5 (%)	28.3

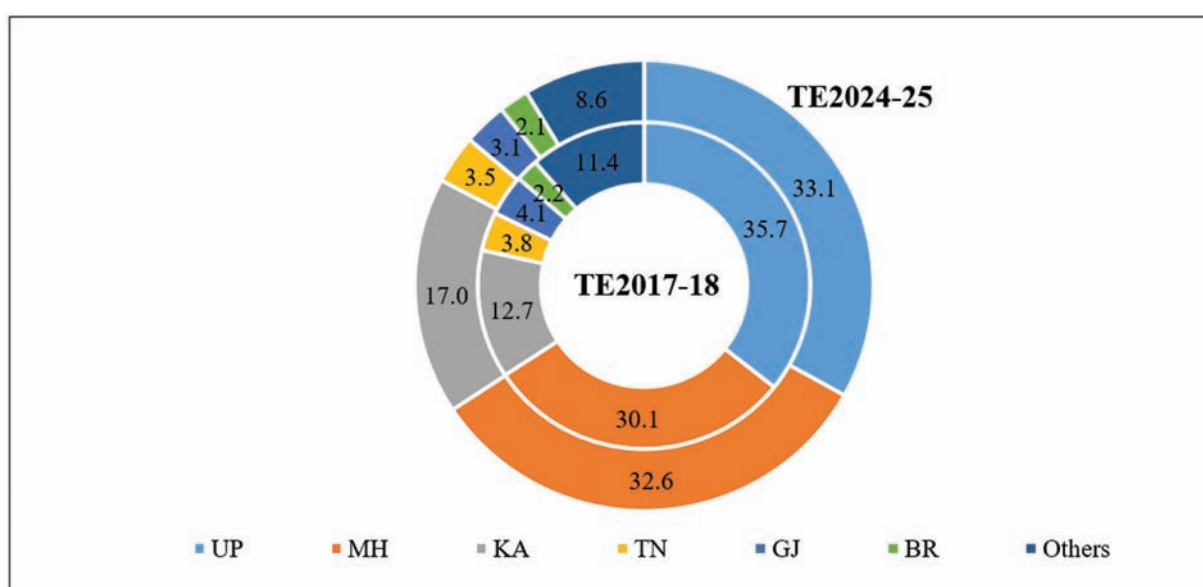
Source: International Sugar Organization



Domestic

- 1.10 Sugar production in India is highly concentrated and top-three producer States account for over 80 percent of total production. Uttar Pradesh and Maharashtra account for around two-third of the total sugar production, with their individual output share being nearly equal in TE2024-25. Maharashtra has marginally improved its position during the last 10 years, while Uttar Pradesh has lost its share in total sugar production. Karnataka has emerged as third largest producer of sugar in the country and increased its share from 12.7 percent in TE2017-18 to 17 percent in TE2024-25 (Chart 1.1). Tamil Nadu, Gujarat and Bihar contributed 8.7 percent to national production in TE2024-25. Annex Table 1.4 gives trends in sugar production in major producing States from 2011-12 to 2024-25.

Chart 1.1: State-wise Production of Sugar in India: TE2017-18 and TE2024-25

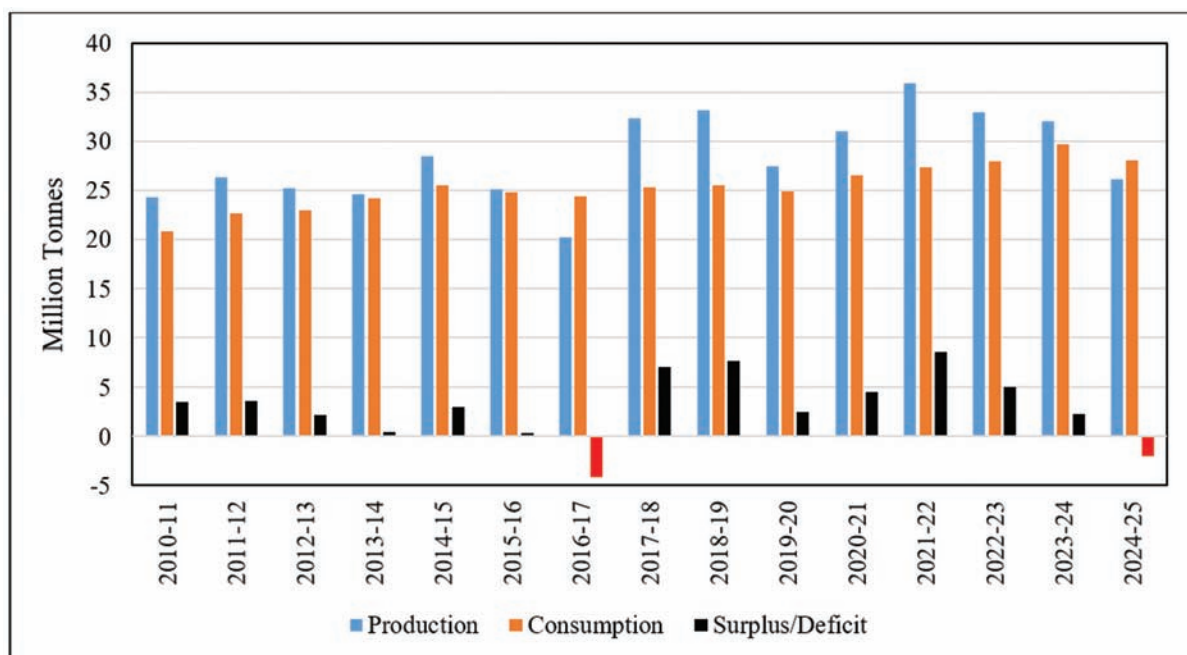


Source: Directorate of Sugar and Vegetable Oils, Department of Food and Public Distribution

- 1.11 Chart 1.2 shows the trends in sugar production, consumption, and surplus/deficit in India from 2010-11 to 2024-25. Both production and consumption recorded a steady growth during the last 15 years, though volatility as measured in terms of coefficient of variation (CV) in production (15.4%) was higher than consumption (9.1%). The sugar production grew at CAGR of 2 percent, from 25.3 million tonnes in TE2012-13 to about 30.4 million tonnes in TE2024-25 and consumption grew from 22.2 million tonnes to 28.6 million tonnes (CAGR 1.9%) during the period. However, gross sugar production (considering diversion for ethanol production) increased at a CAGR of 2.9 percent, from 25.3 million tonnes in TE2012-13 to about 33.7 million tonnes in TE2024-25. India, which had sugar deficit in four years during the 2001-02 to 2009-10 period, faced shortfall only in one year during 2010s and then in 2024-25. The country recorded the largest production surplus of 8.6 million tonnes in 2021-22 due to a significant rise in production. However, diversion of sugar for ethanol production has reduced the problem of production surpluses. Overall, India remains a surplus producer of sugar, but fluctuations in production underscore the need for policies aimed at addressing climate change induced weather variability impact on productivity and acreage, better production planning and management of diseases and pests to ensure market stability.



Chart 1.2: Production, Consumption and Surplus/Deficit of Sugar in India, 2010-11 to 2024-25

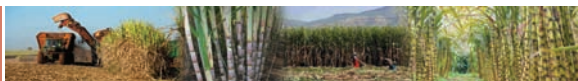
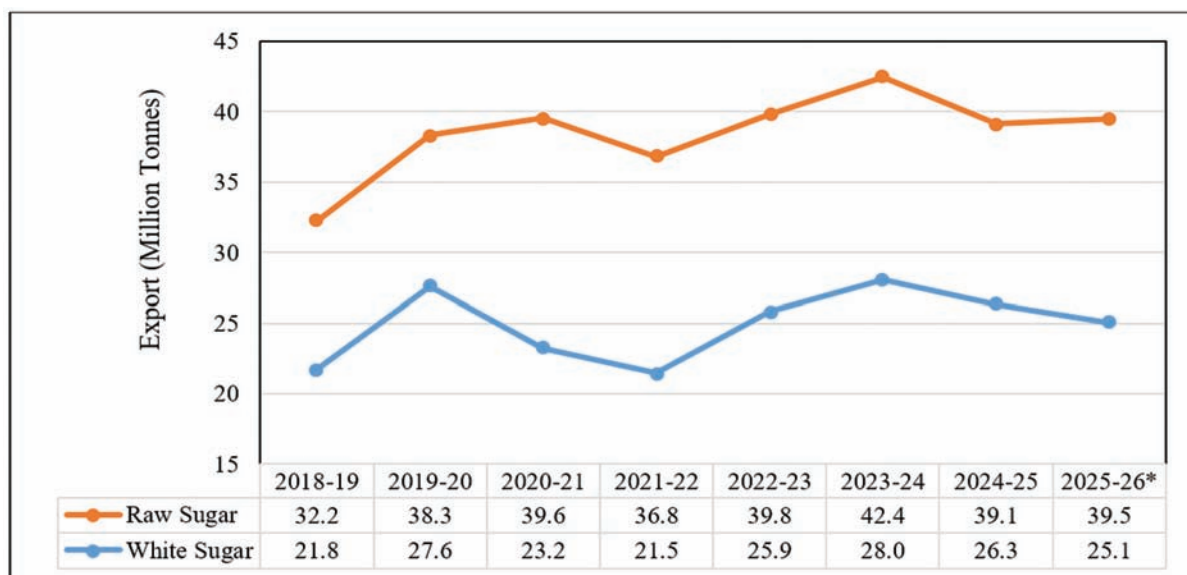


Source: Directorate of Sugar and Vegetable Oils, Department of Food and Public Distribution

Trade Performance

World Sugar Trade

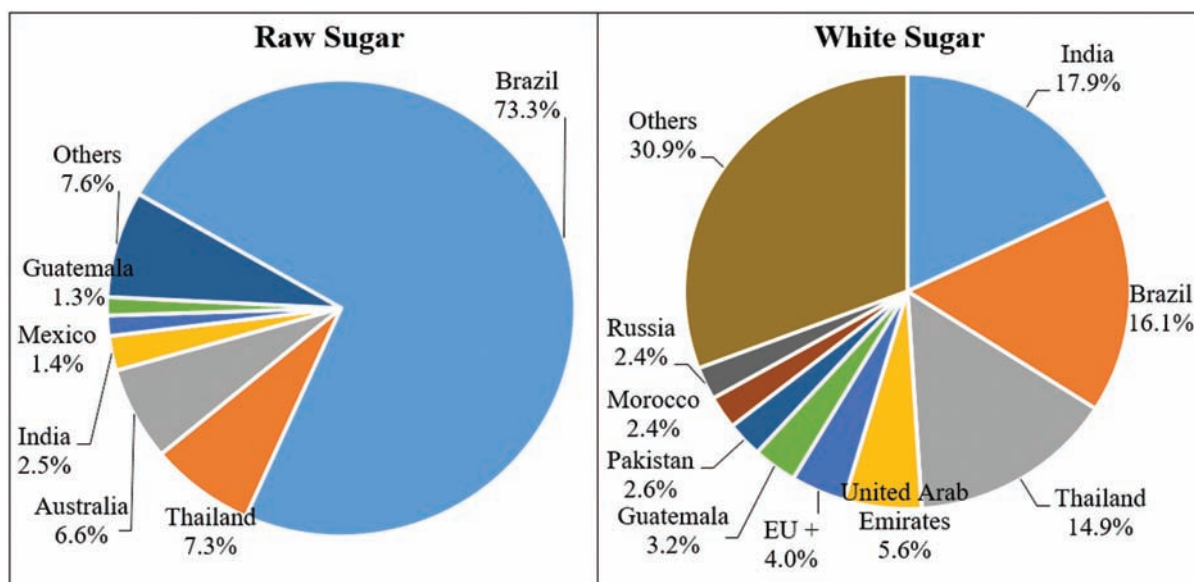
- 1.12 Sugar is traded internationally in raw and white forms and according to International Sugar Organization (ISO), raw sugar accounts for about 60 percent of world sugar market, and white sugar for remaining 40 percent. During TE2024-25, 67.2 million tonnes of sugar (raw and white) entered the global trade and represented nearly 38 percent of world sugar production. Chart 1.3 shows that global exports of raw sugar, which is traded in bulk, remained consistently higher than white sugar, reflecting stronger international demand for raw sugar. Raw sugar exports rose from about 32 million tonnes in 2018-19 to a peak of over 42 million tonnes in 2023-24, and declined to 39.1 million tonnes in 2024-25. White sugar exports, meanwhile, increased from 21.8 million tonnes in 2018-19 to a high of 28 million tonnes in 2023-24, but declined to 26.3 million tonnes in 2024-25. For 2025-26, white sugar exports are forecast to fall, while raw sugar exports are likely to rise marginally. During 2018-19 to 2024-25, white sugar exports grew at annual average growth rate of 4.4 percent compared with 3.7 percent in case of raw sugar.

**Chart 1.3: Trends in Global Exports of Raw and White Sugar: 2018-19 to 2025-26**

Note: *Estimated

Source: International Sugar Organization

- 1.13 Chart 1.4 shows major exporters of raw and white sugar. Raw sugar exports are more concentrated, and Brazil dominates the raw sugar market with 73.3 percent of exports share. Thailand (7.3%) and Australia (6.6%) are two other major exporters of raw sugar. In contrast, white sugar exports are less concentrated and India was the largest exporter of white sugar with 17.9 percent share in TE2024-25, closely followed by Brazil (16.1%) and Thailand (14.9%). Top-three exporters of raw sugar accounted for 87.3 percent of world exports, while in case of white sugar the share was less than half. As per ISO estimates, volume of world trade in sugar in 2025-26 is expected to remain steady with exports marginally lower than last season, while import demand is estimated at the same level as in 2024-25 season.

Chart 1.4: Major Exporters of Raw and White Sugar: TE2024-25

Source: International Sugar Organization

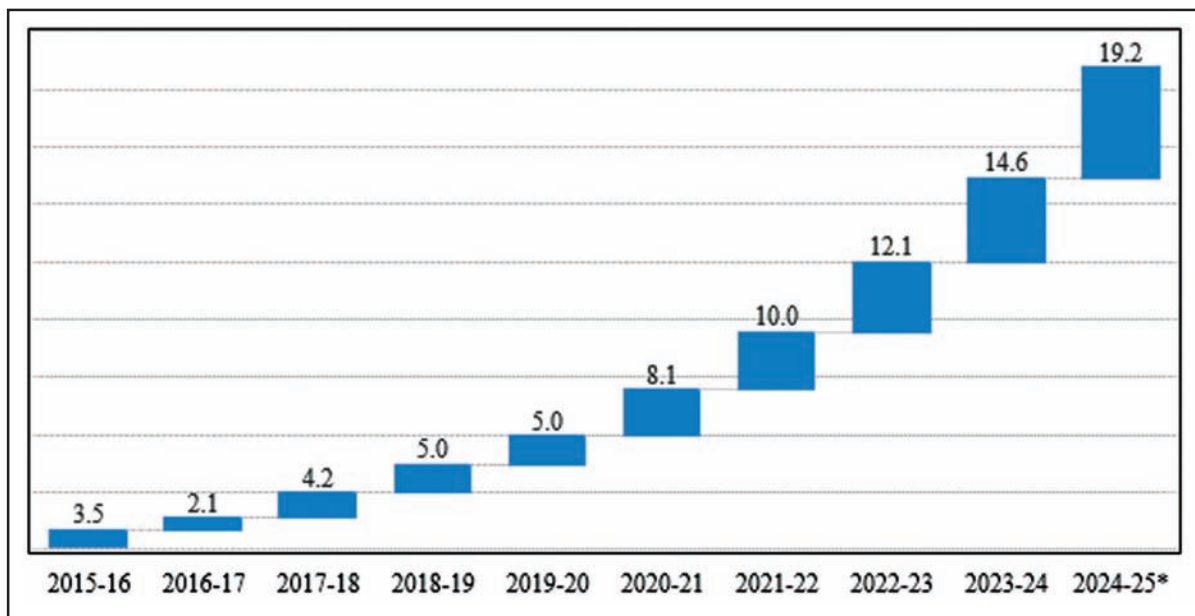


Sugar Industry Diversification: Government Initiatives

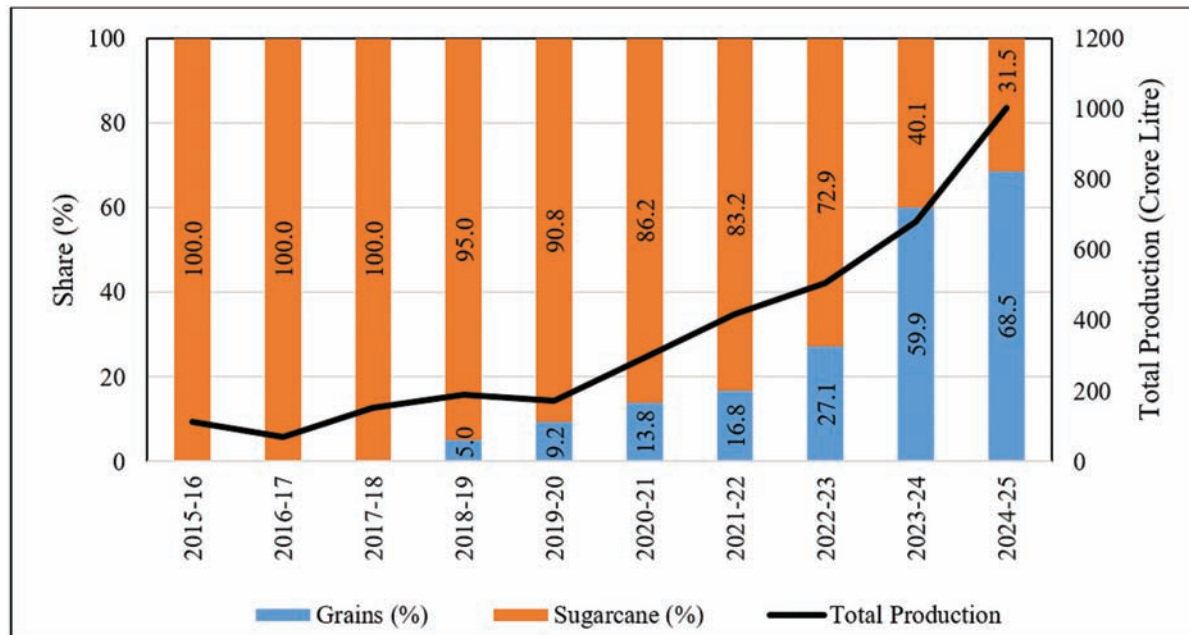
- 1.14 According to OECD-FAO Agricultural Outlook 2025-2034, about 81 percent of the world's sugar crops are used to produce sugar, but this share is expected to decline to 77 percent by 2034. Brazil (37%) and India (23%) are expected to remain the major producers of sugarcane, and contribute 24 percent and 19 percent of world sugar production and 75 percent and 21 percent of global sugarcane-based ethanol production, respectively, by 2034. Ethanol production in Brazil has a long history but major turning point for ethanol came during the oil crisis of 1970s and introduction of flex-fuel vehicles in 2003. Public policies including strong Government support have played a central role in shaping the ethanol industry in Brazil. However, corn-based ethanol has emerged as an important player, and its share has increased from about 5 percent in 2019-20 to 17 percent in 2023-24.
- 1.15 Indian sugar industry has undergone significant transformation in the recent years, initially by diversifying into cogeneration and later through production of ethanol to exploit economies of scale and scope and improve economic efficiency and profitability. During 2017-18 and 2018-19 sugar season, excess sugar production had adversely affected market prices and liquidity position of the sugar mills resulting in accumulation of cane price arrears of farmers. Government took various policy interventions including Minimum Selling Price (MSP) of sugar, creation of buffer stock, financial assistance to sugar mills, remunerative prices for ethanol produced from C-Heavy and B-Heavy molasses and cane juice, financial assistance through banks to molasses based stand-alone distilleries and also to distilleries attached with sugar mills for augmenting ethanol production capacity, soft loans to sugar mills, interest subvention, etc. The National Policy on Biofuels-2018, has been instrumental in helping sugarcane-based industry to diversify and sustain growth, and significantly expanding the EBP programme in the country.
- 1.16 India's success story of ethanol has been driven by the strong policy push by the Government, including financial assistance, interest subvention scheme, guaranteed market access and administered pricing mechanism for ethanol. Total ethanol production in the country has increased from 67.8 crore litre in Ethanol Supply Year (ESY) 2016-17 to about 1003 crore litre in 2024-25, registering a CAGR of 35.3 percent. This has resulted in a sharp rise in ethanol blending in petrol from 3.5 percent in 2015-16 to 19.2 percent in 2024-25 (Chart 1.5). Sugarcane-based ethanol from molasses was the primary driver of EBP programme but grain-based ethanol has become a major driver in ethanol production in the recent years, as shown in Chart 1.6. In 2024-25, grain-based ethanol production reached about 687 crore litre, representing about two-third of total ethanol production in the country, up from five percent in 2018-19. Grain-based ethanol production recorded an average annual growth rate of about 118 percent during the last five years, while sugarcane-based ethanol production witnessed a significantly lower growth rate of around 19 percent. During the last five years, price of ethanol produced from maize, C-Heavy molasses and damaged foodgrains increased by 39.4 percent, 26.9 percent and 24.2 percent, respectively, but the increase in prices was marginal in case of ethanol produced from sugarcane juice/sugar/sugar syrup (4.7%) and B-Heavy molasses (5.4%). Therefore, there is a need to review prices of sugarcane-based ethanol for sustainable growth of sugar industry. The grain-based ethanol production has developed synergy with sugarcane-based biorefineries, and production of dual-feedstock refineries has increased in the recent years.

**Chart 1.5: Progress under Ethanol Blended Petrol Programme in India**

(Blending %)

Note: * As on 12th October 2025

Source: Directorate of Sugar and Vegetable Oils, Department of Food and Public Distribution

Chart 1.6: Evolution of Ethanol Production in India

Source: Ministry of Petroleum & Natural Gas

Sugarcane and Sugar Sector Regulations

- 1.17 Indian sugar industry, which was highly regulated, has undergone a significant transition during the last two and half decades, when licensing requirement for new sugar mills was abolished in 1998 and levy on sugar mills was dispensed with in April 2013.

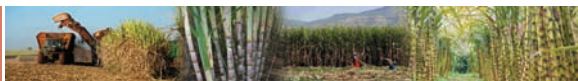


Sugarcane (Control) Order, 1966

- 1.18 The pricing of sugarcane is governed by the statutory provisions of the Sugarcane (Control) Order, 1966 issued under the Essential Commodities Act (ECA), 1955, which was amended in October 2009 and the concept of ‘Statutory Minimum Price’ (SMP) of sugarcane was replaced by the ‘Fair and Remunerative Price’ (FRP). A new clause, ‘*reasonable margins for growers of sugarcane on account of risk and profits*’, was inserted as an additional factor for working out FRP and was made effective from the 2009-10 sugar season. The Rangarajan Committee on deregulation of sugar sector (2012) and the NITI Aayog Task Force (2020) had recommended that cane prices should be linked to the value realized from sugar and its by-products through a Revenue Sharing Formula (RSF) with different sharing proportions and creation of Price Stabilisation Fund (PSF) to protect farmers from receiving prices below the FRP. The Rangarajan Committee had also recommended that the States should be urged to desist from announcing State Advised Price (SAP) unless the States are willing to bear additional cost of SAP and not forcing the mills to bear the burden of sugarcane price above FRP. However, the recommendation of the Committee was left to the State Governments for adoption and implementation, as considered appropriate by them. Only two States, Maharashtra and Karnataka, have adopted the Revenue Sharing Formula. Several sugarcane-growing countries such as Brazil, Thailand, Australia, and Mauritius use a similar cane price payment system to promote long-term viability and financial stability for both farmers and the sugar industry.

Cane Area Reservation and Minimum Distance Criteria

- 1.19 The Sugarcane (Control) Order, 1966 empowers State Governments to reserve cane-growing areas for a factory to ensure sufficient cane availability for sugar mills, and sugar mills are mandated to purchase cane from farmers within the reserve area. Maharashtra is the only major sugarcane-growing State where there is no cane area reservation system, while other States such as Uttar Pradesh, Punjab, Bihar, Karnataka, and Tamil Nadu have the system of cane area reservation (Annex Table 1.5).
- 1.20 The minimum distance between two sugar mills was introduced in July 1980 and 30 km was fixed to ensure adequate availability of sugarcane for the existing capacity as well as future expansion. The distance was increased to 40 km in January 1987, which was reduced to 25 km in November 1991 and further to 15 km in January 1997. Indian sugar industry, which was de-licensed in August 1998 and the requirement of license for setting up new sugar mills was dispensed with, continued with a minimum distance criteria of 15 km to avoid competition among sugar mills to procure sugarcane. The Sugarcane (Control) Order, 1966 was amended vide notification dated 10th November 2006 to give statutory backing to the norm of keeping a minimum distance of 15 km between two mills. The State Governments were authorized to notify a minimum distance higher than 15 km, in case they considered it necessary and expedient in public interest to do so, subject to prior approval from the Central Government. On the request of the Government of Punjab, Haryana and Maharashtra, Central Government has approved increasing the minimum distance from 15 km to 25 km in these States.
- 1.21 The Rangarajan Committee had recommended that States should encourage development of market based long-term contractual arrangements, and phase out cane reservation area and bonding, and as



and when a State does away with cane reservation area and bonding, the minimum distance criteria may be dispensed with. However, the Government of India left these recommendations to the State Governments for adoption and implementation, as considered appropriate by them.

- 1.22 Over the last two and half decades, Indian sugar industry has expanded significantly, initially due to push for cogeneration and later through ethanol production. With setting up of new sugar mills and modernization/expansion of existing mills, Indian sugar industry is operating with excess capacity, resulting in low-capacity utilization. The installed production capacity in terms of sugar production has more than tripled since delicensing of the industry, from about 128 lakh tonnes in TE1996-97 to 548 lakh tonnes in TE2024-25 and average capacity per factory has risen from 28,935 tonnes to 71,547 tonnes during the period. However, capacity expansion has not been matched by a corresponding increase in cane availability, resulting in low-capacity utilization. The average capacity utilisation has declined from about 115 percent in TE1996-97 to 55.6 percent in TE2024-25. The average capacity utilization at all-India level was 47.3 percent in 2024-25 season with large inter-State variations. The widening gap between installed capacity and utilization underscores emerging structural imbalances in the sector and need for reviewing minimum distance criteria.

Sugar (Control) Order, 2025

- 1.23 Sugar (Control) Order, 1966, which authorizes the Central Government to regulate production, storage, sale, movement, quality, and other matters, was comprehensively reviewed and the Sugar (Control) Order, 2025 was notified on 1st May 2025. Major changes in the Order were inclusion of the *khandsari* sugar units having crushing capacity more than 500 tonnes of cane per day, raw sugar in total stocks of sugar and incorporation of clause related to regulation of MSP of sugar in place of Sugar Price (Control) Order, 2018. Sugar (Control) Order, 2025 is expected to help in simplification and streamlining of the regulatory framework governing sugar sector under the changing industry dynamics and technological advancements.

Stagnation in Sugarcane and Sugar Yield

- 1.24 Both sugarcane yield and sugar recovery in the country, which witnessed a marked improvement during the 2010s, experienced stagnation during the last five years. The spread of high-yielding and high-sugar content variety (Co 0238) has played an important role in increasing sugarcane and sugar yield in the sub-tropical region. The share of Co 0238 variety in the region has increased from less than two percent in early-2010s to around 84 percent in 2020-21. However, the variety has become susceptible to diseases and insect pests, particularly to red rot disease, resulting in lower cane yield and sugar recovery rate.
- 1.25 In some States, share of ratoon crop is significantly high and ratoon is grown through multiple ratoon cycles, which result in gradual decline in yield with each successive harvest. For example, in Karnataka, the share of ratoon crop was about two-third during the last ten years. In addition, sugarcane yield levels vary substantially across different producing regions and are significantly lower than the potential yield, resulting in yield gaps of different magnitudes, which represent an opportunity for increasing sugarcane production. The Varietal Replacement Rate (VRR) and Seed Replacement Rate (SRR) in



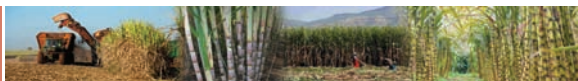
sugarcane are very low due to high seed cost and non-availability of quality seed cane and planting material. Therefore, there is a need to develop high yielding varieties and promote varietal diversification to improve VRR and SRR.

Mechanization of Sugarcane Cultivation

- 1.26 Given the fact that sugarcane is a labour-intensive crop with human labour accounting for more than half of the cost of cultivation, labour is becoming scarce, and wage rates are increasing rapidly, it is necessary to respond to such situation by promoting mechanization in sugarcane cultivation. However, high cost of machinery/equipment and small and fragmented operational holdings are the major constraints in adoption of mechanisation, particularly in harvesting operations. Therefore, sugarcane harvesters need to be customized according to Indian requirements. The Commission, in its earlier reports, had recommended that sugar mills be encouraged to purchase harvesting machines as well as promote Custom Hiring Centres (CHCs), Farm Machinery Banks, etc. to provide services to the farmers on rental basis. There is a provision of financial assistance up to 50 percent on sugarcane machinery to farmers, CHCs and machinery banks under Sub-Mission on Agricultural Mechanization (SMAM) but allocation for sugarcane mechanization is low in most States. Concerted efforts are required to formulate a comprehensive strategy for mechanising sugarcane cultivation, particularly harvesting operations and a separate budgetary provision should be made for sugarcane mechanisation.

Sugarcane and Sugar Outlook

- 1.27 FAO's preliminary forecast for 2025-26 indicates bumper sugarcane harvest in key producing countries, namely Brazil, India and Thailand, due to favourable growing conditions and expansion in area. In India, sugarcane production in 2025-26 is estimated at 475.6 million tonnes, an increase of 4.6 percent over previous year driven by area expansion.
- 1.28 The International Sugar Organization (ISO) report (August 2025) projects global sugar production to rise by 3.1 percent to reach 180.6 million tonnes in 2025-26, while consumption is expected to marginally grow (0.4%) and is estimated at 180.8 million tonnes. The ABARES forecast estimates world sugar production to grow by 4 percent and consumption at a marginal rate of 0.2 percent in 2025-26. FAO forecasts an increase of 5.5 percent in production but a modest increase of 1.3 percent in consumption. As per ISO estimates, both ending stocks and Stock-to-Use Ratio (SUR) are expected to marginally fall, but FAO and ABARES forecast an increase in both closing stocks and SUR. The FAO Sugar Price Index in October 2025 witnessed the second consecutive monthly decline and was 5.3 percent lower than in September, 27.4 percent below its level in October 2024 and the lowest level since December 2020. The decline was mainly driven by expectations of ample global supplies from Brazil, Thailand and India, and adverse impact of lower international crude oil prices on biofuel sector.
- 1.29 India's sugar production is estimated at 26.1 million tonnes in 2024-25, registering a negative growth rate of (-) 18.4 percent, while consumption is estimated at 28.1 million tonnes, a decline of 5.4 percent from the previous year. For 2025-26 season, India's sugar production is estimated to increase by 17.2 percent at 30.6 million tonnes, while consumption is forecast to increase by 1.4 percent. High production



coupled with stagnant consumption of sugar in 2025-26 season will result in substantial increase in both closing stocks and SUR, putting sugar prices under downward pressure in the coming season.

Structure of the Report

- 1.30 Chapter 1 presents an overview of sugarcane and sugar sector covering key trends, issues and outlook. Chapter 2 discusses demand-supply situation of sugar in domestic and world market, trends in domestic and international prices of sugar, prices of primary by-products and policy issues related to sugarcane and sugar. In Chapter 3, trends in productivity and production and factors affecting productivity are analysed. Chapter 4 provides an overview of developments in Indian and global sugar markets with respect to production, trade, prices and market outlook. Chapter 5 analyses the costs and returns of sugarcane and discusses inter-crop parity issue. A summary of key considerations and recommendations of FRP for sugarcane and non-price measures are given in Chapter 6.



CHAPTER 2

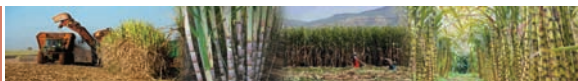


Demand-Supply Scenario and Pricing Policy

- 2.1 Sugarcane is the primary source for sugar production, with Brazil and India being the top two producers of global sugarcane, sugar and sugarcane-based ethanol production. About 80 percent of world's sugarcane production is converted into sugar, while the rest is used for biofuel. Therefore, sugarcane prices are closely linked to the demand and supply dynamics of sugar in domestic and global markets and energy prices. Globally, about 85 percent of sugar is produced from sugarcane, primarily in Asia and South/Central America, while remaining 15 percent is made from sugar beet. Historically, the world sugar market has been highly volatile and one of the most distorted commodity markets. Prices are influenced by factors such as production costs, energy prices, climatic conditions, global trade trends, domestic policies, etc. This chapter provides an overview of sugarcane and sugar production, demand-supply situation, price trends in domestic and global markets and cane pricing issues.

Sugarcane and Sugar Production

- 2.2 Globally, about 2026 million tonnes of sugarcane was produced in 2023, with Brazil accounting for highest production share (38.6%), followed by India with over 24 percent share, as per Food and Agriculture Organization (FAO). Over the last 10 years, sugarcane production in India has increased by about 35 percent from 344.8 million tonnes in TE2017-18 to 466.1 million tonnes in TE2024-25. Among major producing States, Maharashtra (65.6%), Karnataka (54.9%) and Uttar Pradesh (42.9%) recorded significant increase in sugarcane production, while Tamil Nadu (-23.8%), Haryana (-10.1%) and Bihar (-6.4%) recorded a decline in production between TE2017-18 and TE2024-25. The production and production shares of major sugarcane producing States in India during TE2017-18 and TE2024-25 are presented in Table 2.1. Uttar Pradesh continues to be the largest producer with increase in share from 44.7 percent in TE2017-18 to 47.3 percent in TE2024-25. Among the other States, Maharashtra and Karnataka recorded an increase in production share, while Gujarat, Tamil Nadu, Bihar, Haryana, Punjab, Uttarakhand, and Andhra Pradesh witnessed a decline.

**Table 2.1: Sugarcane Production and Share of Major Producing States in India**

State	Production (Million Tonnes)		Share (%)	
	TE2017-18	TE2024-25	TE2017-18	TE2024-25
Uttar Pradesh	154.2	220.3	44.7	47.3
Maharashtra	69.6	115.3	20.2	24.7
Karnataka	32.1	49.7	9.3	10.7
Gujarat	11.7	13.9	3.4	3.0
Tamil Nadu	20.5	15.6	6.0	3.4
Bihar	13.2	12.3	3.8	2.6
Haryana	8.2	7.4	2.4	1.6
Punjab	7.3	7.4	2.1	1.6
Madhya Pradesh	5.1	6.7	1.5	1.4
Uttarakhand	6.2	6.4	1.8	1.4
Andhra Pradesh	8.3	2.4	2.4	0.5
Others	8.3	8.6	2.4	1.8
All India	344.8	466.1	-	-

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 2.3 India has emerged as the world's second largest sugar producer and production has increased from about 25.9 million tonnes in TE2017-18 to 30.4 million tonnes in TE2024-25. Indian sugar production remains volatile due to various factors including erratic weather conditions, while domestic consumption has largely remained stable and tended upwards. Sugar production grew at an average annual growth rate of 4.2 percent during the last 10 years, while consumption recorded an average annual growth rate of 1.5 percent.
- 2.4 The production and production shares of major sugar producing States in TE2017-18 and TE2024-25 are presented in Table 2.2. Uttar Pradesh had the largest share in total sugar production at 33.1 percent in TE2024-25 but a decline of about 2.5 percentage points from TE2017-18. Maharashtra, the second largest producer, witnessed an increase in share from 30 percent in TE2017-18 to 32.6 percent in TE2024-25, while share of Karnataka increased from 12.7 percent to 17 percent during the period. Gujarat, Tamil Nadu, Bihar, Haryana, Punjab, Telangana and Andhra Pradesh witnessed a decline in production share between TE2017-18 and TE2024-25.

**Table 2.2: Sugar Production and Share of Major Producing States in India**

State	Production (Lakh Tonnes)		Share (%)	
	TE2017-18	TE2024-25*	TE2017-18	TE2024-25
Uttar Pradesh	92.1	100.5	35.6	33.1
Maharashtra	77.8	99.1	30.0	32.6
Karnataka	33.0	51.7	12.7	17.0
Gujarat	10.5	9.3	4.0	3.1
Tamil Nadu	10.5	10.6	4.0	3.5
Bihar	5.8	6.5	2.2	2.1
Haryana	6.8	6.3	2.6	2.1
Punjab	7.2	6.1	2.8	2.0
Madhya Pradesh	3.9	5.3	1.5	1.8
Uttarakhand	3.5	3.9	1.3	1.3
Telangana	2.2	2.0	0.9	0.7
Andhra Pradesh	4.7	1.5	1.8	0.5
Others	1.2	1.3	0.5	0.4
All India	259.2	304.1	-	-

Note: * Excludes diversion of ethanol

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution

Sugar Industry Balance Sheet and Outlook

Domestic

- 2.5 Table 2.3 presents the domestic demand-supply situation of sugar for TE2020-21, last four sugar seasons (October-September) and estimates for 2025-26. Sugar production witnessed a continuous fall during the last three seasons, from 35.9 million tonnes in 2021-22 to 26.1 million tonnes in 2024-25, a steep decline of 27.3 percent, while consumption went up by about 2.4 million tonnes from 27.3 million tonnes in 2021-22 to 29.7 million tonnes in 2023-24 but declined to 28.1 million tonnes in 2024-25. Due to a significant drop in sugar production during the 2024-25 season, mainly attributed to adverse weather conditions, Indian sugar sector recorded a deficit of two million tonnes. Exports, which reached a record high of 11 million tonnes in 2021-22, declined to one lakh tonnes in 2023-24 due to export restrictions. However, exports rose to nine lakh tonnes in 2024-25. Closing stocks and Stocks-to-Use Ratio (SUR) significantly increased in 2023-24, but were still below TE2020-21 level. Both closing stocks and SUR registered a steep decline in 2024-25 due to lower production, higher



diversion for ethanol and exports. In 2025-26, domestic output is projected to increase by over 17 percent to reach 30.6 million tonnes and exports are estimated at 1.5 million tonnes. Consumption is projected to reach 28.5 million tonnes. Both closing stocks and SUR are projected to improve in 2025-26 at 5.6 million tonnes and 19.6 percent, respectively. Both National Federation of Cooperative Sugar Factories Ltd. (NFCSF) and Indian Sugar & Bio-energy Manufacturers Association (ISMA) estimates for sugar season 2025-26 also indicate almost the similar trends in demand-supply, stocks and SUR (Annex Table 2.1 and 2.2).

Table 2.3: Domestic Demand-Supply Situation and Outlook of Sugar

(Lakh Tonnes)

Particulars	TE2020-21	2021-22	2022-23	2023-24	2024-25	2025-26 ^(E)
Opening Stocks	120.3	85.0	70.0 ^{\$}	57.0	79.0	50.0
Production [#]	305.3	359.0	330.0	320.0	261.0	306.0
Imports	0.0	0.0	0.0	0.0	0.0	0.0
Exports	56.0	110.0	63.0	1.0	9.0	15.0
Net Availability	369.7	334.0	337.0	376.0	331.0	341.0
Consumption	256.3	273.0	280.0	297.0	281.0	285.0
Closing Stocks	113.3	61.0	57.0	79.0	50.0	56.0
Stocks-to-Use Ratio (%)	44.2	22.3	20.4	26.6	17.8	19.6

Note: 1. ^E Estimated, ^{\$} As per actual stock verification by Cane Commissioners

2. [#] After discounting diversion of 36 LMT, 43 LMT, 24 LMT of sugar to ethanol production during sugar season 2021-22, 2022-23, 2023-24 and 34 LMT in 2024-25 and 2025-26 each.

3. Stocks-to-Use Ratio (SUR) shows closing stocks as percentage of consumption

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

World Sugar Industry

- 2.6 Global sugar production and consumption over the last five seasons is presented in Chart 2.1. It is evident from the Chart that consumption has outpaced production in all years resulting in net deficit. During the last four years (2021-22 to 2024-25), global sugar production expanded by 0.6 percent per annum and consumption grew by 0.8 percent per annum. According to the ISO report (August 2025), global sugar production is estimated to increase by over 5 million tonnes to 180.6 million tonnes in 2025-26, while consumption is forecast to increase marginally by about one million tonnes. The agency expects a small deficit of 0.2 million tonnes in 2025-26 compared to 4.9 million tonnes in 2024-25.



Chart 2.1: Global Sugar Production and Consumption



Source: International Sugar Organization

- 2.7 Table 2.4 presents the global sugar balance-sheet for TE2022-23, last two seasons and forecast for 2025-26 as per by Food and Agriculture Organization (FAO), International Sugar Organization (ISO), United States Department of Agriculture (USDA) and Australian Bureau of Agricultural and Resource Economics and Science (ABARES).
- 2.8 FAO's latest Food Outlook Report (November 2025) forecasts world production to increase by 5.5 percent in 2025-26 at 185.3 million tonnes, reaching a new record level, while global utilization which recorded a steady rise during the last two years is forecast to increase by 1.3 percent at 180.2 million tonnes. Exports are projected to increase by 3 percent and reach 65.2 million tonnes in 2025-26. As a result, ending stocks and SUR are forecast to increase to 126.3 million tonnes and 70.1 percent, respectively in 2025-26.
- 2.9 According to ISO (August 2025), global sugar production is forecast to grow by 3.1 percent to 180.6 million tonnes, while a modest consumption growth of 0.4 percent is projected in 2025-26. Imports, which declined in 2024-25, are set to remain at the same level (63.8 million tonnes) in 2025-26, while exports are expected to marginally decline. Both the ending stocks and SUR, which declined in 2024-25, are forecast to continue the declining trend and are estimated at 92.1 million tonnes and 51 percent, respectively, in 2025-26.



- 2.10 As per the USDA report in May 2025, world sugar production in 2025-26 is projected to increase by 4.7 percent to 189.3 million tonnes. Imports and total supply are forecast to increase marginally, while exports are estimated to decline and consumption is anticipated to increase by about 1.4 percent to 177.9 million tonnes in 2025-26. Ending stocks and SUR are forecast to increase but still remain below the levels of 2023-24 and TE2022-23.
- 2.11 The ABARES has projected a marginal increase (0.2%) in consumption in 2025-26 but a significant increase of 4 percent in production. Contrary to ISO and USDA estimates, ABARES forecasts exports of sugar to rise by 4 percent in 2025-26. The closing stocks and SUR, which recorded a decline in 2024-25, are forecast to increase in 2025-26.
- 2.12 All the estimates indicate larger increase in global sugar production compared to consumption in 2025-26, resulting in higher surplus. FAO, USDA and ABARES estimates indicate higher closing stocks and SUR in 2025-26, putting downward pressure on sugar prices. As per FAO, in October 2025 international prices of sugar declined to the lowest level since December 2020 due to higher production in Brazil, India and Thailand.

Table 2.4: Global Demand-Supply Situation and Outlook of Sugar

(Million Tonnes)

Particulars	TE2022-23	2023-24	2024-25 ^(E)	2025-26 ^(F)
FAO				
Production	174.1	182.7	175.6	185.3
Utilization	172.6	176.1	177.8	180.2
Exports	62.0	67.3	63.3	65.2
Ending Stocks	113.3	122.8	120.5	126.3
Stocks-to-Use Ratio (%)	65.6	69.7	67.8	70.1
ISO				
Production	172.1	181.2	175.2	180.6
Consumption	174.2	181.6	180.1	180.8
Surplus/Deficit	-2.1	-0.4	-4.9	-0.2
Imports Demand	67.0	70.1	63.8	63.8
Exports Availability	67.0	68.8	64.2	63.9
Ending Stocks	99.4	97.8	92.5	92.1
Stocks-to-Use Ratio (%)	57.1	53.8	51.4	51.0
USDA				
Beginning Stocks	48.6	46.2	45.1	38.3
Production	180.1	175.7	180.8	189.3



Particulars	TE2022-23	2023-24	2024-25 ^(E)	2025-26 ^(F)
Imports	57.7	59.9	56.6	57.4
Total Supply	286.3	281.8	282.4	285.0
Exports	63.7	59.9	68.0	65.2
Consumption	173.7	176.1	175.4	177.9
Ending Stocks	48.0	45.1	38.3	41.2
Stocks-to-Use Ratio (%)	27.7	25.6	21.8	23.1
ABARES				
Production	183.8	192.4	186.2	193.6
Consumption	185.1	190.9	191.4	191.7
Exports	67.7	73.9	69.1	71.9
Closing Stocks	98.7	103.7	99.3	101.2
Stocks-to-Use Ratio (%)	53.3	54.4	51.9	52.8

Note: 1. Stocks-to-Use Ratio (SUR) shows ending stocks as percentage of consumption
2. E- estimated, F: forecasted

Source: 1. Food & Agriculture Organization as on 13th November, 2025

2. International Sugar Organization as on August 2025

3. United States Department of Agriculture as on 22nd May, 2025

4. Australian Bureau of Agricultural and Resource Economics and Sciences as on September quarter 2025

Price Trends

Wholesale, Retail and Ex-mill Prices of Sugar

- 2.13 The trends in domestic average wholesale, retail and ex-mill prices of sugar from 2017-18 to 2024-25 are presented in Chart 2.2. Retail and wholesale prices followed almost similar trend with retail prices being higher than wholesale prices. Both retail and wholesale prices declined in 2018-19 due to higher production and stocks, but improved in 2019-20 due to various initiatives of the Government including increase in Minimum Selling Price (MSP) of sugar and higher exports. However, average wholesale price of sugar marginally fell (-0.2%) during 2020-21, while retail prices rose by 0.5 percent. Both retail and wholesale prices followed an upward trajectory during the last four seasons and registered an average annual growth rate of 3.4 percent and 3.7 percent, respectively. The weighted ex-mill prices followed almost a similar trend with a fall in prices during 2018-19 and 2020-21. However, ex-mill prices showed a rising trend during the last four years with an average annual growth rate of 4.8 percent. Despite continuous decline in domestic sugar production during the last three seasons, sugar prices have remained fairly stable with sugar inflation below 4 percent during the last two seasons. All-India monthly average retail and wholesale prices of sugar during last five years are presented in Annex Table 2.3 and 2.4.

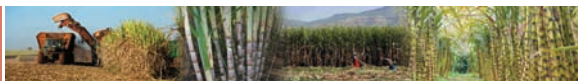
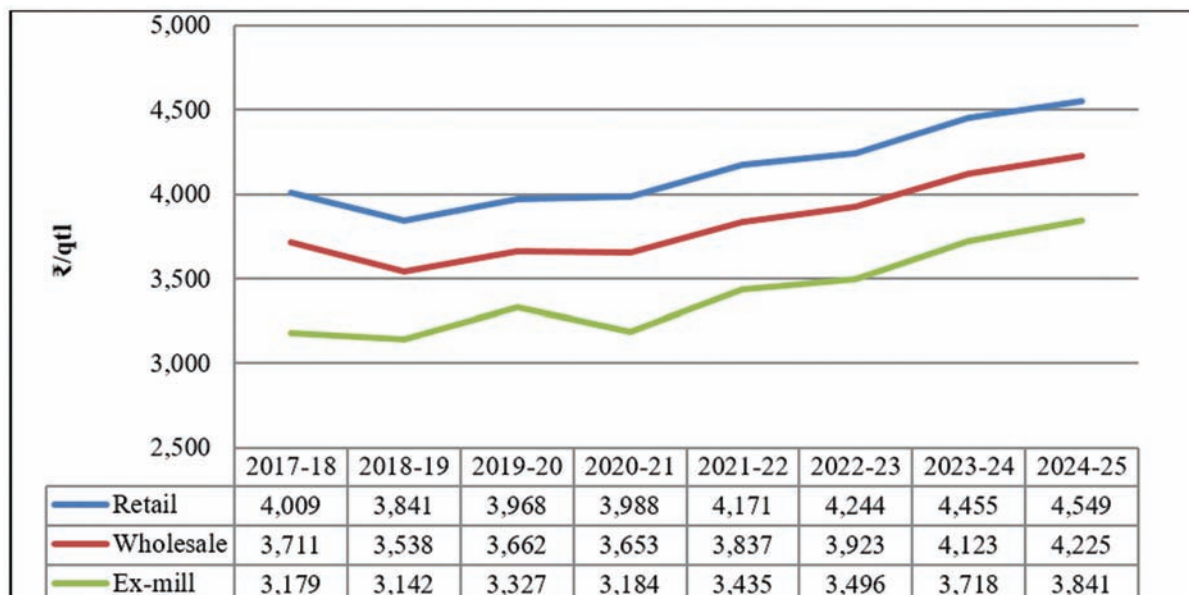


Chart 2.2: Domestic Wholesale, Retail and Ex-mill Prices of Sugar



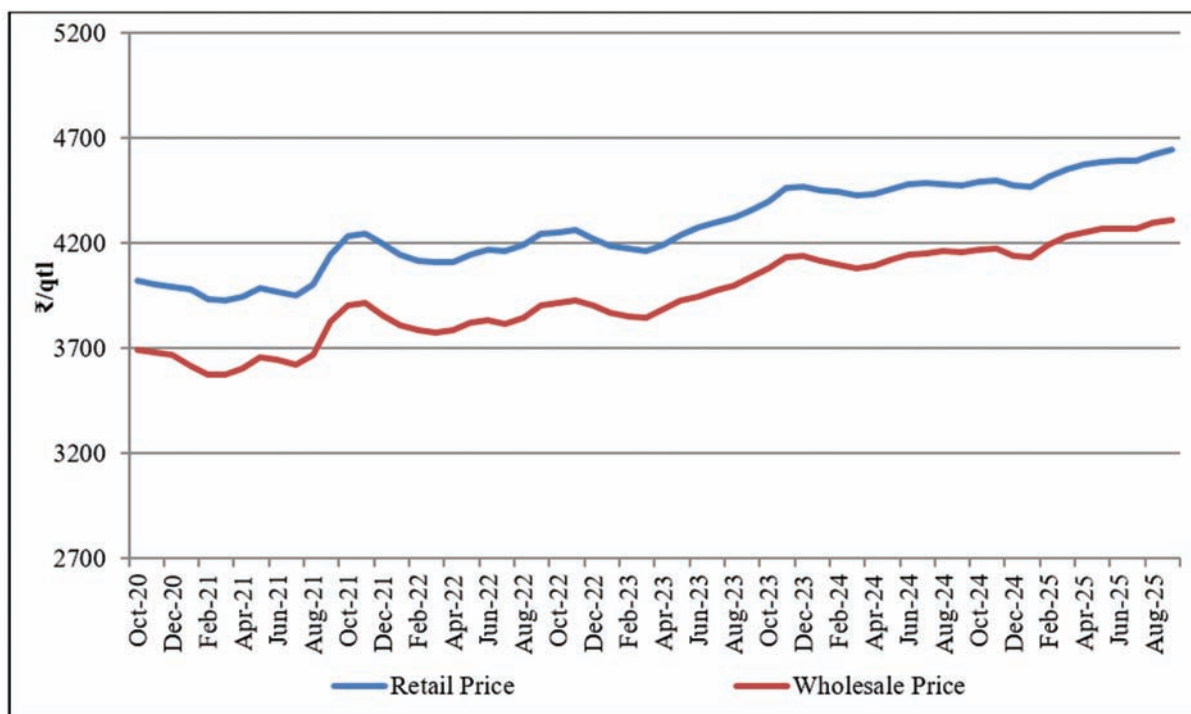
Note: 1. Retail, Wholesale and Ex-mill Prices of sugar are for respective sugar season (October-September)
 2. State-wise Ex-mill prices are presented in Annex Table 2.5

Source: 1. Directorate of Sugar & Vegetable Oils, Department of Food and Public Distribution, Ministry of Consumer Affairs, Food & Public Distribution
 2. Price Monitoring Division, Department of Consumer Affairs, Ministry of Consumer Affairs, Food and Public Distribution
 3. Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare

- 2.14 Trends in monthly wholesale and retail prices of sugar during last five seasons have been presented in Chart 2.3. There has been a co-movement of retail and wholesale prices of sugar and prices have shown less volatility over the last five seasons. However, sugar prices exhibited clear seasonality driven by the seasonal nature of sugarcane production and varying demand patterns mainly due to festivals. The domestic prices of sugar were generally lower between December and March, peak crushing season. The gap between retail and wholesale prices has decreased over the years from 8.4 percent in 2020-21 to 7.1 percent in 2024-25.



Chart 2.3: Domestic Monthly Wholesale and Retail Prices of Sugar



Source: Department of Consumer Affairs, Ministry of Consumer Affairs, Food & Public Distribution

Inflation

- 2.15 The annual WPI inflation rates for sugar and major by-products, namely, molasses, bagasse and gur from 2022-23 to 2025-26 are presented in Chart 2.4. Among sugar and by-products, sugar has the highest weightage of 1.06 percent, followed by molasses (0.04%), bagasse (0.03%) and gur (0.02%). The wholesale inflation rate for sugar rose sharply from 2.6 percent in 2022-23 to 6.5 percent in 2023-24, but declined to 3.2 percent in 2024-25 and marginally rose to 3.9 percent in 2025-26 (April-September). In case of molasses, the annual inflation rate has recorded a steady decline from 8.3 percent in 2022-23 to 3.1 percent in 2025-26 (April-September). For bagasse, the annual wholesale inflation rate was very high (11.6%) in 2022-23 but continuously declined thereafter and became negative (-1.1%) in 2024-25 and marginally rose to 0.3 percent in 2025-26 (April-September). The wholesale inflation rate of gur rose from 4.5 percent in 2022-23 to 8.3 percent in 2023-24 but declined during the last two years and was negative (-1.2%) in 2025-26 (April-September).

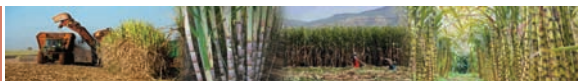
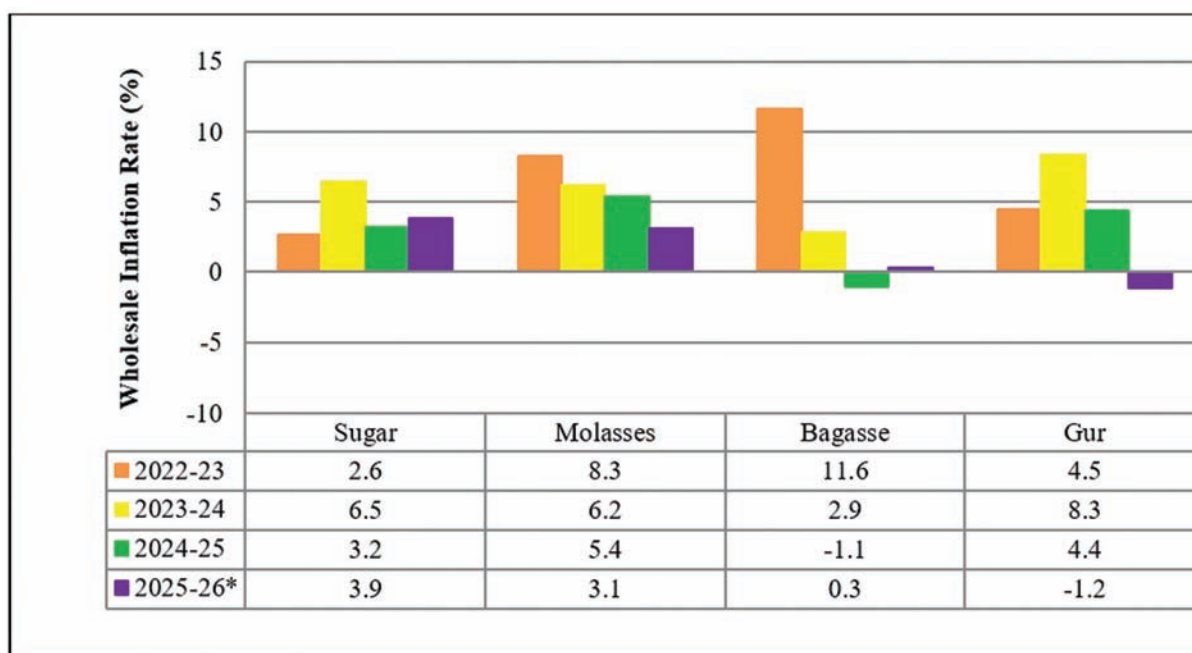


Chart 2.4: Trends in WPI Inflation (Base: 2011-12) for Sugar and By-Products



Note: * upto September, 2025

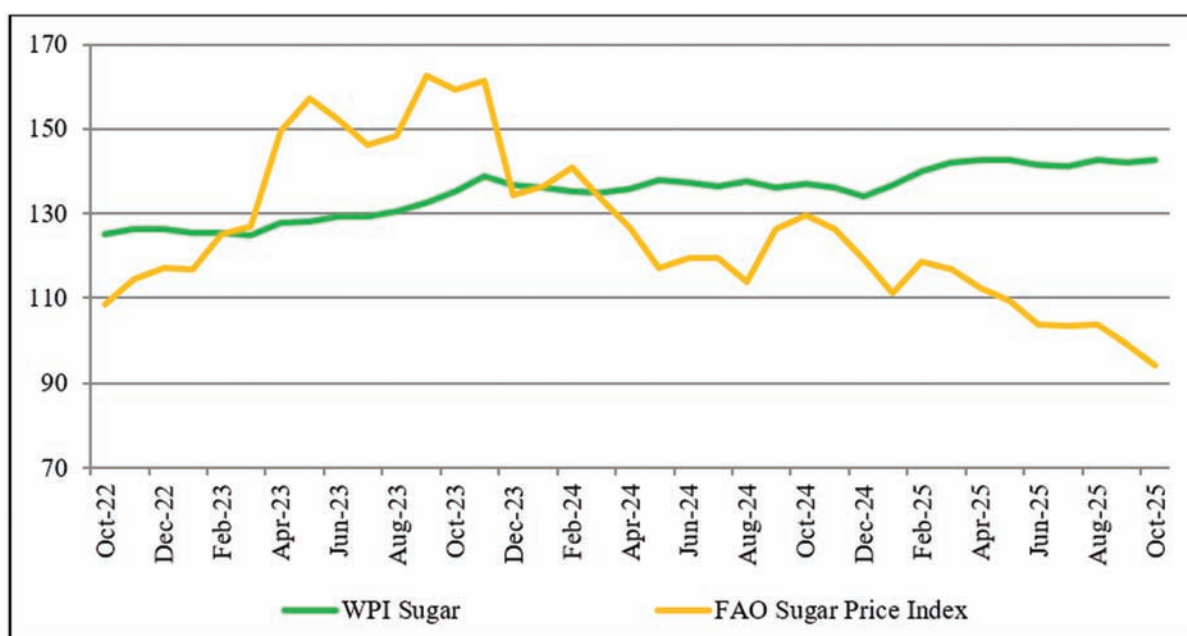
Source: Office of the Economic Advisor, Ministry of Commerce & Industry

Domestic and World Sugar Price Index: Trends and Volatility

- 2.16 Although fluctuations in world prices affect sugar prices in the country, domestic sugar policy continues to drive domestic sugar price movements. Chart 2.5 presents a comparison of Wholesale Price Index (Base 2011-12=100) with FAO Sugar Price Index (Base 2014-16=100). As evident from the Chart, global sugar prices have been characterised by high volatility than domestic prices, particularly during 2022-23. The coefficient of variation of FAO price index was 14.7 percent during October 2022 and October 2025 compared to 4.3 percent in domestic WPI. Domestic wholesale price index for sugar showed a steady upward trend with few dips and increased from 125.1 in October 2022 to 142.6 in October 2025. The FAO Sugar Price Index averaged 94.1 points in October 2025, down 5.3 points from September but 35.5 points below its level a year ago and the lowest level since December 2020. This decline was mainly driven by expectations of ample global supplies and weaker global economic growth. The fall in domestic WPI was much lower during the corresponding period. The WPI of sugar was 142.6 points in October 2025, down 0.5 points from September but 5.7 points above the WPI in October 2024. The FAO sugar price index recorded the highest increase (48.3% y-o-y) in September 2023 and the highest fall (-27.4% y-o-y) in October 2025 during the last three seasons. In case of domestic WPI, the highest rise (10.1%) was recorded in November 2023 and the lowest decline (-2.1%) was recorded in November 2024. These trends clearly indicate that domestic sugar prices have remained more stable than global prices.



Chart 2.5: Trends in Wholesale Price Index of Sugar in India and FAO Sugar Price Index

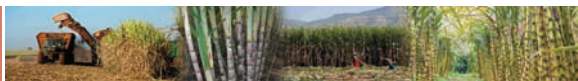


Note: 1. Wholesale Price Index (Base: 2011-12); FAO Sugar Price Index (Base: 2014-16)
 2. Month-wise FAO Price Index of Sugar & Wholesale Prices Indices of Sugar, Molasses, Bagasse and Gur are presented in Annex Table 2.6

Source: 1. Office of the Economic Advisor, Ministry of Commerce & Industry
 2. Food and Agriculture Organization

Prices of Sugar Primary By-Products

- 2.17 The profitability of sugar sector is also influenced by prices of by-products such as molasses, bagasse, press mud, etc. Table 2.5 presents the weighted average prices of primary by-products, namely, molasses, bagasse and press mud. Prices of molasses have shown a continuous and significant upward trend during the last 5 years largely due to rising use of molasses in ethanol production. Among the primary by-products, press mud witnessed the highest average annual growth rate of 12.1 percent, followed by C-Heavy molasses (7.5%), bagasse (7.1%) and the lowest (6.4%) in B-Heavy molasses. In 2024-25, average price of C-Heavy molasses increased by 9.1 percent, B-Heavy molasses by 8.8 percent and bagasse by 2.5 percent but press mud prices declined by a whopping 11.1 percent. The removal of export duty on cane molasses in November 2025 is expected to result in increase in prices of molasses and exports.

**Table 2.5: Weighted Prices of Sugar Primary By-Products**

(₹/tonne)

Particulars	2020-21	2021-22	2022-23	2023-24	2024-25
Molasses B-Heavy	10,047	10,784	11,393	11,843	12,886
Molasses C-Heavy	7,031	7,251	8,210	8,587	9,367
Bagasse	1,924	2,101	2,398	2,462	2,525
Press Mud	291	339	421	500	444

Note: 1. Weighted prices of sugar primary by-products have been calculated by taking sugar production in respective States as weights

2. State-wise weighted prices of sugar primary by-products are given in Annex Table 2.10

Source: 1. State Replies

2. Directorate of Sugar & Vegetable Oils, Department of Food and Public Distribution, Ministry of Consumer Affairs, Food & Public Distribution

Ethanol Prices

2.18 The Ethanol Blended Petrol (EBP) programme was launched by the Government of India in January 2003 for sale of 5 percent ethanol blended petrol in nine States and four UTs, which was extended to 20 States and four UTs with effect from 1st November 2006. However, programme showed a slow progress with average blending ranging from 0.1 percent to 1.5 percent till 2013-14. The National Policy on Biofuels - 2018 marked a significant milestone in advancing biofuel production while supporting the sugar industry and sugarcane farmers. Government has allowed ethanol production using sugarcane-based raw material viz. sugarcane juice, C & B heavy molasses, sugar/sugar syrup, surplus rice with FCI, maize, sugar beet, sweet sorghum, and damaged foodgrains (DFG) such as broken rice and wheat. The current ethanol production capacity in the country has increased from 685 crore litres in ESY2019-20 to 1953 crore litres (49.8 percent of molasses-based and 50.2 percent of grain-based) in ESY2024-25 and blending percentage has increased from 5 percent in ESY2019-20 to 19.2 percent in 2024-25. State-wise ethanol production capacity during last three years is given in Annex Tables 2.9 (a-c).

2.19 Government introduced Administered Pricing Mechanism for ethanol from ESY2014-15 and prices of ethanol produced from sugarcane-based feedstocks such as B-Heavy molasses (BHM), C-Heavy molasses (CHM), sugarcane juice/sugar/sugar syrup are fixed by the Government, whereas prices of ethanol from damaged foodgrains (DFG), FCI rice and maize are fixed by the Oil Marketing Companies (OMCs). Table 2.6 presents the price trends of ethanol from various feedstocks from ESY2020-21 to ESY2024-25. In ESY2021-22, prices of ethanol from all feedstocks witnessed a significant increase ranging from 1.3 percent in sugarcane juice to 2.7 percent in maize. However, prices of ethanol from sugarcane juice, B-Heavy molasses and rice from FCI have not been revised during the last three years. The price of ethanol produced from C-Heavy molasses and maize increased in ESY2024-25 but remained unchanged for DFG. In ESY2025-26, the Government increased the price of FCI rice-based ethanol but there was no revision in price of ethanol produced from sugar-based feedstocks. The EBP programme has significantly contributed towards stability of India's sugar sector, thereby,



reducing problem of cane price arrears. However, non-revision of sugarcane-based ethanol prices might adversely affect Indian sugar industry and sugarcane farmers.

Table 2.6: Prices of Ethanol from various Feedstocks

(₹/litre)

Feedstock	ESY 2020-21	ESY2021-22		ESY2022-23		ESY2023-24		ESY 2024-25	ESY 2025-26 [@]
		Regular Price	Relief amount wef 01 st June 22	Regular Price	Relief amount wef 07 th Aug 23	Regular Price	Relief amount		
Cane Juice	62.65	63.45	1.60	65.61	0.00	65.61	0.00	65.61	65.61
BHM	57.61	59.08	1.49	60.73	0.00	60.73	0.00	60.73	60.73
CHM	45.69	46.66	1.18	49.41	0.00	49.41	6.87 [#]	57.97	57.97
DFG	51.55	52.92	2.34	55.54	8.46	64.00	0.00	64.00	64.00
FCI Rice	56.87	56.87	1.44	58.50	0.00	58.50	0.00	58.50	60.32
Maize	51.55	52.92	2.34	56.35	9.72	66.07	5.79 ^{##}	71.86	71.86

Note: 1. ESY from December-November for ESY2020-21 and ESY2021-22, from December-October for ESY2022-23 and from November-October for 2023-24 onwards

2. [@]Upto C1 Cycle

3. [#] w.e.f 01.11.2023

4. ^{##} w.e.f 05.01.2024

Source: 1. Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution
2. Ministry of Petroleum and Natural Gas for ESY2025-26

Growth in FRP of Sugarcane and Prices of Sugarcane-based Products and By-Products

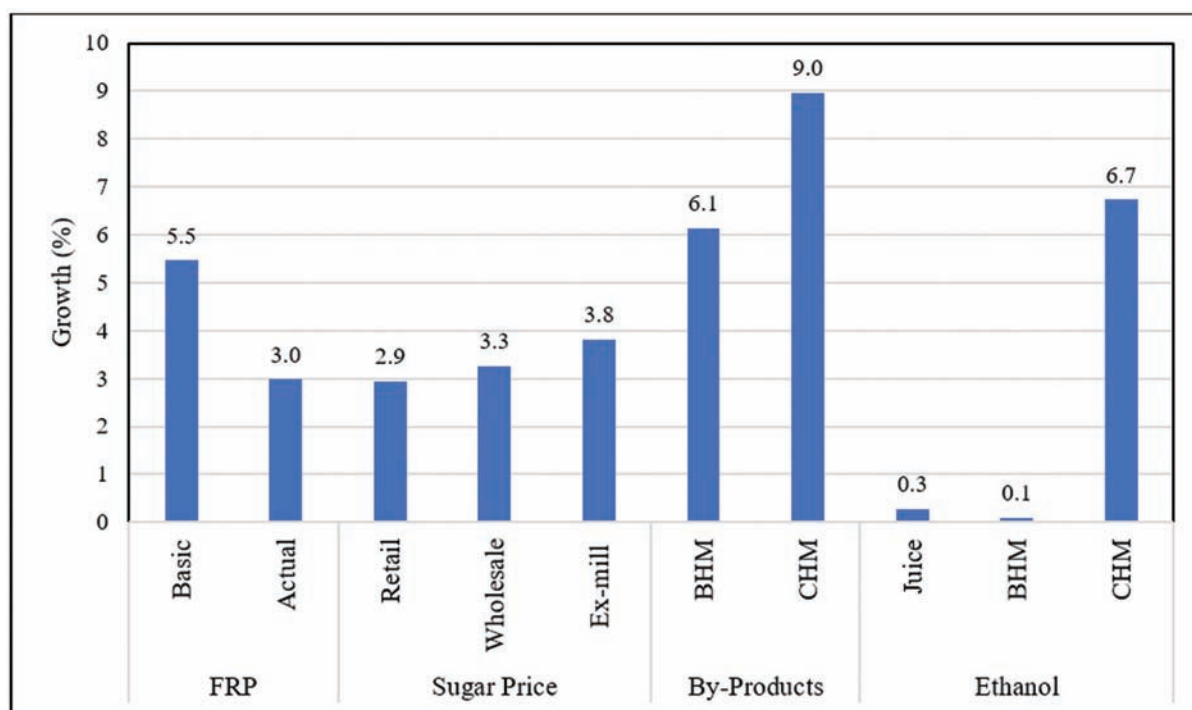
2.20 Chart 2.6 shows average annual growth rate in prices of sugarcane and sugarcane-based products and by-products during the last three sugar seasons. The FRP of sugarcane at a basic recovery rate of 10.25 percent grew at an average annual rate of 5.5 percent but only 3 percent at actual sugar recovery rate due to reduction in recovery rate during the last two seasons. On the contrary, retail, wholesale and ex-mill prices of sugar grew at a much slower pace than the FRP at basic recovery rate. In case of ethanol, except C-Heavy molasses, prices of both B-Heavy molasses and sugarcane juice derived ethanol recorded a negligible increase during the last three seasons due to competition from grain-based ethanol. However, molasses prices witnessed a much higher growth rate, and grew by 9 percent in case of C-Heavy molasses and 6.1 percent in B-Heavy molasses. Since cost of cultivation of sugarcane is rising and both cane productivity and sugar recovery have either stagnated or declined in some areas, it is imperative to reduce costs and improve cane and sugar productivity.

2.21 As sugar industry is facing problem of rising cost of production and low-price realization from sugar and ethanol, possibility of dual pricing of sugar should be explored given the fact that industrial/commercial sector uses significant quantity (60-65%) of total production. Various stakeholders, including sugar



industry, industry associations and some State Governments, have long been suggesting introduction of dual pricing for sugar.

Chart 2.6: Average Annual Growth Rate in FRP of Sugarcane, Sugar and By-Product Prices, and Ethanol Prices



Source: 1. Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution
 2. Price Monitoring Division, Department of Consumer Affairs, Ministry of Consumer Affairs, Food and Public Distribution
 3. Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare
 4. Ministry of Petroleum and Natural Gas
 5. State replies

Distortions in Sugarcane Pricing: State Advised Price

- 2.22 The Central Government announces the Fair and Remunerative Price (FRP) for sugarcane based on the recommendations of the Commission for Agricultural Costs and Prices (CACP), which is paid by sugar producers or their agents to farmers. The FRP is linked to a basic sugar recovery rate with additional premium paid to farmers for higher sugar recovery, thereby promoting efficiency and productivity. However, some States declare State Advised Price (SAP), higher than the Fair and Remunerative Price (FRP), which creates distortions in the sugar sector and results in high cost structures and financial strain on sugar mills.

State Advised Price (SAP)

- 2.23 Four States namely, Haryana, Punjab, Uttar Pradesh and Uttarakhand, announce SAP for different varieties of sugarcane, which is not linked to sugar recovery. Consequently, it neither incentivizes higher sugar recovery nor penalises lower recovery, thereby does not promote efficiency. A comparison of all-India FRP at the basic recovery rate, effective FRP (FRP based on actual average sugar recovery



in the State) and SAP in Haryana, Punjab, Uttar Pradesh and Uttarakhand is presented in Table 2.7. The all-India FRP increased by 17.2 percent from ₹290 per quintal in 2021-22 to ₹340 per quintal in 2024-25, while effective FRP increased by 9.2 percent from ₹319 per quintal to ₹348 per quintal over the same period due to increase in base recovery rate from 10 percent in 2021-22 to 10.25 percent in 2022-23 and slower growth in average sugar recovery rate in the recent years. The SAP in 2024-25 was highest (₹393/quintal) in Haryana, followed by Punjab (₹391/quintal), Uttarakhand (₹365/quintal) and the lowest in Uttar Pradesh (₹360/quintal). The SAP for different varieties during the last five seasons is given in Annex Table 2.7. During the last four seasons, Uttarakhand witnessed the highest increase (20.1%) in effective FRP between 2021-22 and 2024-25 due to improvement in sugar recovery rates, followed by Haryana (12.4%), Punjab (11.9%) and Uttar Pradesh (6.6%) against 9.2 percent increase in effective FRP at all-India level. The effective FRP was the highest for Uttarakhand, followed by Uttar Pradesh, Haryana and Punjab in 2024-25. The highest increase in SAP of sugarcane between 2021-22 and 2024-25 was in Haryana (10.7%) from ₹355 per quintal to ₹393 per quintal, followed by Punjab (8.6%), Uttar Pradesh (5.9%) and Uttarakhand (5.8%).

- 2.24 Recently, the Government of Uttar Pradesh increased SAP of sugarcane by ₹30 per quintal for 2025-26 season. The SAP of early-maturing varieties in Uttar Pradesh has been raised from ₹370 per quintal to ₹400 per quintal, while the price for common varieties has been increased from ₹360 to ₹390 per quintal. Government of Haryana has also announced ₹15 per quintal hike in SAP of sugarcane for 2025-26. The SAP for early-varieties has increased from ₹400 to ₹415 per quintal and for late-varieties from ₹393 to ₹408 per quintal. The SAP announced by Haryana and Uttar Pradesh are significantly higher than the FRP for 2025-26 season and will put additional financial burden on sugar mills in the State.
- 2.25 The Commission has been arguing for discontinuing SAP by State governments as adoption of FRP would encourage and incentivise higher sugar recovery, remove inter-State price distortions and also help in improving financial health of sugar mills. Those States which want to continue with SAP, should pay the difference between FRP and SAP directly to farmers through Direct Benefit Transfer (DBT) to reduce financial burden on sugar mills and help in timely cane payment to farmers. It is worth mentioning that in Punjab, State Government pays 100 percent differential amount between SAP and FRP in case of co-operative sugar mills and two-third in case of private sugar mills, while in Haryana, State Government provides subsidy to sugar mills based on sugar recovery rate to meet the gap between SAP and FRP. The Commission recommends that Uttar Pradesh and Uttarakhand should also adopt DBT mechanism for differential between SAP and FRP.

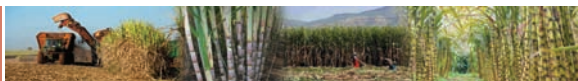


Table 2.7: Effective FRP and SAP of Sugarcane during Last Four Years

(₹/qtl)

Year	FRP* All India	Effective FRP ^s					State Advised Price (SAP)			
		HR	PB	UP	UK	All India	HR ^α	PB [@]	UP [#]	UK [#]
2021-22	290	286	282	332	301	319	355	360	340	345
2022-23	305	301	315	339	301	333	365	370	340	345
2023-24	315	323	310	353	320	338	379	381	360	365
2024-25	340	321	315	354	362	348	393	391	360	365
2025-26	355	-	-	-	-	-	408	NA	390	NA

- Note: 1. *At 10% sugar recovery rate for 2021-22 and 10.25% for 2022-23, 2023-24 and 2024-25
 2. ^s Effective FRP calculated at State-specific sugar recovery rate inclusive of sugar diverted to ethanol
 3. ^α State Advised Price is for Mid & Late variety
 4. [@] State Advised Price is for Mid variety
 5. [#] State Advised Price is for Common variety

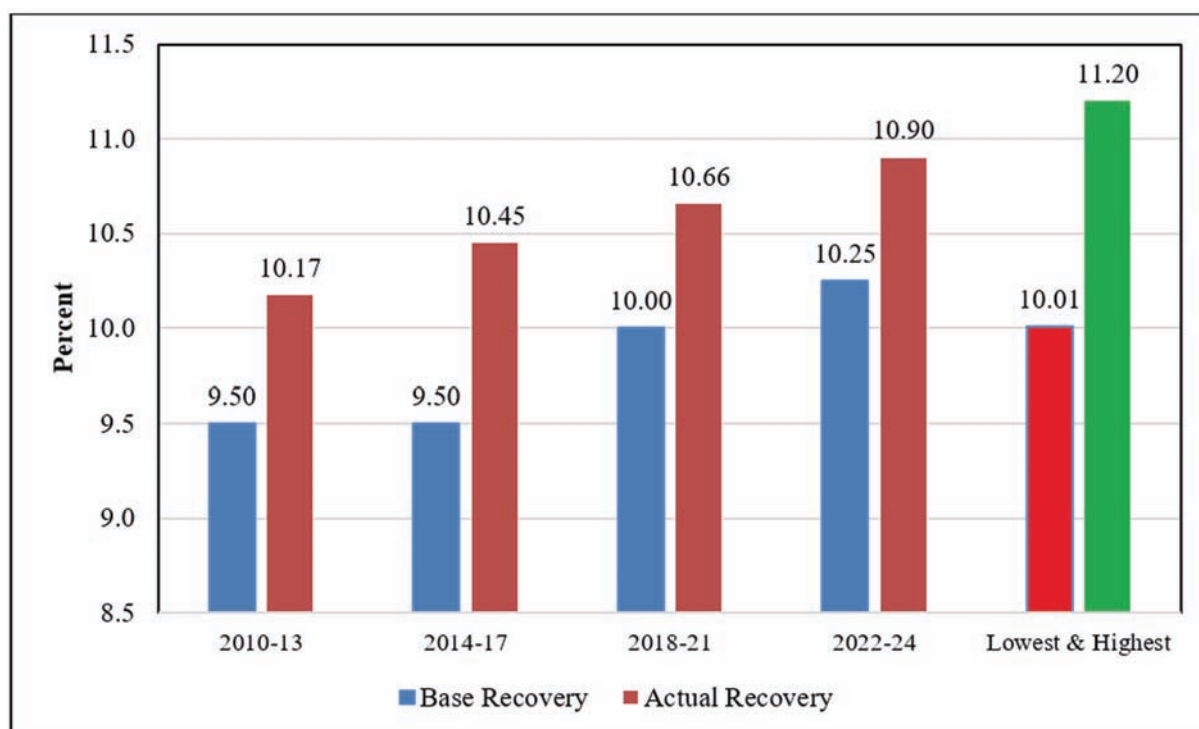
Source: 1. State replies
 2. Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution

Trends in Sugar Recovery Rate

- 2.26 The trends in basic and actual sugar recovery rate at all-India level from 2010-11 to 2024-25 are presented in Chart 2.7. The basic sugar recovery rate has been revised three times during the last 16 years, in 2009-10 (from 9.0% to 9.50%), 2018-19 (10.0%) and 2022-23 (10.25%), which is in line with notable improvement in actual recovery rate over the past one and half decade, from an average of 10.17 percent in 2010-11 to 2013-14 to 10.45 percent in 2014-15 to 2017-18 season and was 10.90 percent during 2022-23 to 2024-25 season. However, the average recovery rate witnessed a continuous decline during the last two seasons from 11.2 percent in 2022-23 to 11 percent in 2023-24 and 10.5 percent in 2024-25 due to *El-Niño* induced adverse weather conditions and incidence of pests and diseases. Over the last 10 seasons, FRP of sugarcane at basic recovery rate has increased at an annual average growth rate of 4.5 percent, but actual all-India sugar recovery rate has increased very marginally (0.1%). Therefore, efforts should be made to develop climate resilient and disease & pest resistant high-yielding sugarcane varieties. Such efforts would not only stabilize but also enhance sugar recovery and cane yield, contributing to greater efficiency and sustainability in sugar sector.



Chart 2.7: Trends in Basic and Actual Recovery of Sugar from Sugarcane

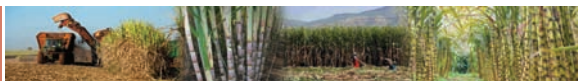


Note: 1. All-India Sugar Recovery Rate for 2021-22 to 2024-25 includes sugar diverted to ethanol production
 2. State-wise sugar recovery rate from 1999-00 to 2024-25 is presented in Annex Table 2.8

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

Cane Price Payment to Farmers

- 2.27 There has been significant increase in cane payment to farmers due to increase in FRP and improvement in productivity (cane yield and sugar recovery) over the last decade. Total cane payment to farmers has increased from ₹55,464 crore in 2015-16 to over one lakh crore in the recent years. However, excess availability of sugar in the country coupled with subdued prices of sugar during the second half of last decade resulted in accumulation of sugar stocks, which adversely affected liquidity of sugar mills and resulted in cane price arrears to farmers. To address these issues, the Central Government introduced various measures, including the Ethanol Blended Petrol (EBP) Programme under the National Biofuel Policy - 2018, financial support to offset cane costs and maintain buffer stocks, export facilitation, soft loans, Minimum Selling Price (MSP) for sugar, interest subvention to boost ethanol production, etc. As evident in Table 2.8, these measures have resulted in addressing the problem of cane price arrears and improving financial health of sugar sector. In 2019-20, cane price arrears were as high as 16.4 percent of total cane price payable, which showed a continuous declining trend and were only 3.4 percent of price payable in 2024-25 season.
- 2.28 The reduction in cane price arrears can be largely attributed to increased diversion of sugar to ethanol production from 3 lakh tonnes in 2018-19 to 43 lakh tonnes in 2022-23 season, which declined during the last two seasons due to ban/restrictions on use of sugarcane juice, sugar syrup, and B-Heavy molasses for ethanol production for the ESY2023-24 and 2024-25 to ensure adequate sugar availability for domestic consumption and keep prices in check. However, Government has removed all restrictions



on ethanol production from sugarcane juice, syrup, and all types of molasses for ESY2025-26. Blending percentage in petrol has increased from 5 percent in 2018-19 to 19.2 percent in 2024-25, which is in line with Government's target of 20 percent blending of ethanol with petrol by ESY2025-26. These initiatives have helped stabilize sugar prices, achieve blending target and timely cane payments to farmers. Continued efforts to expand ethanol production programme and timely payment to farmers are vital for sustainable growth of sugar sector, but sharp increase in share of corn in ethanol production in recent years might adversely affect sugar sector.

Table 2.8: Cane Price Arrears, Sugar Production and Diversion to Ethanol Blending

Season	Price (₹ crore)		Arrears* (₹ crore)		Sugar (lakh tonnes)		Blending (%)	Quantity Supplied (crore litre)
	Payable	Paid*	Total	Percent	Production	Diversion to Ethanol		
2018-19	86,357	76,903	9,454	10.9	332.0	3.0	5.0	188.6
2019-20	75,444	63,048	12,396	16.4	274.0	9.0	5.0	173.0
2020-21	91,676	85,418	6,258	6.8	310.0	22.0	8.1	295.6
2021-22	1,18,117	1,12,207	5,910	5.0	359.0	36.0	10.0	418.7
2022-23	1,14,494	1,09,014	5,480	4.8	330.0	43.0	12.1	506.4
2023-24	1,11,615	1,07,755	3,860	3.5	320.0	21.6	14.6	679.0
2024-25	1,02,447	98,998	3,449	3.4	261.1	34.9**	19.2**	1,003.1

Note: * as on 30th September

** upto 12th October 2025

Source: 1. Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution
2. Ministry of Petroleum and Natural Gas

- 2.29 Table 2.9 presents the State-wise cane price arrears for 2024-25 season (as on 30th September, 2025). Among top-producers, Uttar Pradesh has the highest arrears amounting to ₹2,401 crore, followed by Gujarat (₹310 crore), Maharashtra (₹261 crore), Punjab (₹236 crore), Tamil Nadu (₹149 crore), while Karnataka has the lowest arrears. In Gujarat, cooperatives sugar mills make payment to member farmers in three equal instalments: first instalment is paid at the time of supply of sugarcane, second in the month of April, and balance one-third after the sugar factory accounts are finalised. These trends clearly indicate that various policy interventions by the Central Government have resulted in significant reduction in cane price arrears and ensured timely payment to farmers.



Table 2.9: State-Wise Cane Price Arrears for 2024-25 Season

State	Cane Price (in ₹ crore)		Arrears* (in ₹ crore)	
	Payable	Paid*	Total	Percent
Chhattisgarh	218	186	32	14.7
Punjab	2,532	2,296	236	9.3
Gujarat	3,514	3,204	310	8.8
Uttar Pradesh	34,948	32,547	2,401	6.9
Andhra Pradesh	412	384	28	6.8
Tamil Nadu	2,412	2,263	149	6.2
Uttarakhand	1,450	1,430	20	1.4
Maharashtra	31,599	31,338	261	0.8
Haryana	2,225	2,214	11	0.5
Karnataka	18,222	18,221	1	neg.
All India	1,02,447	98,998	3,449	3.4

Note: 1. * As on 30th September 2025

2. neg. : negligible

3. Bihar, Madhya Pradesh, Odisha and Telangana have nil arrears in 2024-25

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

Capacity Expansion and Utilization Trends

- 2.30 Over the last three decades, sugar production capacity in the country has significantly expanded due to various policy initiatives like deregulation of sugar sector in 1998 and Government's strong support for promotion of biomass-based cogeneration and ethanol blending in sugar mills, etc. However, there are large variations in total capacity and capacity utilization of sugar factories across major producers. Maharashtra has the highest installed capacity (176 lakh tonnes), followed by Uttar Pradesh (160.9 lakh tonnes), Karnataka (102.2 lakh tonnes) and the lowest in Gujarat as is evident in Chart 2.8. However, capacity utilization was highest in Gujarat (65.9%), followed by Uttar Pradesh (62.5%), Maharashtra (56.3%) and lowest in Tamil Nadu (50%). The reduced availability of cane in some States and increase in capacity expansion of sugar mills have led to low capacity utilization.

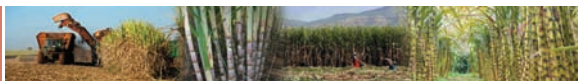
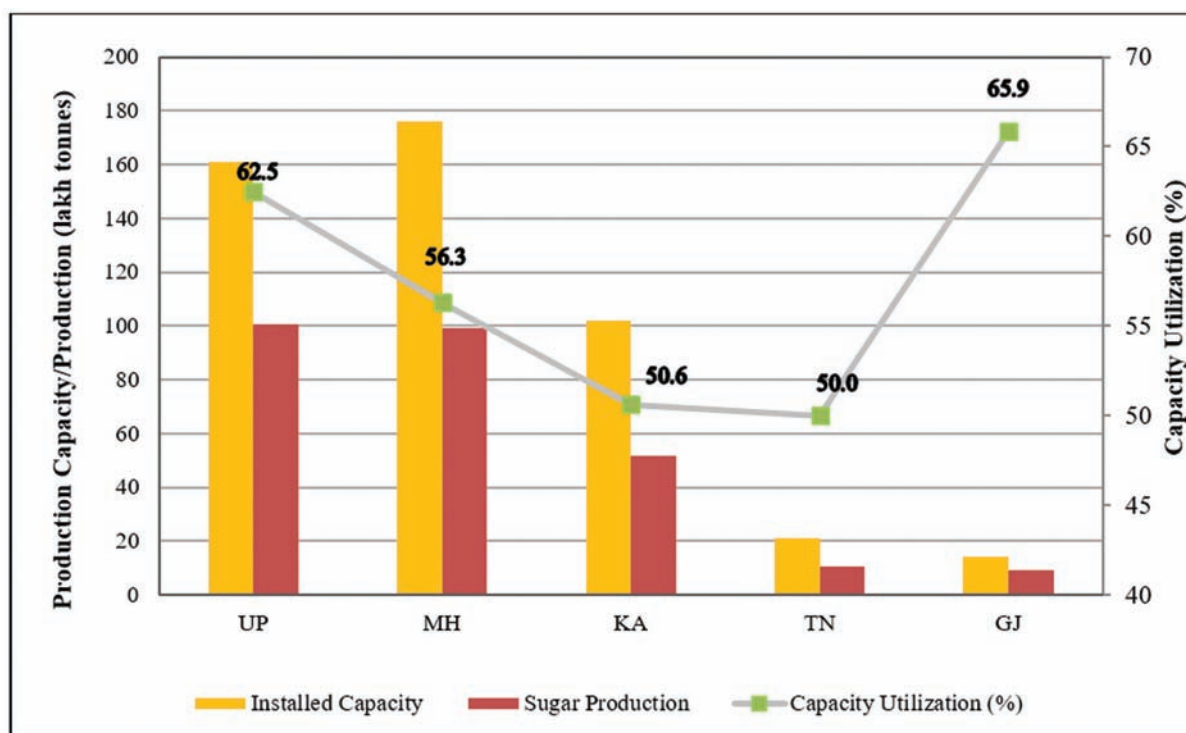


Chart 2.8: Installed Capacity, Sugar Production & Capacity Utilization in Major Sugar Producing States for TE2024-25



Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

- 2.31 Table 2.10 presents trends in installed capacity, capacity utilization and average capacity per plant from TE1996-97 to TE2024-25. Total number of sugar factories in the country has increased from about 443 in pre-delicensing period (TE1996-97) to 765 in TE2024-25, nearly 73 percent increase. On the contrary, the number of operational factories increased by 31.2 percent during the period, indicating surplus capacity. The installed capacity increased by more than 4 times from 128.1 lakh tonnes in TE1996-97 to about 548 lakh tonnes in TE2024-25 but capacity utilization significantly fell from about 115 percent to 55.6 percent during the period. The average capacity per factory increased by 2.5 times and was estimated at 71,547 tonnes. The increase in number of factories and their capacity has resulted in low capacity utilization primarily due to reduced availability of cane.
- 2.32 The cane area reservation and minimum distance criteria are intended to assure cane availability to sugar mill and avoid unhealthy competition among sugar factories. However, in view of expansion in capacity of sugar mills and increase in number of sugar factories in the recent years, it is necessary to re-visit the minimum distance criterion of 15 km and increase it to minimum 25 km to improve capacity utilization and avoid unhealthy competition. The number of total sugar mills, operational mills, sugar production and recovery rates of sugar in major sugar producing States during 2024-25 season are given in Annex Table 2.11.



Table 2.10: Installed Capacity, Capacity Utilization and Average Capacity per Plant in Sugar Mills: TE1996-97 to TE2024-25

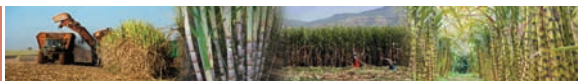
Years	Number of Factories		Capacity(Lakh Tonnes)		Average Capacity per Plant (Tonnes)
	Total	Operational	Installed	Utilization (%)	
TE1996-97	443	407	128.1	114.9	28,935
TE2009-10	628	495	230.4	86.8	36,681
TE2017-18	731	511	338.3	76.5	46,287
TE2024-25	765	534	547.6	55.6	71,547

Source: 1. National Federation of Cooperative Sugar Factories Ltd.

2. Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

Recapitulation

- 2.33 The domestic output of sugar continued the declining trend in 2024-25 and decreased by over 18 percent leading to reduced availability and lower closing stocks despite fall in consumption and higher opening stocks. For 2025-26 season, preliminary estimates indicate bumper harvest of sugarcane due to favourable weather conditions and expansion in acreage leading to strong rebound in sugar production. By contrast, consumption is forecast to rise marginally, resulting in larger ending stocks and higher SUR. On the global front, FAO, ISO, USDA and ABARES forecasts significant increase in global production but marginal rise in consumption in 2025-26. As a result, FAO, USDA and ABARES project ending stocks to rise and SUR to be higher in 2025-26.
- 2.34 Domestic wholesale, retail and ex-mill prices of sugar continued with a steady upward trend during the last five seasons. In contrast, global sugar prices have been more volatile and FAO Sugar Price Index tended downward since February 2025. Significant increase in SAP has been observed in all four States during the last four seasons. Government interventions have resulted in significant reduction in cane price arrears in recent years and nearly 97 percent of cane arrears have been cleared for 2024-25 season. The sugar industry expanded its capacity and continues to operate with excess capacity which needs to be addressed. There are substantial variations in capacity utilisation across major sugar producing States.



CHAPTER 3



Productivity and Production Performance: Trends and Issues

- 3.1 Increasing sugarcane productivity and sugar recovery is crucial to meet rising demand, minimising cost of production and ensuring higher returns for farmers. Yield enhancement is also vital for sustainable supply of ethanol for blending with petrol and promoting environmentally sustainable production, given land scarcity and sugarcane being water-intensive crop. Higher sugarcane productivity would help in improving global competitiveness of Indian sugar industry.
- 3.2 This Chapter analyses the trends in sugarcane area, production and productivity at national, sub-national and disaggregated levels. India's sugarcane yield is compared with major sugarcane producing countries and with respect to potential yield obtained under Frontline Demonstrations. Factors affecting sugarcane yield and key initiatives aimed at achieving sustainable yield improvements amid challenges of resource scarcity and climate change are also discussed in this Chapter.

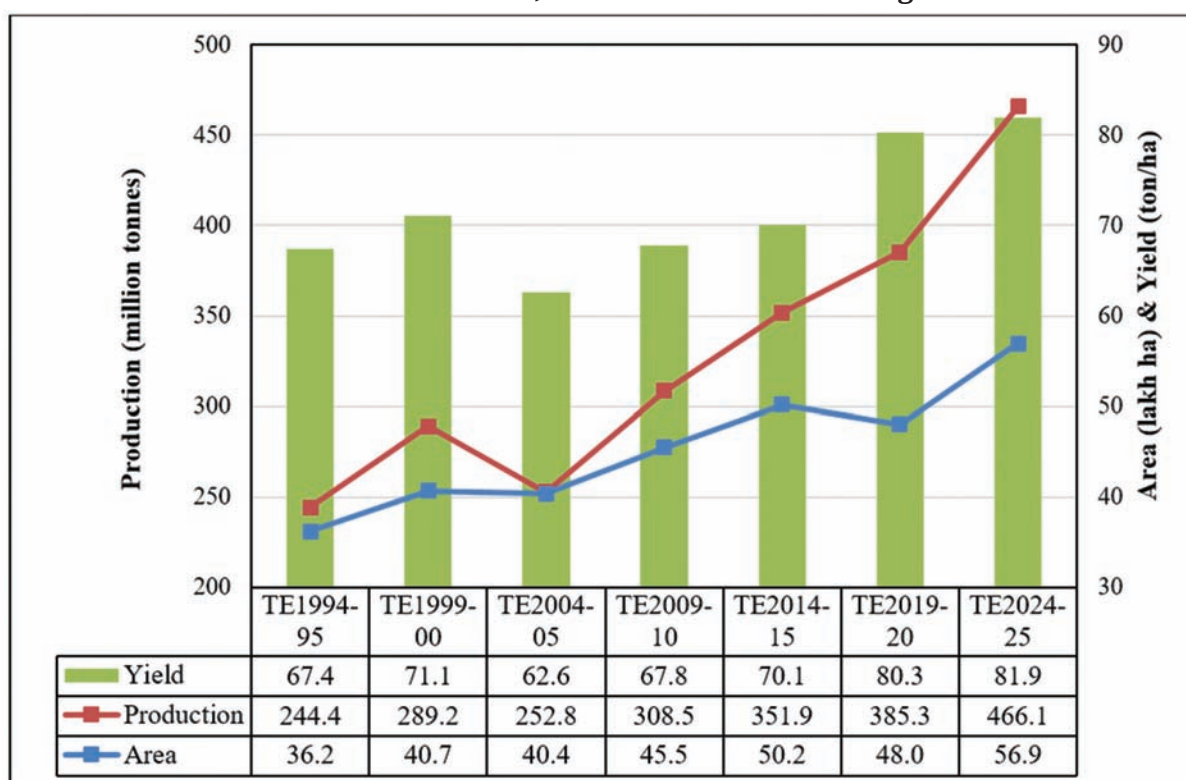
Trends in Area, Production and Yield

All-India

- 3.3 Chart 3.1 demonstrates all-India trends in area, production and yield of sugarcane during the period TE1994-95 to TE2024-25. Sugarcane production, which increased from 244.4 million tonnes in TE1994-95 to 289.2 million tonnes in TE1999-00, declined to 252.8 million tonnes in TE2004-05. Thereafter, production steadily increased, though growth rate followed a cyclical trend.
- 3.4 The average annual growth in production decelerated from 4.5 percent in early-1990s to 1.7 percent in late-1990s and turned negative (-)4.3 percent during the period of 2000-01 to 2004-05 (Table 3.1). This was followed by a significantly high growth of 5.5 percent during 2005-06 to 2009-10, which declined to 4.6 percent in early-2010s and further to 1.2 percent in late-2010s. During last five years, sugarcane production increased at an average annual growth rate of 4.4 percent.



Chart 3.1: All-India Area, Production and Yield of Sugarcane



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.5 It is noteworthy that growth in area has played an important role in increasing sugarcane production. As shown in Table 3.1, growth in area has been higher than the growth in yield during all periods except for 2000-01 to 2004-05 and 2015-16 to 2019-20. During the last five years (2020-21 to 2024-25), area growth (3.7%) was the major source of growth in production, while yield increased at average annual growth rate of 0.8 percent. Since area expansion has its limitations, concerted efforts are required to increase sugarcane yield.

Table 3.1: Growth in Area, Production and Yield of Sugarcane in India

Period	Area	Production	Yield
1990-91 to 1994-95	2.6	4.5	1.7
1995-96 to 1999-00	1.9	1.7	0
2000-01 to 2004-05	-2.6	-4.3	-1.7
2005-06 to 2009-10	3.5	5.5	1.7
2010-11 to 2014-15	4.1	4.6	0.5
2015-16 to 2019-20	-1.6	1.2	2.6
2020-21 to 2024-25	3.7	4.4	0.8

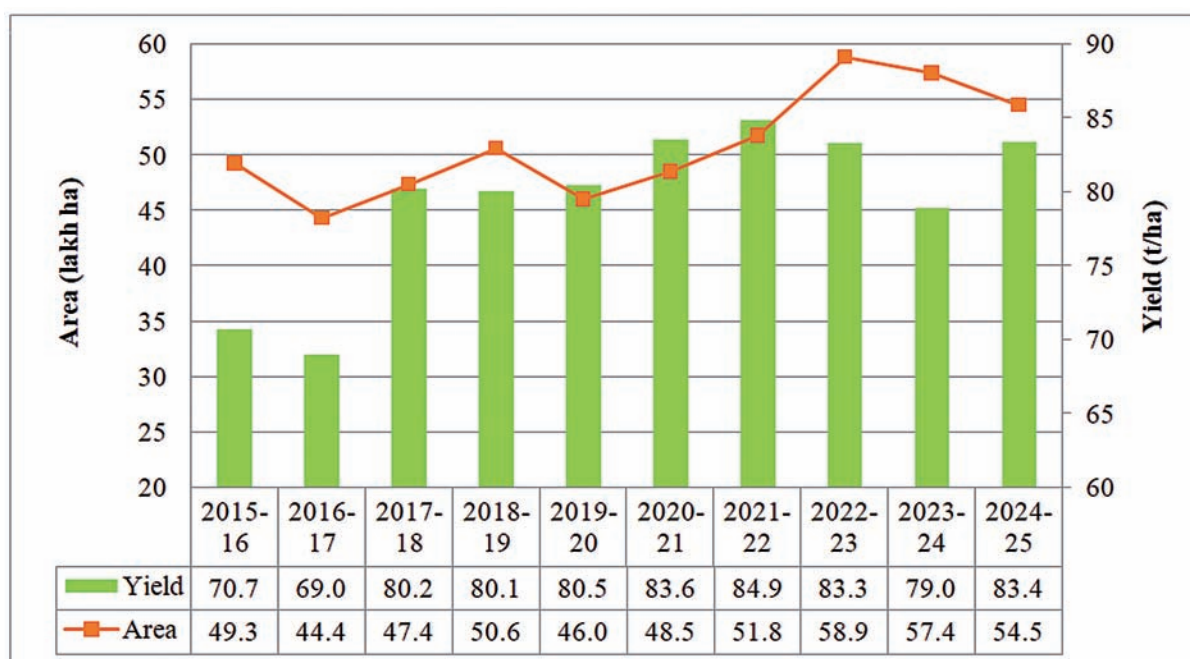
Note : Growth is average of year-on-year growth during the period

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



- 3.6 Area under sugarcane during last ten years varied between 44.4 lakh hectare in 2016-17 to 58.9 lakh hectare in 2022-23 (Chart 3.2). A closer look at trends in area shows that cycle of decline in area is repeating every 3-4 years and thereby affecting production. Sugarcane area, which declined by 10 percent in 2016-17 from 49.3 lakh hectare in 2015-16, registered a growth of 6.8 percent for next two years. However, area declined by more than 9 percent in 2019-20, which registered a positive growth of 5.4 percent in 2020-21 and further improved to 13.7 percent in 2022-23. Area has continuously declined during last two years, by 2.5 percent in 2023-24 and 5 percent in 2024-25.
- 3.7 Sugarcane yield, which marginally declined from 70.7 tonnes per hectare in 2015-16 to 69 tonnes per hectare in 2016-17, increased by more than 16 percent in 2017-18 but remained almost stagnant for the next two years. Yield increased by 3.8 percent in 2020-21 and 1.6 percent in 2021-22 over the previous year but again declined for two consecutive years. In 2024-25, sugarcane yield improved by 5.7 percent.

Chart 3.2: All-India Area and Yield of Sugarcane: 2015-16 to 2024-25



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.8 Though sugarcane in India is largely an irrigated crop with more than 95 percent of area under irrigation, rainfall variation significantly affects both acreage and yield. The back-to-back drought for two consecutive years, 2015 and 2016, significantly affected area planted and yield of sugarcane. In 2015, two-third of top-30 sugarcane-growing districts received deficient rainfall, including one district with large deficient rainfall. In 2016, 13 districts received deficient rainfall, while two districts received excess rainfall (Table 3.2). In 2019, ten districts were under deficient to large deficient category of rainfall, while seven districts were under excess (4) to large excess (3) category. In 2020, 11 districts received excess rainfall and eight districts received deficient rainfall, while in 2021, a large number of districts (14) suffered from excessive rainfall, with one district having large excess (76%). Similarly, 2023 was also an abnormal year as 19 out of 30 districts received either deficient/large deficient rainfall (14) or excessive rainfall (5). Excess rainfall in poor drainage areas adversely affects both cane yield and sugar recovery.



Table 3.2: Rainfall Deviations in Top-30 Sugarcane Producing Districts

(% deviation from Long Period Average)

District	State	Prod. Share (%)	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Kheri	UP	5.9	-15	13	-4	12	-37	-6	52	44	-12	-8
Belagavi	KA	4.8	-31	-16		6	83	49	38	44	-21	17
Solapur	MH	4.7	-46	-6	18	-46	-6	36	34	10	-34	
Bijnor	UP	4.6	-9	-12	-3	10	-4	4	10	-3	42	-7
Muzaffarnagar	UP	3.3	-10	-36	-36		-13	5	22	7	41	-11
Pune	MH	3.3	-21	20	44	5	105	48	20	33	-8	39
Sitapur	UP	3.3	-40	-34	-36	-24	-30	-29	9	-26	-26	-40
Ahmednagar	MH	3.2	-17	12	39	-36			45	37	-18	31
Kolhapur	MH	3.1	-51	-16	-13	-8	65	25	28	22	-21	40
Meerut	UP	2.8	-36	-42	-34	-18	-25	-39	-3	-5	23	-11
Satara	MH	2.5	-29	18	11	-4	57	6	29	27	-37	12
Sangli	MH	2.2	-34	-13	-16	-37	30	39	2	-9	-46	30
Saharanpur	UP	2.0	-23	-35	-21	10	-31	4	-13	-20	9	-37
Pashchim Champaran	BR	1.8	-27	-4	40	4	17	57	76	-4	-38	-7
Bagalkote	KA	1.8	-22	-30	5	-39	37	57		54	-36	6
Bareilly	UP	1.8	-18	22	-7	33		-1	32	9	16	16
Pilibhit	UP	1.7	-55	-51	-60	-33	-53	-24	-46	-20	-46	-47
Shahjahanpur	UP	1.6	-36	-29	-43	2	-15	-39	-15	-41	-13	-23
Hardoi	UP	1.5	-29	-22	-39	-32	-16	-45	-44	-32	-16	-16
Shamli	UP	1.5	-10	-36	-36		-68	-36	-46	-35	-56	-72
Amroha	UP	1.4	-27	-10	-11	-30	-48	-47	-37	-33	-17	-37
Vijayapura	KA	1.4		-29	-2	-42	-1	34	6	35	-31	14
Gonda	UP	1.4	-49	-24	-42	-39	-9	6	24	-8	-22	-13
Haridwar	UK	1.4	-11	-3	-8	13	-23	2	-20	-22	38	-20
Kushi Nagar	UP	1.3	-78	-65	-65	-83	-28		22	-53	-61	-54
Surat	GJ	1.3	-31	-16	-5	-1	43	59	18	35	2	54
Balrampur	UP	1.3	-39	-40	-48	-42		31	44	21	-17	39
Moradabad	UP	1.2	-8	-3	-5	16	-6	-13	11	-16	27	14
Bulandshahr	UP	1.2	-29	-24	-30	-40	-40	-54	-17	-10	-16	6
Beed	MH	1.1	-43		9		7	40	59	22	-28	17

Note: 1. Large excess ($\geq 60\%$)
 Excess (≥ 20 but $< 60\%$)
 Normal ($\geq -20\%$ but $< 20\%$)
 Deficient ($\geq -60\%$ but $< -20\%$)
 Large deficient ($\geq -99\%$ but $< -60\%$)
 No data available

2. Districts arranged in descending order based on sugarcane production in TE2023-24

Source : Rainfall Statistics of India, India Meteorological Department, Ministry of Earth Sciences



Major States

- 3.9 Sugarcane cultivation in the country can be broadly classified into two major agro-climatic regions viz. tropical and sub-tropical. In TE2024-25, the contribution of sub-tropical region, mainly covering Bihar, Haryana, Punjab, Uttar Pradesh and Uttarakhand, was higher than the tropical region, 55.9 percent of total area and 54.5 percent of production. However, yield is relatively higher in tropical States due to favourable climatic conditions.
- 3.10 Trends and average growth in area, production and yield of sugarcane in major producing States of tropical and sub-tropical regions during last 20 years are exhibited in Table 3.3. Uttar Pradesh, followed by Maharashtra, Karnataka, Tamil Nadu, Gujarat and Bihar were the major sugarcane producing States during TE2024-25, together accounting for around 91.7 percent of the total production and more than 90 percent of area. Uttar Pradesh, Maharashtra and Karnataka have seen a steady increase in production during last 20 years, except a marginal decline during TE2009-10 in Uttar Pradesh and during TE2019-20 in Karnataka. Bihar recorded an increase in production up to TE2019-20 but showed a decline of 22.2 percent during TE2024-25. Area and production are also almost stagnant in Gujarat, Punjab and Haryana, while Tamil Nadu experienced substantial decline in both area and production during last 10 years. There were high fluctuations in production growth in all the States, except Uttar Pradesh where production growth remained positive over the periods.
- 3.11 There are wide inter-State variations in sugarcane yield with relatively higher cane yield in tropical region compared to sub-tropical States. In TE2024-25, among the major sugarcane producing States, Tamil Nadu recorded the highest yield (106 t/ha), followed by Maharashtra (87.8 t/ha) and Karnataka (85.3 t/ha), while Bihar recorded the lowest (61.2 t/ha). However, with the adoption of high yielding varieties, the gap is narrowing down over the years. For instance, gap between yield of Maharashtra and Uttar Pradesh, top two producers, has reduced from 31 percent in TE2009-10 to 7.8 percent in TE2024-25. Similarly, the gap between Tamil Nadu (highest yielding State) and Uttar Pradesh reduced from 46.4 percent to 23.7 percent and between Karnataka and Uttar Pradesh from 34.9 percent to 5.2 percent over the same period.
- 3.12 During last five years, Maharashtra recorded the highest growth in yield (2.9%/annum) followed by Bihar (1.0%) and Karnataka (0.6%), while in Tamil Nadu (-1.0%), yield growth was negative. Average yield growth decelerated during last five years compared to preceding five years at all-India level and in all States, except Maharashtra and Karnataka.



Table 3.3: Trends in Area, Production and Yield of Sugarcane in Major Producing States

Region	State	TE2004 -05	TE2009 -10	TE2014 -15	TE2019 -20	TE2024 -25	2000-01 to 2004-05	2005-06 to 2009-10	2010-11 to 2014-15	2015-16 to 2019-20	2020-21 to 2024-25
	Area (lakh ha)					Area Growth (%)					
Sub-tropical	UP	20.4	20.8	21.9	22.2	27.2	-0.5	0.4	1.7	0.6	4.9
	BR	1.1	1.1	2.5	2.3	2.0	1.8	3.1	23.7	-2.5	-3.0
	PB	1.2	0.8	0.9	0.9	0.9	-2.5	-5.1	9.5	-0.5	-0.5
	HR	1.6	1.0	1.0	1.1	0.9	0.1	-9.1	5.8	0.3	-5.1
Tropical	MH	4.5	8.7	9.7	9.6	13.2	-10.5	27.4	7.0	0.4	8.8
	KA	2.7	3.1	4.4	4.2	5.8	-12.0	15.5	7.8	-1.3	6.5
	TN	2.3	3.2	3.1	1.6	1.5	-4.6	6.8	-1.6	-12.6	0.5
	GJ	1.9	2.0	1.9	1.7	1.9	0.2	-3.6	7.0	-4.0	4.8
	IND	40.4	45.5	50.2	48.0	56.9	-2.6	3.5	4.1	-1.6	3.7
	Production (million tonnes)					Production Growth (%)					
Sub-tropical	UP	117.5	117.0	133.4	178.8	220.3	0.8	0.1	2.6	6.7	4.7
	BR	4.3	4.6	13.2	15.8	12.3	1.1	7.5	33.0	2.5	-2.0
	PB	7.0	5.0	6.5	7.7	7.4	-3.3	-4.4	14.2	1.0	0.0
	HR	9.3	6.4	7.4	8.6	7.4	1.9	-5.4	6.3	2.5	-4.7
Tropical	MH	29.6	71.1	77.1	80.7	115.3	-16.2	35.8	6.9	0.4	11.5
	KA	20.9	26.7	39.1	37.2	49.7	-14.4	19.2	8.4	-0.2	7.5
	TN	21.7	33.5	31.5	16.1	15.6	-5.1	7.3	-0.4	-12.4	-0.2
	GJ	13.8	14.4	13.2	11.7	13.9	1.3	-2.7	3.2	-3.6	5.8
	IND	252.8	308.5	351.9	385.3	466.1	-4.3	5.5	4.6	1.2	4.4
	Yield (t/ha)					Yield Growth (%)					
Sub-tropical	UP	57.5	56.3	60.8	80.5	80.9	1.3	-0.2	1.0	5.9	0.0
	BR	41.0	41.1	52.0	69.7	61.2	-1.1	3.2	5.2	5.3	1.0
	PB	58.1	60.0	73.7	81.9	82.8	-0.5	0.6	4.2	1.5	0.5
	HR	58.8	64.1	73.7	81.0	82.4	2.2	3.8	0.5	1.9	0.4
Tropical	MH	65.2	81.6	79.7	84.5	87.8	-6.3	6.5	-0.4	1.2	2.9
	KA	77.0	86.4	88.5	87.7	85.3	-3.0	2.5	0.3	0.4	0.6
	TN	95.1	105.0	102.7	103.5	106.0	-1.2	0.2	1.3	0.6	-1.0
	GJ	71.7	74.2	71.0	70.5	72.9	1.2	1.9	-2.6	1.0	0.3
	IND	62.6	67.8	70.1	80.3	81.9	-1.7	1.7	0.5	2.6	0.8

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

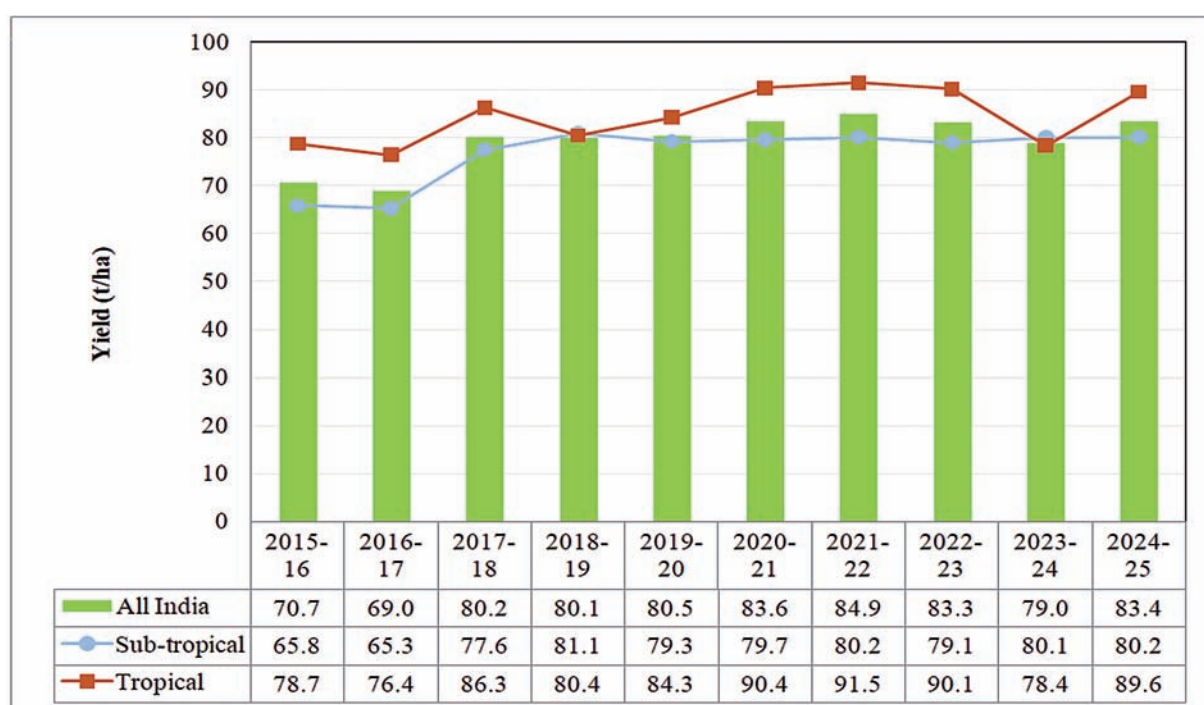
3.13 Chart 3.3 presents region-wise and all-India average yield of sugarcane during last ten years. As is evident from the chart, productivity in the tropical region is significantly higher than in the sub-tropical



region in all years except 2018-19 and 2023-24. It is pertinent to note that major sugarcane-producing districts in the tropical region received deficit rainfall in 2018 and 2023 (Table 3.2). However, the gap in the yield between the two regions is on a declining trend, from 16.4 percent in 2015-16 to 10.5 percent in 2024-25.

- 3.14 Sugarcane yield in sub-tropical region increased from 65.8 tonnes per hectare in 2015-16 to 80.2 tonnes per hectare in 2024-25, registering a higher Compound Annual Growth Rate (CAGR) of 2.0 percent compared to 1.2 percent in tropical region. The wonder variety, Co 0238, has played a vital role in increasing cane productivity in sub-tropical region, particularly Uttar Pradesh. However, due to continuous cultivation, the variety has degenerated over the years and become susceptible to insect pests and diseases, particularly red rot disease, resulting in a decline/stagnation in productivity level in the recent years in sub-tropical region.

Chart 3.3: Sugarcane Productivity by Regions, 2015-16 to 2024-25

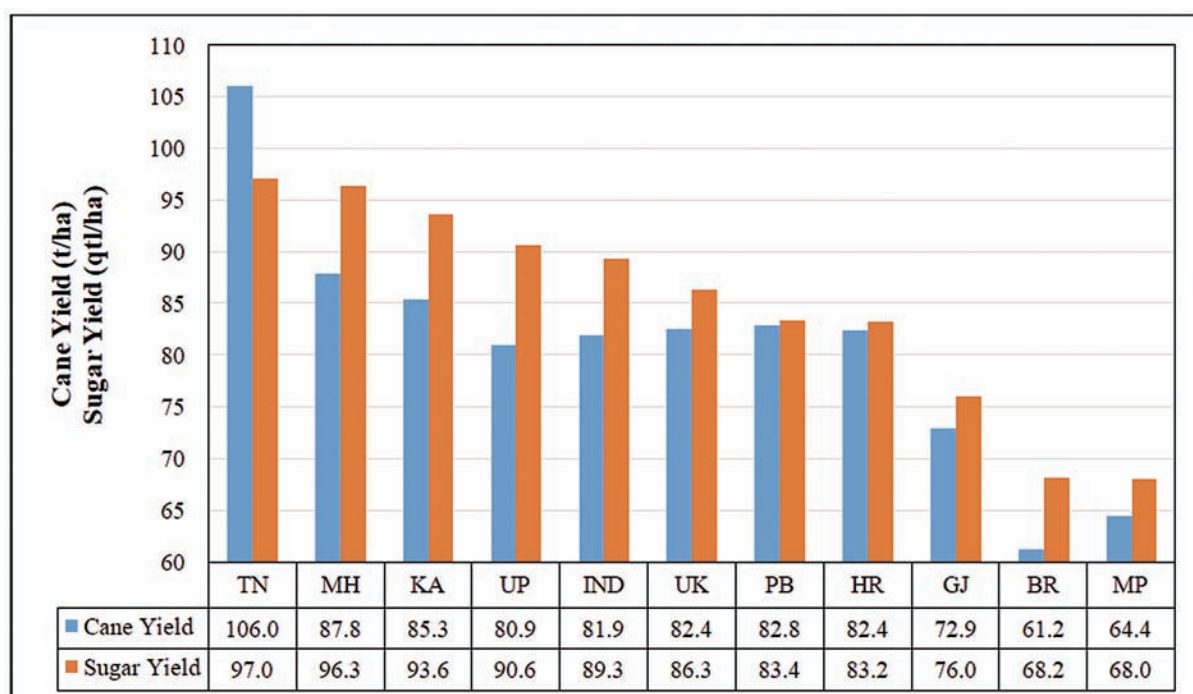


Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.15 Though there are large differences in cane yield of tropical and sub-tropical States, gaps are relatively narrow in terms of sugar yield, due to low sugar recovery rate in tropical States. During TE2024-25, cane yield in Tamil Nadu was 31.0 percent higher than in Uttar Pradesh and 73.3 percent higher than in Bihar but the difference in sugar yield was only 7.1 percent and 42.3 percent, respectively (Chart 3.4). High sugar content varieties play an important role in improving sugar yield, therefore, there is a need to develop high yielding high sucrose content varieties suitable for different agro-climatic condition, particularly for Tamil Nadu.



Chart 3.4: Sugarcane and Sugar Yield of Major Producing States, TE2024-25



Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
2. Directorate of Sugar & Vegetable Oils, Department of Food & Public Distribution

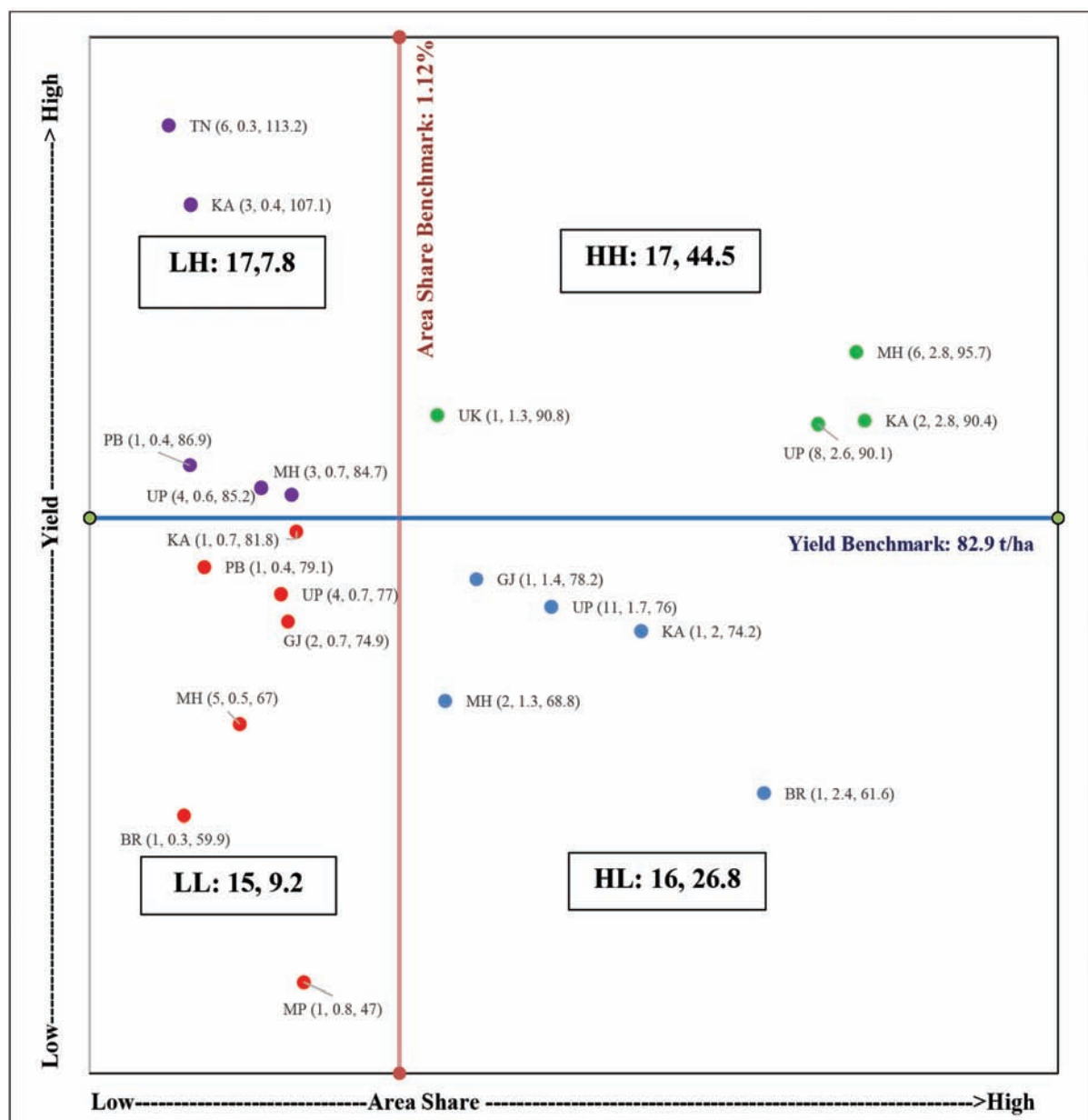
District Level Yield Trends

- 3.16 This section analyses productivity in major sugarcane producing districts during TE2023-24. On the basis of the yield performance and area share at the all-India level, districts contributing around 90 percent of total production have been classified into high and low categories. All-India average yield and median area share (1.12%) for TE2023-24 were taken as benchmark for classification of districts under four categories, high area-high yield (HH), high area-low yield (HL), low area-high yield (LH) and low area-low yield (LL).
- 3.17 There were 17 districts with 44.5 percent area share under the most efficient (HH) category, eight in Uttar Pradesh, six in Maharashtra, two in Karnataka and one in Uttarakhand (Chart 3.5). Another 17 districts with 7.8 percent area were under the LH category, highest in Tamil Nadu (6), followed by Uttar Pradesh (4), Karnataka (3), Maharashtra (3) and Punjab (1). On the other hand, 16 districts with 26.8 percent area share were under the HL category and 15 districts, with 9.2 percent area share, were under the least efficient (LL) category. Majority of HL category districts were from Uttar Pradesh, while number of LL category districts was highest in Maharashtra.
- 3.18 More than one-fourth of the sugarcane area is still under HL category, which indicates potential for increasing production by yield improvement. Hence, concerted efforts should be made to increase yield with adoption of high yielding varieties and best management practices. In LH category districts, interventions should be made to bring more area under sugarcane as productivity is high in these districts, while sugarcane area under LL category should be shifted to other suitable crops for better



resource utilization. Some districts/blocks cultivating sugarcane are under over-exploited, critical and semi-critical category of groundwater extraction and sugarcane being water-intensive crop, it's necessary to promote micro-irrigation for sustainable production. The Commission is of the view that a special component for promoting micro-irrigation in sugarcane under *Pradhan Mantri Krishi Sinchayee Yojana*- Per Drop More Crop (PMKSY-PDMC) should be created and sugar mills should be involved in promoting micro-irrigation in sugarcane.

Chart 3.5: Categorization of Districts based on Sugarcane Area and Yield: TE2023-24



Note: 1. Figures in boxes indicate total number of districts, followed by area share
 2. Figures in parentheses show number of districts, followed by average area share and yield under the respective category

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Sugarcane Yield: International Comparison

- 3.19 Table 3.4 compares India's sugarcane yield with the world average yield and major producing countries between TE2016 and TE2023. Brazil (38.3%) and India (23%) are the two leading producers of sugarcane in the world. In TE2016, sugarcane yield in Brazil was higher than India but India's yield has improved significantly (18.5%) and surpassed Brazil in TE2023. India's yield was about 13 percent higher than Brazil and 14.1 percent higher than the world average. As a result of yield improvement and increased area, India's share in world sugarcane production has increased from 18.8 percent in TE2016 to 23 percent in TE2023. However, yield in Australia, important player in global market, is significantly higher than India. Though remarkable improvement in yield has been recorded in India, it is still lower than Colombia and Australia, indicating scope for further improvement in yield and competitiveness of Indian sugar sector.

Table 3.4: Comparison of Sugarcane Yield among Major Producing Countries

	Yield (t/ha)		Area Share (%)		Production share (%)	
	TE2016	TE2023	TE2016	TE2023	TE2016	TE2023
World	70.6	73.5	-	-	-	-
Brazil	73.3	74.3	38.5	37.9	40.0	38.3
India	70.8	83.9	18.8	20.1	18.8	23.0
China	72.5	80.5	5.8	4.9	6.0	5.4
Pakistan	58.3	70.4	4.4	4.8	3.6	4.5
Thailand	69.8	56.9	5.2	5.6	5.2	4.3
Mexico	73.2	68.9	2.9	3.1	3.0	2.9
Indonesia	59.7	71.3	1.7	1.8	1.5	1.8
Colombia	89.9	91.8	1.5	1.4	1.9	1.8
Australia	81.4	90.0	1.5	1.3	1.7	1.6
USA	79.8	80.5	1.3	1.4	1.5	1.6

Source: Food and Agriculture Organization (Data accessed from <https://www.fao.org/faostat/> on 17th October 2025)

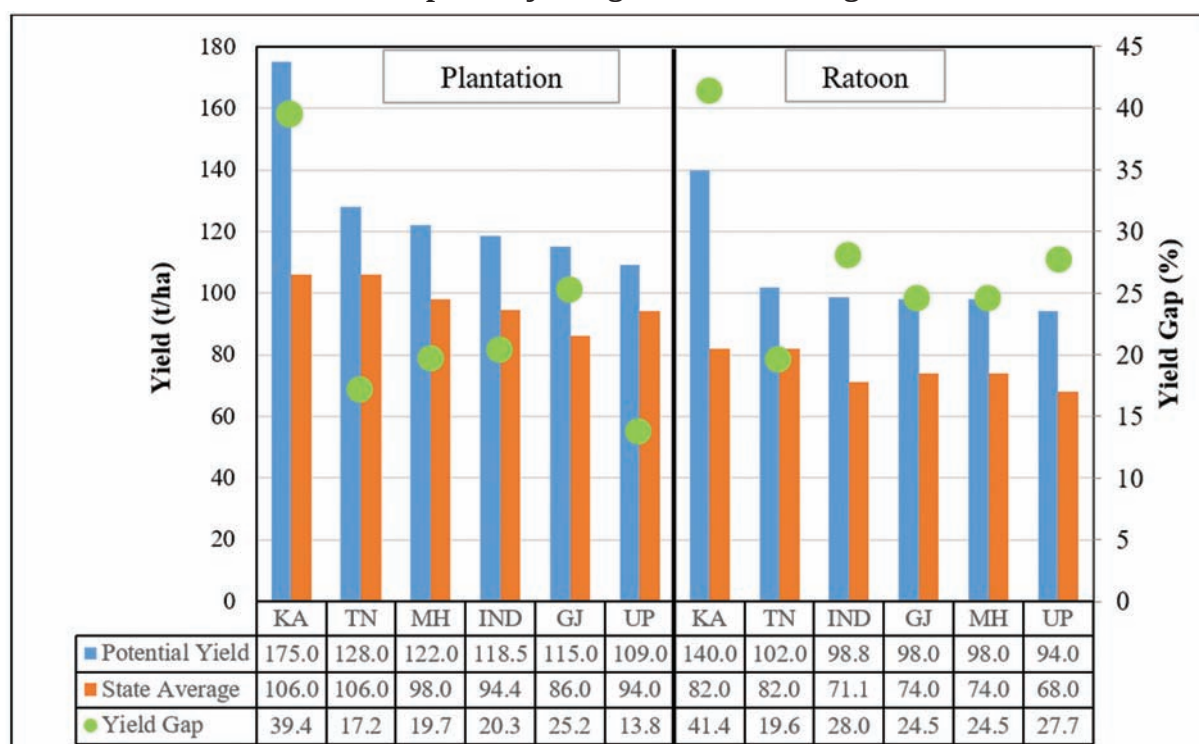
Yield Gaps Analysis

- 3.20 Chart 3.6 shows potential yield and State average yield in five major States, Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu and Gujarat, and all-India average under plantation and ratoon crop. The all-India average yield gap was 28 percent under ratoon and 20.3 percent under plantation system. Gap was relatively higher in Karnataka under both plantation (39.4%) and ratoon system (41.4%), while gap was lowest in Uttar Pradesh (13.8%) under plantation and Tamil Nadu (19.6%) under ratoon system. Though cost of cultivation in ratoon crop is relatively lower than plantation system, yield



significantly decreases in repetitive ratooning, depending on management practices. Generally, one ratoon is recommended after fresh plantation but in some States, particularly in southern region, repeated ratooning is prevalent. For example, share of ratoon crop in Karnataka remained more than 57 percent during the last ten years (2014-15 to 2023-24) and reached up to 97.4 percent in 2016-17, indicating repeated ratooning in the State (Annex Table 3.1). The incidences of insect pests and diseases are high in ratoon system. Sugarcane is high-nutrient-absorbing crop and continuous ratooning results in deterioration of soil health. Therefore, optimizing plantation to ratoon crop ratio and proper management of ratoon are important for improving yield and containing pests and diseases.

Chart 3.6: Yield Gap in Major Sugarcane Producing States, 2024-25



Source: ICAR-Indian Institute of Sugarcane Research, Lucknow

Growth Drivers of Productivity

High Yielding Varieties

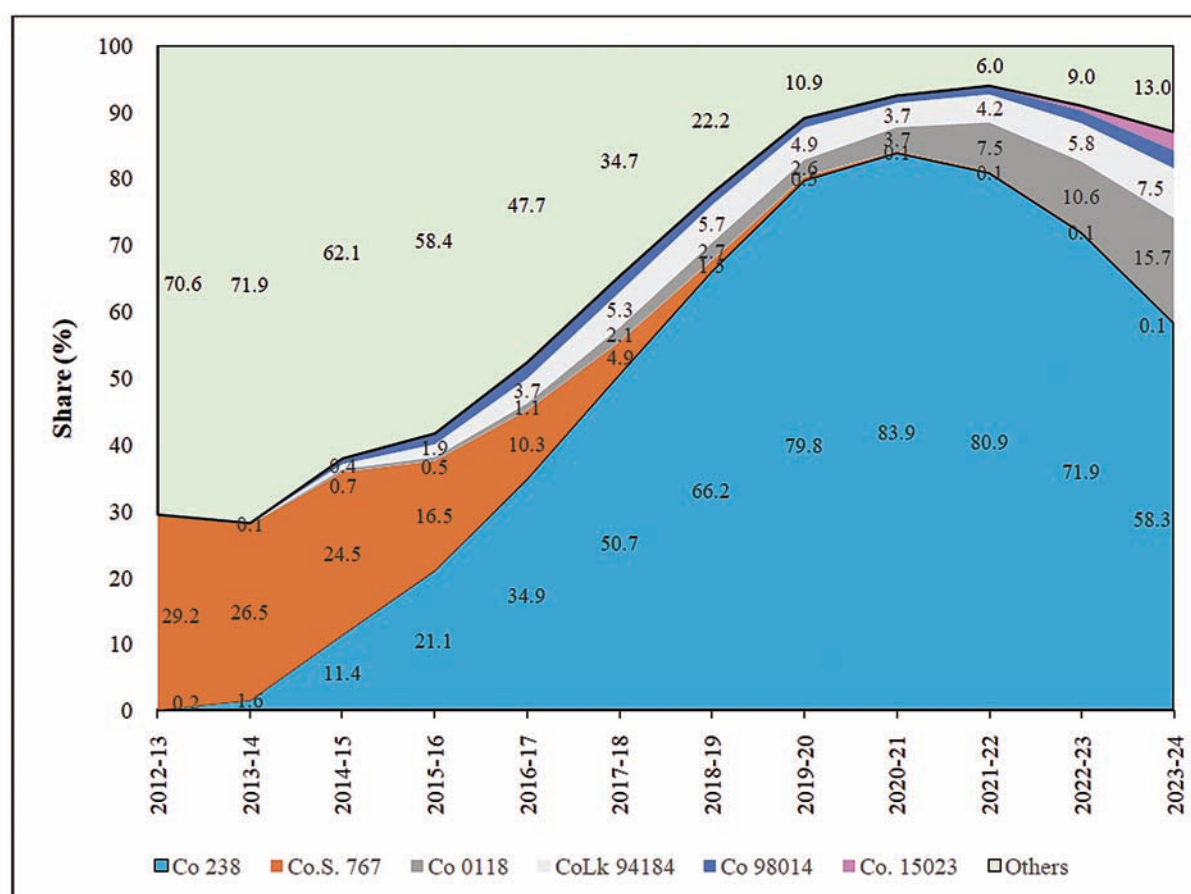
- 3.21 High yielding varieties play an important role in enhancing productivity of sugarcane and sugar recovery. Several region-specific high-yielding and pest and diseases tolerant varieties have been developed by the research institutes and State Agricultural Universities but only a few of these are popular among farmers. High-yielding and high-sugar content variety Co 0238 developed by Indian Council of Agricultural Research-Sugarcane Breeding Institute (ICAR-SBI), Regional Centre Karnal, Haryana, released in 2009 for sub-tropical region replaced CoS 767 variety and has become a dominant variety in the region for more than 15 years. The share of Co 0238 in the region has increased from 11.4 percent in 2014-15 to a peak of around 84 percent in 2020-21 but its share is on declining trend since 2021-22 due to incidence of red rot disease (Chart 3.7). However, this variety still dominates in the



region and occupies more than 58 percent of area in the region in 2023-24. Mono-cropping of Co 0238 has made it susceptible to pests and diseases, particularly red rot disease, and needs to be replaced with newer varieties.

- 3.22 The area under some varieties like Co 0118, CoLk 94184, Co 98014 and Co 15023 has increased in the region during recent years. In 2023-24, Co 0118 was the second most popular variety with area share of 15.7 percent, followed by CoLk 94184 (7.5%), Co 98014 (2.8%) and Co 15023 (2.7%) (Chart 3.8). Other varieties, particularly CoLk 14201, CoS 13235, CoLk 16202, CoS 17231, and CoS 18231 are gaining acceptance among farmers in recent years and replacing Co 0238.
- 3.23 In the tropical region, Co 86032, CoM 268 and CoV 09356 are the major varieties grown by farmers, and Co 86032, developed by ICAR-SBI, is the most popular among these. Though share of this variety has declined, but still 48.7 percent of the total area in Tamil Nadu and 45.4 percent in Maharashtra in TE2023-24 was under Co 86032. CoM 265 is popular mainly in Maharashtra and CoV 09356 is cultivated in Tamil Nadu (Annex Table 3.2).

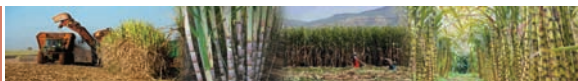
Chart 3.7: Area under Major Varieties in Sub-tropical Region



Note: Data pertains to Bihar, Haryana, Punjab, Uttar Pradesh and Uttarakhand States only

Source: State replies

- 3.24 The above analysis indicates that despite a large number of varieties released for cultivation, the varietal diversity is very low and top-two varieties cover more than two-third of total area in major States.



Continuous cultivation of a variety makes it susceptible to pests and diseases, thereby increases risk of crop failure. Therefore, there is a need to promote varietal diversity for sustainable sugarcane production. Intensive research efforts are required to develop biotic and abiotic stress resistant varieties.

Seed Replacement Rate and Seed Treatment

- 3.25 Seed-borne diseases, particularly red rot, can be minimised by seed treatment and improving Seed Replacement Rate (SRR). However, SRR is critically low in sugarcane due to requirement of large quantity of seed (6-7 t/ha), which involves significant cost. Studies show that at all-India level, around 10 percent of seed is replaced every year and formal sources may cater only a part of the total seed requirement. Therefore, there is a need to improve SRR by increasing participatory seed production at farmers' fields under strict monitoring. Advancements have been made in reducing seed rate through improved practices such as single bud sett planting and optimum spacing but adoption of these methods needs to be improved. Sugar factories need to take up three-tier seed production system, invariably following seed treatment.

Intercropping for Improving Soil Health and Farmers' Income

- 3.26 Sugarcane is considered one of the best suitable crop for intercropping being long-duration and wide spacing crop. Hence, short-duration crops like pulses, oilseeds, vegetables and spices can be grown as intercrops in the initial growth stages of sugarcane. Intercropping can also help in improving soil health and increasing organic matter in the soil through incorporation of crop residues as well as supplementing income of the farmers.
- 3.27 ICAR-Indian Institute of Maize Research, Ludhiana has demonstrated that short-duration maize can be promoted as an intercrop to increase land productivity, ensure round-the-year feedstock supply for dual-feed ethanol distilleries and enhance farmer income. Various interventions in sugarcane-based intercropping such as frontline demonstrations, supply of seed and planting material, trainings and awareness programmes, etc. are being undertaken under the National Food Security and Nutrition Mission. Government of Uttar Pradesh is promoting intercropping of mustard, potato, field pea, lentil, chickpea, etc. with sugarcane and several progressive farmers are successfully practicing intercropping and earning additional income.

Efficient Water Management

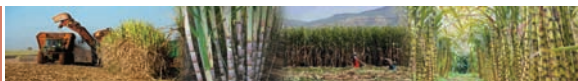
- 3.28 Sugarcane is a water intensive crop and total water requirement varies from 1,200 to 3,000 mm, depending on season, agro-climatic conditions, soil type, method of planting, etc. Moisture stress, especially in the initial stages, adversely affects sugarcane yield and sugar content. Therefore, efficient irrigation water management is important, as groundwater is the major source of irrigation in most of sugarcane cultivating districts and stage of groundwater extraction in many of these is very high. List of districts along with status of groundwater extraction, level of micro-irrigation and sugarcane yield is presented in Annex Table 3.3.
- 3.29 The coverage of micro-irrigation in all the high groundwater extracting districts, except for four districts in Tamil Nadu and two in Karnataka, was below 10 percent of total irrigated area. For example, although groundwater extraction was alarmingly high in two major cane-growing districts in Punjab,



namely, Gurdaspur (136.4%) and Hoshiarpur (116.2%), which account for more than 50 percent of cane production in the State, micro-irrigation coverage was less than 2 percent. Intensive sugarcane cultivation without efficient water use can pose serious threat to sustainability. Therefore, there is an urgent need to promote efficient water management practices like skip-furrow irrigation system and micro-irrigation in water-stressed sugarcane-producing regions.

Mechanization

- 3.30 With rising wages and increasing labour scarcity, enhanced mechanization in sugarcane cultivation has become essential not only to facilitate timely field operations but also for improving operational efficiency and profitability. Studies show that mechanization is particularly low in operations like planting, harvesting, seed treatment, etc.
- 3.31 Research institutes have developed various machinery such as deep furrow sugarcane cutter planter (adjustable) developed by the ICAR-Sugarcane Breeding Institute (ICAR-SBI), Coimbatore and the ICAR-Indian Institute of Sugarcane Research (ICAR-IISR), Lucknow, which requires about 11 times less labour compared to the conventional method of planting and saves cost by about 53 percent. Tractor operated sugarcane ratoon management devices reduce labour requirement up to 90 percent and minimise costs significantly. Sugarcane trash mulcher-cum-stubble shaver developed by the ICAR-IISR facilitates trash shredding, mulching, chemical spraying and stubble shaving into one operation, thereby reducing human labour requirement. The tractor-operated whole cane harvester developed by ICAR- Central Institute of Agricultural Engineering (ICAR-CIAE), Bhopal, in collaboration with ICAR-SBI and ICAR-IISR, can harvest on an average 0.11 hectare of sugarcane per hour and significantly saves cost of harvesting as compared to manual harvesting.
- 3.32 High cost of machinery/equipment and small and fragmented operational holding are the major constraints in adoption of mechanization. The ‘Mini Tractor-Operated Sugarcane Settling Transplanter’ recently developed by ICAR-SBI and ICAR-CIAE saves considerable labour and time and ensures uniform planting. The planter is suitable for small and medium farmers and saves seed and irrigation water. Improvements have also been made in sugarcane harvesting machines, but high extraneous matter in the harvested cane is still a challenge. Therefore, concerted efforts are required in developing improved machinery and increasing adoption among the small farmers through collective ownership. There is also a need to promote mechanisation in low-cost yet very important operations like seed treatment for improving yield and containing spread of diseases by ensuring availability of ‘Seed Treatment Devices’ at community level.
- 3.33 Under the Sub-Mission on Agricultural Mechanization (SMAM), which aims to increase level of mechanization in the country by providing subsidy for purchase of agricultural machinery, setting up of Custom Hiring Centres (CHCs), and organising training and awareness programmes, financial assistance of up to 50 percent is provided on sugarcane machinery to the farmers, including cane harvesters for Custom Hiring Centres. However, the level of mechanization in operations like harvesting is quite low. Therefore, to increase the level of mechanization in sugarcane cultivation, special provisions need to be made for sugarcane under SMAM. Moreover, advanced technologies such as precision agriculture, remote sensing, data-driven Artificial Intelligence (AI) powered tools, etc. can play a vital role in increasing the efficiency and yield of sugarcane farming in India.



- 3.34 The recent reduction in GST on tractors and farm machinery is a significant move to make the sugarcane machinery more affordable and accessible to farmers, hence accelerating sugarcane mechanization. However, initial investment for some of the machines like sugarcane transplanter, harvesters, etc., is still very high to be affordable for small farmers. Unavailability of improved machinery suitable for small farms is also a constraint. Therefore, research and development efforts should be made for developing improved low-cost machinery and increasing adoption among the small farmers through collective ownership.

National Food Security and Nutrition Mission- Commercial Crops

- 3.35 The National Food Security and Nutrition Mission-Commercial Crops (NFSNM-CC) is operating in 13 major sugarcane-producing States and aims to increase sugarcane production through area expansion, adoption of different cropping patterns and productivity enhancement. Under this Scheme, thrust is given on availability of quality planting material, demonstration on improved technologies, assistance for breeder seed production, etc. However, trends in allocation, release and expenditure of funds under NFSNM-Sugarcane during last ten years show that the allocation under the scheme is too low. The average allocation under the programme during last five years has been less than ₹10 crore, while actual expenditure was about ₹5 crore. Since, sugarcane is an important commercial crop, there is a need to substantially increase allocation under the programme.

Recapitulation

- 3.36 At the all-India level, sugarcane production has significantly increased over the past three decades and area expansion has been the major driver. However, high fluctuations were observed both in area and yield growths, resulting in cyclical trend in production. The introduction of high yielding varieties like Co 0238 in sub-tropical region and Co 86032 in tropical region has played a pivotal role in improving sugarcane yield and production. During the last five years, production increased at an impressive average annual growth rate of 4.4 percent, but yield growth was less than one percent. More than one-third of the area remains under high potential category offering potential for targeted interventions. Sugarcane yield in India was higher than Brazil and the world average in TE2023, but lower than Australia and Colombia, pointing to the need for sustained research and technological advancement.
- 3.37 Extensive mono-cropping varieties, particularly of Co 0238 in sub-tropical region, has increased vulnerability to various pests and diseases, causing yield stagnation. Therefore, promotion of alternative high-yielding, high-sucrose content and disease-resistant varieties is essential to enhance resilience and sustainability. There were large yield gaps, particularly under ratoon system due to repetitive ratooning. Therefore, optimum plantation to ratoon crop ratio and proper management of ratoon should be promoted for improving yield and containing pests and diseases. Given that sugarcane is a water-intensive crop, there is a need to promote efficient water management technologies and practices such as micro-irrigation, skip-furrow system, etc. Further, farmers should be encouraged to adopt intercropping for efficient use of resources, improving soil health and earning additional income. Small-farm mechanization, particularly in labour-intensive operations like sowing and harvesting, remains a challenge. Therefore, concerted efforts are required to develop improved machinery suitable for small farms and to increase adoption through collective ownership.



CHAPTER 4



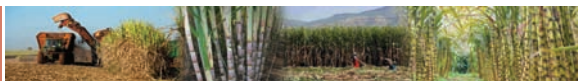
Trade Performance, Patterns and Prospects

- 4.1 Sugar is a major agro-based commodity, widely produced and consumed globally. About 85 percent of total sugar output is derived from sugarcane, cultivated predominantly in tropical regions, while the remaining share is produced from other sources, mainly sugar beet grown largely in northern temperate regions of the world. The international sugar economy is highly complex and volatile, due to climatic risks, distortionary policies and evolving consumer preferences. Global sugar production and trade are concentrated in a few countries, the trade and support policies of which influence the global sugar trade and prices significantly. India occupies a pivotal position among the world's leading producers, consumers and exporters of sugar and its policies have notable implications for the global sugar market. India's sugar trade policy is based on balancing the twin objectives of maintaining adequate domestic supply for consumption and ethanol production and sustaining competitiveness in international markets. This chapter provides an overview of global sugar production, trade, and price dynamics, with particular emphasis on market outlook and India's comparative position in the world sugar economy.

Global Scenario

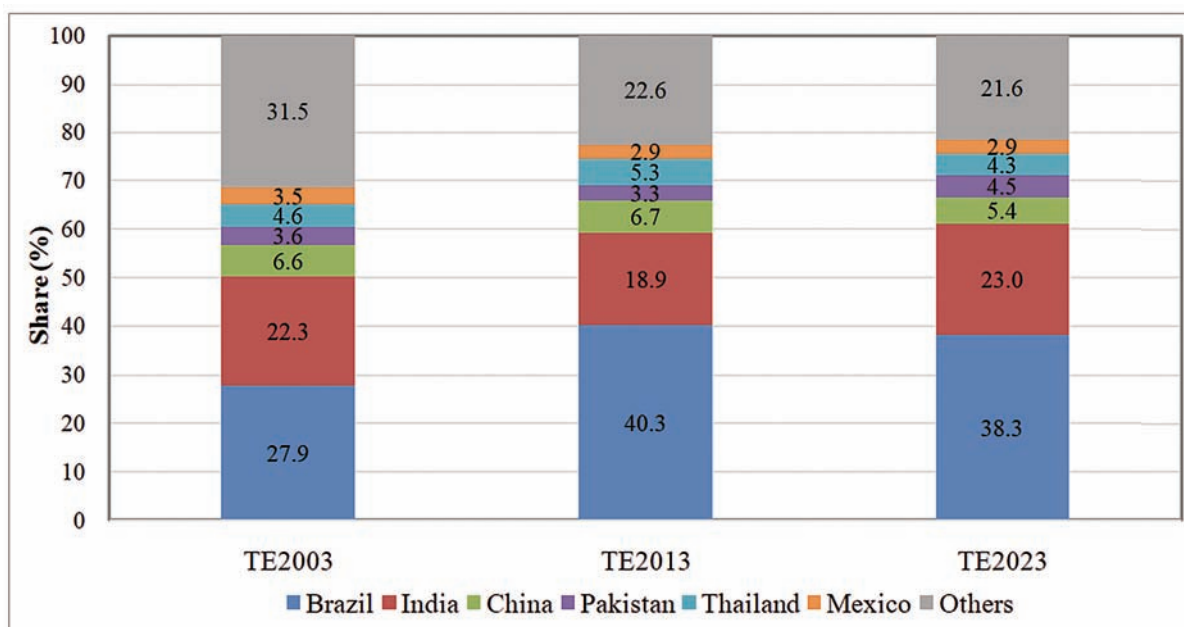
Sugarcane Production

- 4.2 As per USDA, among the major sugarcane producing countries in the world, Brazil, which is a global leader in ethanol production from sugarcane, allocates about half of its sugarcane for ethanol production. The ethanol production in Brazil for 2025 is estimated at 39.7 billion liters, an increase of 22.2 percent over 2024 due to increasing global demand for renewable energy and supportive government policies to use sustainable fuels for implementing the low carbon programmes.
- 4.3 According to Food and Agriculture Organization (FAO), global sugarcane production in 2023 was estimated at 205.8 million tonnes, an increase of 5.2 percent over previous year. The changes in the production shares of major sugarcane producing countries during last two decades are shown in Chart 4.1. It is observed that Brazil, the dominant sugarcane producer in the world, had a global share of 27.9 percent in TE2003, which significantly increased to 40.3 percent in TE2013, but marginally declined to 38.3 percent in TE2023. The share of India, the second largest producer in the world, has mostly remained stagnant during past two decades. India had a global share of 22.3 percent in TE2003,



which declined to 18.9 percent in TE2013, but rose to 23 percent in TE2023. The other important producers of sugarcane in the world in TE2023 were China (5.4%), Pakistan (4.5%), Thailand (4.3%) and Mexico (2.9%). Brazil and India, the top two producers, have increased their combined share from 50.2 percent in TE2003 to 61.2 percent in TE2023. As per USDA, due to lower yields caused by scarcity of water in Brazil, sugarcane production is estimated to fall by 2.6 percent in 2025-26, from 677.8 million tonnes in 2024-25 to 660 million tonnes in 2025-26. On the contrary, sugarcane production in India is expected to improve from 435 million tonnes in 2024-25 to 465 million tonnes in 2025-26 due to favourable weather conditions and increase in area under cultivation.

Chart 4.1: Changing Shares of Major Sugarcane Producers in the World



Source: Food and Agriculture Organization

Sugar Beet Production

- 4.4 Sugar beet is a more efficient and sustainable source of sugar production as compared to sugarcane due to high concentration of sucrose (16-20%). World sugar beet production in 2023 was estimated at about 281.2 million tonnes, registering a growth rate of 8.2 percent over previous year, as per FAO estimates. With a share of 17.4 percent, Russia was the largest producer in the world, followed by USA (11.4%), Germany (11.2%), France (10.9%), Turkey (9%) and Poland (6%). It is worth noting that Russia's share in global production of sugar beet has remarkably increased during the last two decades surpassing France, USA, Germany and Turkey. With top five sugarcane producers accounting for over 75 percent of global share as compared to nearly 60 percent in case of sugar beet in TE2023, world sugarcane production is relatively more concentrated.

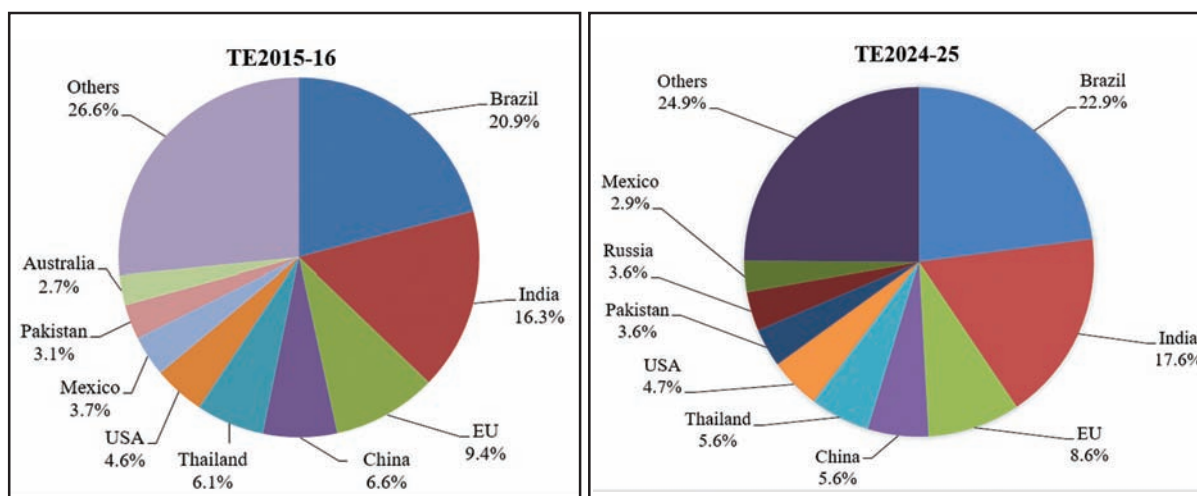
Sugar Production

- 4.5 According to USDA, global sugar production is estimated at 180.8 million tonnes in 2024-25, an increase of 2.9 percent over 2023-24. Chart 4.2 presents the share of major sugar producers in the world in TE2015-16 and TE2024-25. Brazil remains the leading producer of sugar with an increase in



share from 20.9 percent in TE2015-16 to 22.9 percent in TE2024-25. India, the second largest producer, also increased its share between TE2015-16 and TE2024-25. Thailand, the second largest exporter of sugar in the world, marginally lost its share in world production between two time periods. The share of EU declined from 9.4 percent to 8.6 percent and China from 6.6 percent to 5.6 percent. The top two producers Brazil and India accounted for 40.6 percent of the world sugar production in TE2024-25, up from 37.2 percent in TE2015-16. The initial estimates for global sugar production in 2025-26 by International Sugar Organisation (ISO) indicate about 3.1 percent increase in production over 2024-25 season, but marginally below the record production of 181.2 million tonnes in 2023-24. World sugar consumption is projected to marginally increase by 0.77 million tonnes in 2025-26.

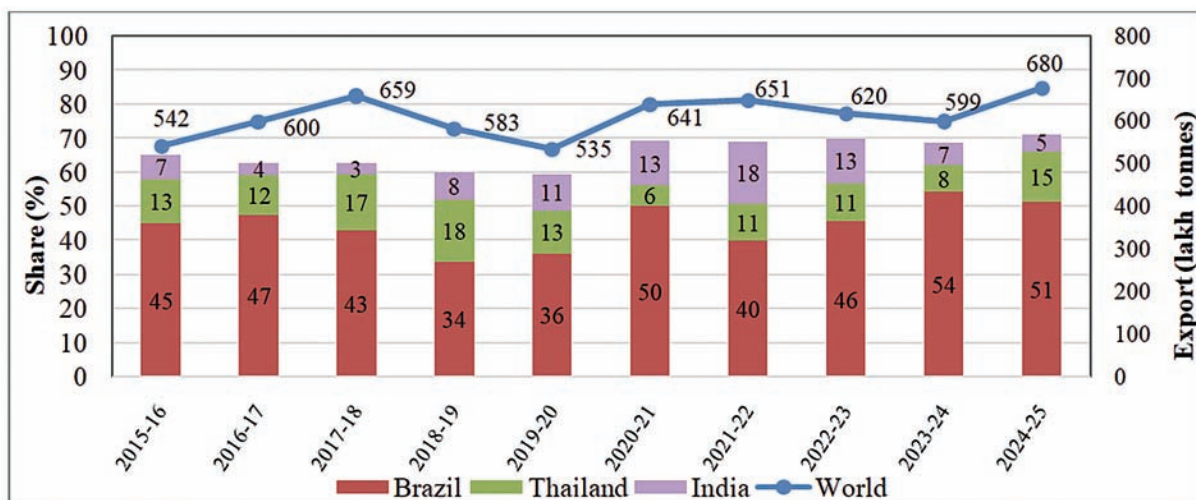
Chart 4.2: Major Producers of Sugar in the World



Source: Sugar: World Markets and Trade Report, United States Department of Agriculture

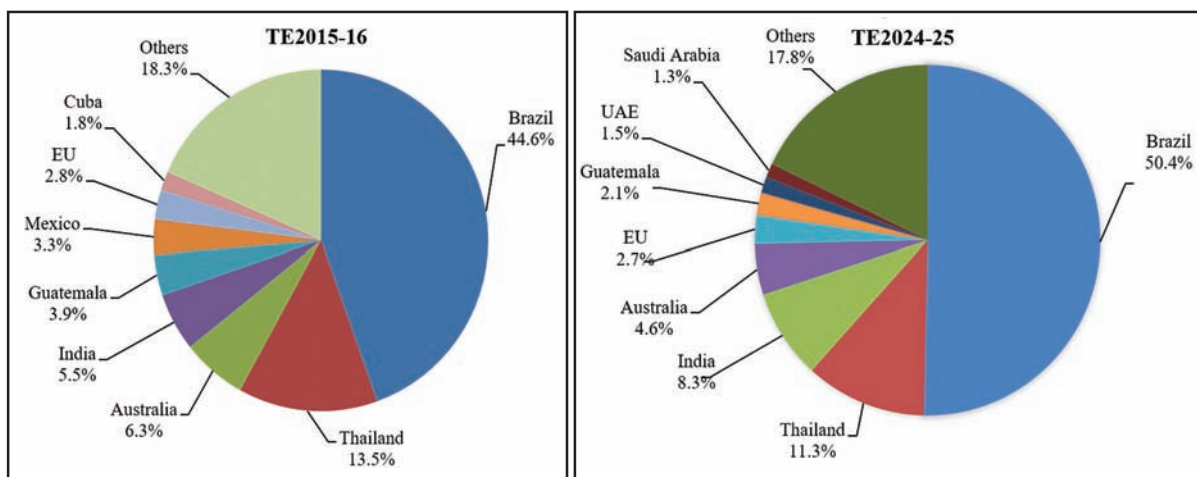
Sugar Trade

- 4.6 According to USDA estimates, global sugar exports, which declined from 65.1 million tonnes in 2021-22 to 59.9 million tonnes in 2023-24, increased to 68 million tonnes in 2024-25 (Chart 4.3). Among the major exporters, India's sugar exports registered a decline of 11.6 percent in 2024-25 over last year due to restrictions on sugar exports by the Government since June 2022. However, Brazil, which saw a significant decline in sugarcane production in 2020-21 due to adverse weather conditions, continued to recover thereafter and recorded 7.4 percent increase in sugar exports in 2024-25 over previous year. As per ISO estimates, volume of world trade in sugar in 2025-26 is expected to remain steady with exports marginally lower than last season. Import demand for sugar in 2025-26 is estimated at the same level as in 2024-25 season.
- 4.7 World sugar exports and share of top three exporters, Brazil, India and Thailand during last 10 years are shown in Chart 4.3. It is observed that Brazil has remained the top exporter of sugar in the world and increased its share from 45 percent in 2015-16 to 54 percent in 2023-24, which marginally fell to 51 percent in 2024-25. While the share of Thailand in world sugar exports, which was higher than India during the period 2015-16 to 2019-20, fell below India's share during 2020-21 to 2022-23 due to larger exports from India. However, Thailand surpassed India in 2023-24 and 2024-25 due to reduced exports from India in 2023-24 and 2024-25 and higher exports by Thailand in 2024-25.

**Chart 4.3: World Sugar Exports & Share of Brazil, India and Thailand**

Source: Sugar: World Markets and Trade Report, United States Department of Agriculture

- 4.8 Chart 4.4 illustrates the share of major sugar exporting countries in the world in TE2015-16 and TE2024-25. Among top five exporters, Brazil and India improved their share in world exports from 50.1 percent (44.6% Brazil and 5.5% India) in TE2015-16 to 58.7 percent (50.4% Brazil and 8.3% India) in TE2024-25. While, other producers such as Thailand, Australia and EU recorded a decline in world exports share and decline was steeper for Thailand and Australia. It is worth noting that the top three exporters accounted for 70 percent of the world sugar exports in TE2024-25 up from 64.3 percent in TE2015-16, which indicates rising concentration of world sugar exports.

Chart 4.4: Major Exporters of Sugar in the World

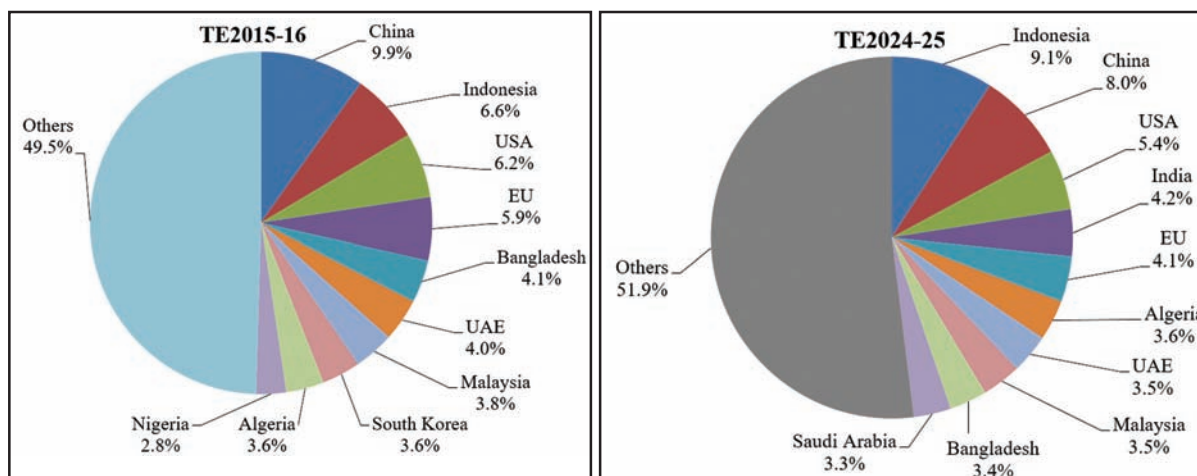
Source: Sugar: World Markets and Trade Report, United States Department of Agriculture

- 4.9 Chart 4.5 shows the major importers of sugar in the world in TE2015-16 and TE2024-25. Unlike exports, global sugar imports are more diversified across countries. During both the periods, Indonesia, China and USA remained the top three importers and accounted for over 22 percent of total imports. In TE2015-16, China was the largest importer, accounting for 9.9 percent of global imports, followed by Indonesia (6.6%), USA (6.2%), EU (5.9%), and Bangladesh (4.1%). However, in TE2024-25, Indonesia (9.1%) emerged as the largest importer of sugar in the world, followed by China (8%), USA



(5.4%), India (4.2%) and EU (4.1%). The top five sugar importers accounting for less than one-third of total imports indicate that import demand is widely dispersed globally.

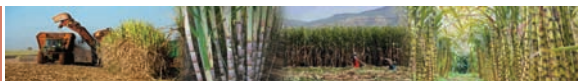
Chart 4.5: Major Importers of Sugar in the World



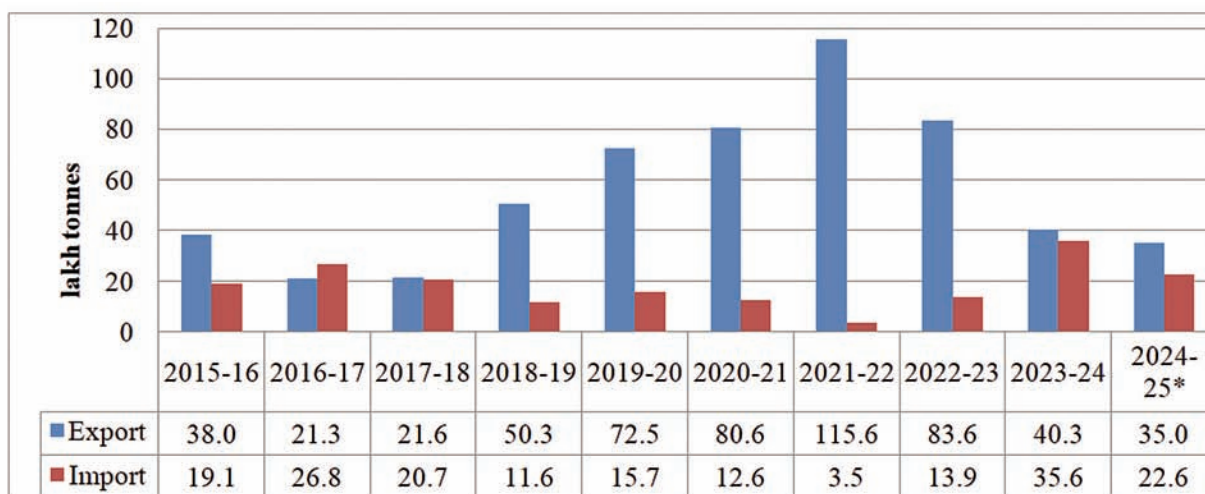
Source: Sugar: World Markets and Trade Report, United States Department of Agriculture

India's Sugar Trade

- 4.10 The export and import trends of sugar in India during the last ten sugar seasons are shown in Chart 4.6. India has become a surplus producer of sugar, which has made the country a net exporter of sugar during last decade. India's sugar exports, which remained depressed during 2016-17 and 2017-18, increased significantly and reached a record level of about 11.6 million tonnes in 2021-22, driven by higher domestic production, strong international demand, favorable global sugar prices and lower exports by Brazil. In contrast, sugar exports recorded a significant decline during the last three seasons and were 3.5 million tonnes in 2024-25 due to lower domestic production and restrictions on exports imposed by the Government. On the other hand, sugar imports remained volatile and significantly lower than exports except in 2016-17, when imports were higher than exports. Sugar imports rose sharply in 2023-24 but declined in 2024-25.



**Chart 4.6: India's Exports and Imports of Sugar (in volume),
Sugar Season 2015-16 to 2024-25**



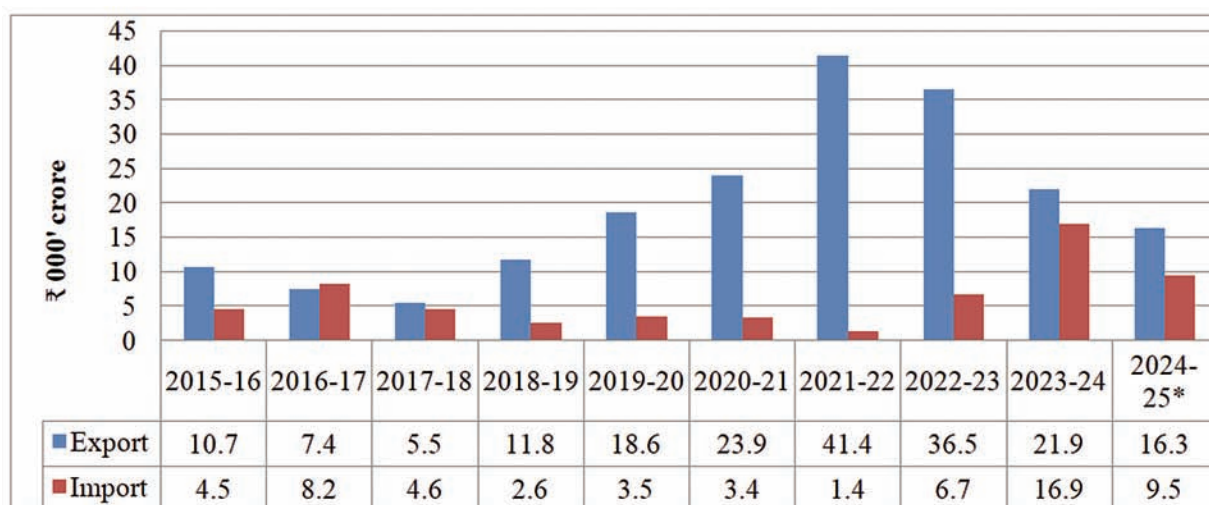
Note: 1. Exports/Imports data include data on sugar re-exports/re-imports under Advance Authorization Scheme (AAS) of DGFT also and hence may differ from sugar Export/Import data of Ministry of Consumer Affairs, Food and Public Distribution
2. *For the period October 2024-August 2025

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce & Industry

- 4.11 India's exports and imports of sugar, both in value and volume, followed almost a similar trend during the last ten seasons as is evident from Chart 4.7. India imports raw sugar primarily from Brazil, the largest producer in the world. India imported sugar worth ₹ 9.5 thousand crore in sugar season 2024-25, out of which, 98 percent was sourced from Brazil, followed by UAE and Germany. The value of sugar imports, which increased during 2022-23 and 2023-24, fell by 43.6 percent in 2024-25, while export earnings from sugar recorded a continuous decline during the last three seasons. The trends during last ten years clearly indicate that India has become a net exporter of sugar in the recent years.



**Chart 4.7: India's Exports and Imports of Sugar (in value),
Sugar Season 2015-16 to 2024-25**



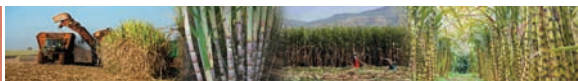
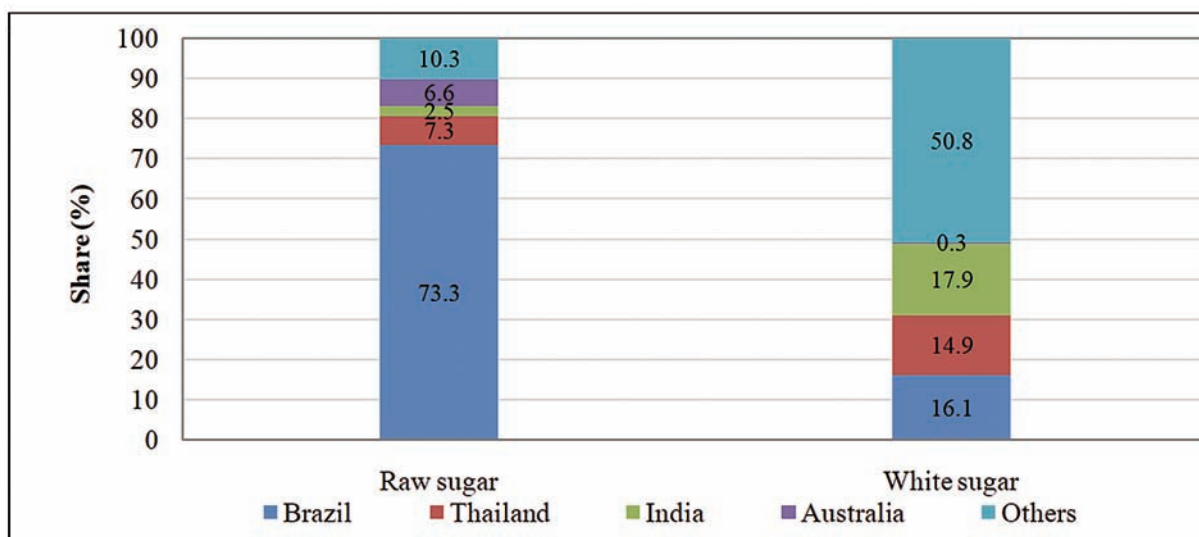
Note: 1. Exports/Imports data include data on sugar re-exports/re-imports under Advance Authorization Scheme (AAS) of DGFT also and hence may differ from sugar Export/Import data of Ministry of Consumer Affairs, Food and Public Distribution
2. * For the period October 2024-August 2025

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce & Industry

- 4.12 India exports sugar to a diverse range of countries across Asia, Africa and Middle East. However, a notable shift in export destinations of sugar from India has been observed during the last five years (Annex Table 4.1 and 4.2). Indonesia, the world's major importer of sugar, was the primary destination for Indian sugar in 2020-21 and 2021-22, but exports declined significantly during the last 3 years, and Sudan remained an important destination for India's exports of sugar with 14.1 percent share, followed by Libya (10.8), Somalia (10.5%), Sri Lanka (8.7%) and Djibouti (8.1%) in 2024-25. The top five importers of Indian sugar accounted for 52.2 percent of total sugar export earnings in 2024-25. The top ten export and import destinations of Indian sugar during sugar season 2020-21 to 2024-25 are presented in Annex Table 4.1 and 4.2.

Composition of Sugar Exports

- 4.13 Raw sugar accounts for 60 percent of world exports in TE2024-25 and Brazil was the largest exporter of raw sugar. White sugar accounted for 40 percent of exports and India was major exporter of white sugar. Brazil was the largest exporter of raw sugar, with a share of 73.3 percent in TE2024-25, followed by Thailand (7.3%), Australia (6.6%), and India (2.5%) (Chart 4.8). India was the largest exporter of white sugar (17.9%) in TE2024-25, followed by Brazil (16.1%) and Thailand (14.9%).

**Chart 4.8: Major Exporters of Raw and White Sugar: TE2024-25**

Source: Quarterly Market Outlook, August 2025, International Sugar Organisation

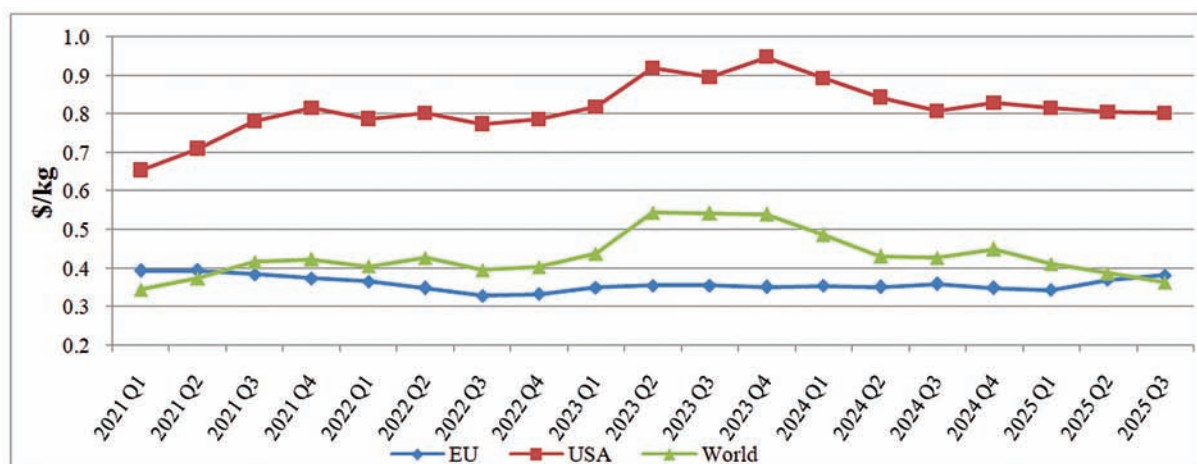
Domestic and World Sugar Prices

Trends in World Prices

- 4.14 International sugar prices have historically been the most volatile due to energy prices and various policy instruments such as subsidies, trade measures, minimum price level etc.. In USA, sugar sector is protected through a combination of price support mechanisms, domestic marketing allotments, production controls and TRQs, which insulate domestic producers from international competition. The sugar sector in EU, which was highly protected through production quotas and minimum price guarantees till 2017, has undergone a significant structural transformation in recent years.
- 4.15 It can be seen from Chart 4.9 that sugar prices in USA have remained significantly higher than the world prices, while in EU, sugar prices were below the world prices during the period 2021(Q3) to 2025(Q2). A significant spike in both world and USA sugar prices was observed in 2021, driven by fall in production in Brazil caused by drought and frost. Subsequently, an increasing trend, albeit with some volatility, was noticed in both the prices in 2023 due to disruption in global sugar production by *El Niño* weather pattern and diversion of sugarcane towards ethanol production by major producing countries. However, a steeper decline in prices was recorded during 2024 due to improved production in Brazil, EU and Thailand. Sugar price in USA and world market continued a declining trend during the first three quarters of 2025.



Chart 4.9: Trends in Sugar Prices in the USA, EU and World, 2021-2025(Q3)



Source: World Bank Commodity Price Data (The Pink Sheet)

World Raw and Refined Sugar Prices

- 4.16 The comparative trends in international prices of raw and refined sugar are shown in Chart 4.10. It is observed that the prices of raw and refined sugar are highly correlated with a correlation coefficient of 0.98 during the last decade. The average price of refined sugar was about 26 percent higher than price of raw sugar, due to cost of refining.
- 4.17 World sugar prices for both raw and refined variant exhibited a cyclical trend. Market prices followed an upward trend in 2016, but the trend reversed thereafter and prices declined significantly till 2019 (Chart 4.10). However, a strong and steady rebound in sugar prices was witnessed from 2020, except for April-May 2020, when prices fell sharply due to COVID-19 induced demand shock. Between August 2022 and November 2023, there was record increase in world sugar prices, driven by tightening global supplies, slow start of harvest in Brazil caused by heavy rains and appreciation of Brazilian Real against the US dollar. Since December 2023, world sugar prices have generally followed downward trend with minor fluctuations due to rise in sugar exports from Brazil and Thailand.

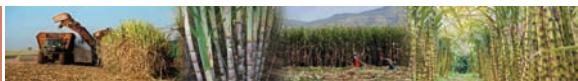
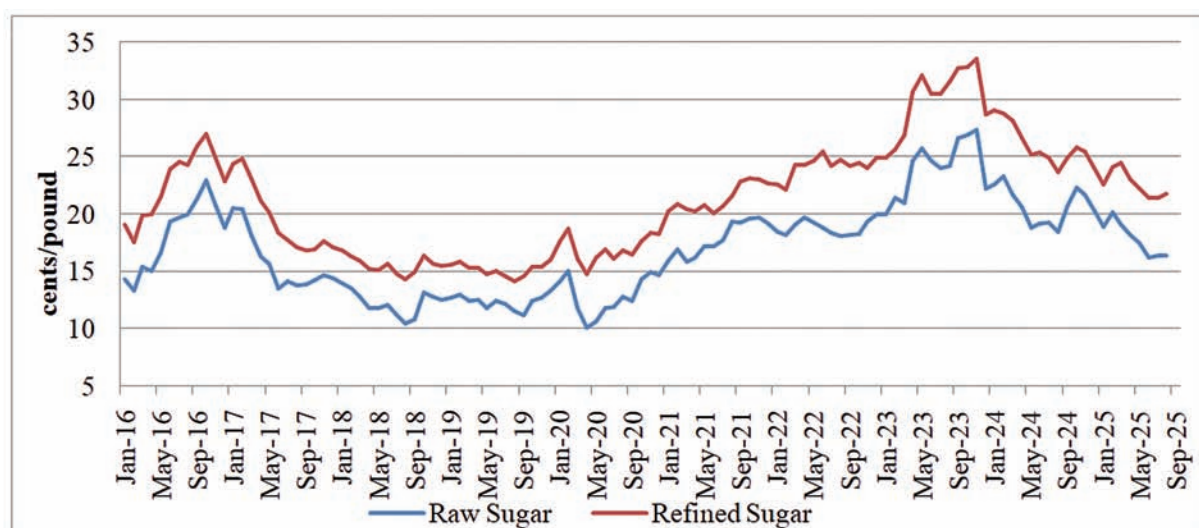


Chart 4.10: World Market Prices of Raw Sugar and Refined Sugar

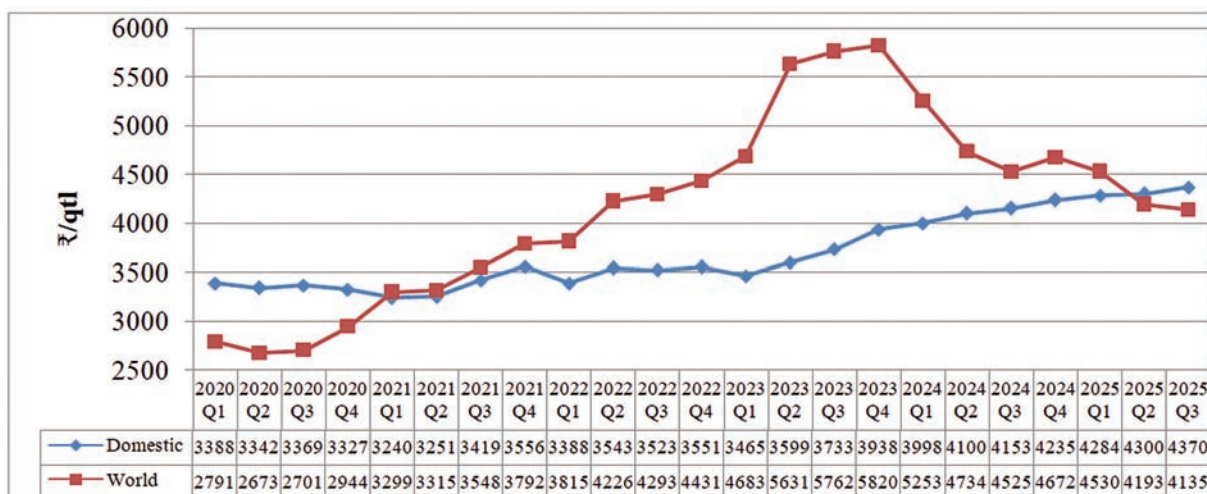


Note: 1. Data not available for September 2025
 2. Raw Sugar (ICE Contract 11 nearby futures price), Refined Sugar (Contract No.5, London Daily Price)

Source: Intercontinental Exchange (ICE) (Data accessed from <https://www.ers.usda.gov/>)

Domestic vis-à-vis World Prices

- 4.18 The trends in refined sugar prices in domestic and world market at Intercontinental Exchange (ICE) during the period 2020 to 2025(Q3) are shown in Chart 4.11. It is observed that domestic prices of sugar in India remained significantly higher than the international prices during 2020 with 21 percent mark-up between two prices. Higher domestic prices of sugar made Indian sugar exports less competitive in the global market. However, international prices moved upward and surpassed domestic prices in 2021(Q1), making Indian sugar exports competitive in the global market. World prices remained significantly higher than domestic prices during 2021 to 2025(Q1), 3.6 percent in 2021 to 48.6 percent in 2023. However, from 2024(Q1), while international price witnessed a steep decline, domestic prices witnessed moderate increase and hence the gap remained 5.7 percent in 2025(Q1). Subsequently, domestic prices surpassed international prices in 2025(Q2) and remained above global price in 2025(Q3). Domestic prices recorded a steady increase from 2023(Q2) till 2025(Q3), while world prices showed a continuous fall since 2024(Q4).

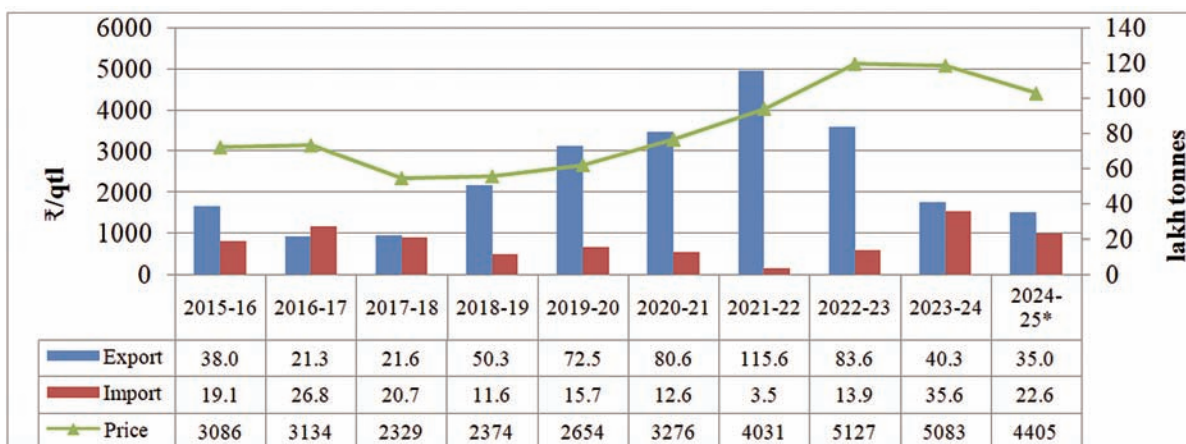
**Chart 4.11: International and Domestic Wholesale Prices of Sugar, 2020-2025(Q3)**

Note: 2025(Q3) for world prices is average for the month of July and August 2025

Source: 1. Intercontinental Exchange (ICE) for International Prices

2. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic Wholesale Price in India (Mumbai center)

- 4.19 The trend of India's sugar exports and imports vis-à-vis international prices of sugar are presented in Chart 4.12. India's sugar exports are driven by both domestic demand-supply situation and international prices of sugar. Sugar exports showed an increasing trend during 2018-19 to 2021-22 due to rising global prices and higher domestic production. Although international prices were high in 2015-16, 2016-17, 2022-23 and 2023-24, but India could not capitalize the benefit of high prices due to lower domestic supplies of sugar, which led to reduced exportable surplus. India's sugar exports declined sharply to 40.3 lakh tonnes in 2023-24 and 35 lakh tonnes in 2024-25 due to export restrictions imposed by the Government. These trends clearly indicate that domestic demand-supply situation is a major driver of exports and imports of sugar.

Chart 4.12: India's Sugar Exports, Imports and International Prices of Sugar, 2015-16 to 2024-25

Note: * For the period October 2024-August 2025

Source: 1. Intercontinental Exchange (ICE) for International Prices

2. Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce & Industry



World Sugar Outlook

- 4.20 As per International Sugar Organisation (ISO) and Food and Agriculture Organisation (FAO) estimates, global sugar production and consumption are forecast to increase in 2025-26. While ISO projects a stable global trade in sugar, FAO forecasts a marginal increase. As regards ending stocks, ISO projects for a marginal fall, but FAO forecasts a marginal increase.
- 4.21 As per OECD-FAO Agricultural Outlook 2025-34, global sugar production is projected to increase by 15 percent over the period 2025-34, while consumption is projected to expand by 12 percent due to population and income growth (Annex Table 4.3). Sugar trade is expected to remain robust and dominated by raw sugar. Exports are projected to remain highly concentrated with dominant share of Brazil (73%) and Thailand (10%). Global sugarcane production is projected to increase by 1.2 percent per annum and Brazil, India and Thailand are expected to contribute significantly driven by relatively higher crop yields in India and Thailand and area expansion in Brazil. The share of sugar crops in world sugar production is expected to decline from about 80 percent to 77 percent in 2034. Brazil and India will account for 96 percent of global sugarcane-based ethanol production.

Sugar Price Outlook

- 4.22 Table 4.1 shows global prices (real and nominal) of raw and white sugar over the period 2025-2034. Sugar prices are projected to decline in real terms over the next decade, driven by productivity gains and improvement in global export availabilities. However, constant real global crude oil prices will partially offset downward pressure on real prices and result in higher diversion of sugar crops for ethanol production. However, nominal prices are projected to follow a moderate upward trend due to anticipated robust sugar trade for the outlook period. White sugar premium is expected to rise in nominal terms due to reduced availability and exports of refined sugar compared with raw sugar.

Table 4.1: World Raw and White Sugar Prices: 2025-2034

Price forecast (US\$/tonne)										
	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Real Prices										
Raw Sugar	376.2	384.6	382.5	381.5	379.7	376.6	372.7	368.9	365.2	361.0
Refined/ White Sugar	479.6	487.4	484.6	486.2	485.3	481.1	476.4	473.1	471.2	467.6
Nominal Prices										
Raw Sugar	383.8	400.0	405.1	411.5	417.1	422.0	426.0	429.9	434.0	437.5
Refined/ White Sugar	489.3	507.0	513.3	524.4	533.0	539.1	544.5	551.4	560.0	566.6

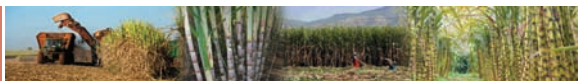
Note: Raw sugar world price, Intercontinental Exchange contract No.11 nearby futures price; Refined Sugar world price, Euronext Liffe, Futures Contract No. 407, London, Europe (October/September)
Real prices year basis are deflated using USA GDP base year 2024=1

Source: OECD-FAO Agricultural Outlook 2025-2034



Overview of India's Sugar Trade Policy

- 4.23 India's sugar trade policy is driven by the need to balance interests of key stakeholders, including farmers, mills, and consumers. Recent changes in India's sugar export and import policies are summarized below and details are provided in Annex Table 4.4 (a) & (b).
- (i) In March 2018, Government removed the customs duty on sugar exports to encourage exports due to surplus production. Thereafter, in September 2018, a Minimum Indicative Export Quota (MIEQ) of 5 million tonnes was set and a Scheme was notified in October 2018 to provide subsidies for transport and handling costs to encourage sugar exports. To manage the large surplus of sugar stock, Government introduced a subsidy of ₹10,448 per tonne in September 2019 to support export of 6 million tonnes under the Maximum Admissible Export Quantity (MAEQ) Scheme, to assist mills cover export-related costs and clear dues to sugarcane farmers. Subsequently, Government approved an export subsidy of ₹6,000 per tonne in December 2020, which was later reduced to ₹4,000 per tonne in May 2021 to help sugar mills offset marketing and transport cost.
 - (ii) Due to fall in Brazil's sugarcane production, global prices surpassed domestic prices in 2021 and Indian sugar exports became competitive without government subsidies. However, export policy of sugar was amended in May 2022 and export of sugar (raw, refined and white sugar) was changed from 'Free' to 'Restricted' category from 1st June 2022 in order to maintain domestic availability, except sugar being exported under the Tariff Rate Quota (TRQ) to the USA and the CXL quota to the EU. Government allocated 6 million tonne export quota proportionally to mills based on past production for 2022-23. The export restrictions were extended on 28th October 2022 and 31st October 2023 until further order. Government has allocated 5,841 MT of sugar to be exported to EU from India under TRQ Scheme for the year 2025-26 (October 2025 to September 2026) on 1st August 2025. Additionally, Government has also allowed exports of pharma grade sugar up to a total 25,000 MTs per financial year to bonafide pharma exporters against a Restricted Export Authorization on 17th June 2025.
 - (iii) On 14th November 2025, Government allocated export quota of 15 LMT of sugar for the sugar season 2025-26 *vide* notification No.1(4)/2025-SP. The export quota has been allocated on pro-rata basis amongst operational sugar mills by taking into their average production of sugar during last three sugar seasons.
 - (iv) Molasses, a by-product of sugarcane, is a key raw material for ethanol production. Global ethanol policies have significantly increased the commercial importance and demand for molasses in world market in recent years. Countries like India, Brazil, Thailand and Indonesia use a substantial share of their domestically produced molasses to meet ethanol production targets. India is a leading exporter of molasses in the world and exported molasses worth ₹1,687 crore in 2020-21 sugar season, which increased to ₹1,902 crore in 2021-22, but the value of exports declined to ₹484 crore in 2023-24 due to 50 percent export duty imposed on molasses



in January 2024 in order to ensure domestic supply for ethanol production. During October 2024 to August 2025, India's molasses exports were worth ₹180 crore mainly to Bangladesh, Bhutan and UK. However, on 14th November 2025, Government removed the 50 percent export duty on molasses *vide* notification No.48/2025-Customs, which is expected to improve Indian exports. The top five export destinations of Indian molasses during sugar season 2020-21 to 2024-25 are presented in Annex Table 4.5.

- (v) In order to stabilize domestic prices and discouraging imports, Government raised import duty on sugar from 40 percent to 50 percent in July 2017, which was further raised to 100 percent in February 2018. At present, sugar can be imported freely with 100 percent import duty.

Recapitulation

- 4.24 India is the world's second largest producer of sugar and ranks among the leading consumers and exporters and hence plays a pivotal role in shaping global sugar market dynamics. As per USDA, due to lower yields caused by scarcity of water in Brazil, sugarcane production is estimated to fall by 2.6 percent in 2025-26, but in India, it is expected to improve due to favourable weather conditions and increase in area under cultivation. As per USDA, global sugar production is estimated at 180.8 million tonnes in 2024-25, an increase of 2.9 percent over 2023-24. The global sugar exports are estimated at 68 million tonnes in 2024-25 compared to 59.9 million tonnes in 2023-24. Brazil and India improved their share in world sugar exports from 50.1 percent in TE2015-16 to 58.7 percent in TE2024-25. Sugar exports from India recorded a significant decline during the last three seasons due to lower domestic production and restrictions on exports. Since December 2023, world sugar prices have generally followed downward trend with minor fluctuations due to rise in sugar exports from Brazil and Thailand. Both ISO and FAO project an increase in global sugar production and consumption in 2025-26, but ISO anticipates a stable, while FAO projects marginal expansion in trade. As regards global ending stocks, ISO forecasts a marginal fall, but FAO anticipates a marginal increase in 2025-26.
- 4.25 As per OECD-FAO Agricultural Outlook 2025-34, over the next decade sugar production, consumption and trade are projected to increase, but the share of sugar crops in world sugar production is expected to decline. Global sugarcane production is projected to increase and Brazil, India and Thailand are expected to contribute significantly driven by relatively higher crop yields in India and Thailand and area expansion in Brazil. Sugar prices are projected to decline in real terms, but nominal prices are projected to follow a moderate upward trend over the next decade.

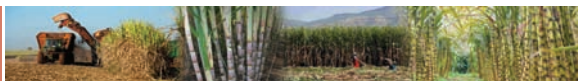


CHAPTER 5



Costs, Returns and Inter-Crop Parity

- 5.1 In determining the Fair and Remunerative Price (FRP) of sugarcane, the Commission has considered various factors such as cost of production of sugarcane, return to sugarcane farmers from the alternative crops, price trend of other agricultural commodities, availability of sugar to consumers at a reasonable price, recovery of sugar from sugarcane, realization made from sale of by-products viz. molasses, bagasse and pressmud or their imputed value and reasonable margins for sugarcane growers on account of risk as per the mandate given under Sugarcane (Control) Order, 1966. To incentivize higher sugar recovery, the FRP is linked to a basic recovery rate, with a premium for higher recovery.
- 5.2 The Commission uses cost estimates of sugarcane compiled by the Economics, Statistics & Evaluation Division (ES&E Division), Ministry of Agriculture and Farmers Welfare, Government of India under the 'Comprehensive Scheme (CS) for studying the Cost of Cultivation of Principal Crops in India'. As the cost estimates are available upto the crop year 2023-24, cost projections are made for sugar season (October to September) 2026-27.
- 5.3 The projected CoC of sugarcane for sugar season 2026-27 is derived from the latest available cost estimates for Triennium Ending (TE) 2023-24. The projected CoC captures movement in overall input cost separately over each of the preceding three years, viz. 2021-22, 2022-23 and 2023-24, for each of the projected State. An assessment of likely changes in input cost for sugar season 2026-27 with reference to each of the crop year 2021-22, 2022-23 and 2023-24, is made by constructing the Composite Input Price Indices (CIPIs) (base 2011-12=100) for each of the projected State. These CIPIs are based on latest prices of different farm inputs like human labour, bullock labour, machine labour, fertilisers, manure, seeds, pesticides and irrigation, available from different sources like Labour Bureau, Ministry of Labour and Employment, State Governments, Office of the Economic Adviser, Ministry of Commerce and Industry, etc. Based on the CIPIs, the Commission projects per hectare CoC A_2 , $A_2 + FL$ and C_2 of sugarcane for major sugarcane growing States.
- 5.4 State-wise Cost of Production (CoP) of sugarcane, i.e. A_2 , $A_2 + FL$ and C_2 , are estimated from the respective projected CoC estimates, ratio of main product to Gross Value of Output (GVO) and estimated yield. Subsequently, all-India CoP is estimated using weighted average of CoP of the States with their production share in total sugarcane production as weights. These projected all-India CoPs are then considered by the Commission for formulating price policy recommendations.



- 5.5 The Commission has undertaken cost projection exercise under the assumptions that the fixed cost components of sugarcane would not, in all likelihood, undergo any significant change during the period for which actual cost estimates are available and for which cost projections are being made. Further, to minimize the impact of inherent variability of yield, three-year average yield for latest TE2023-24 has been used for estimating CoP from projected CoC.

Costs and Returns during TE2023-24

- 5.6 The average CoC, GVO, gross returns and percentage returns over CoC for sugarcane during TE2023-24 for major sugarcane growing States and at all-India level are given in Table 5.1. The GVO of sugarcane is calculated based on the prevailing market prices of main product and by-products during the harvest season.

Cost of Cultivation

- 5.7 During TE2023-24, all-India average CoC A_2 was estimated at ₹1,00,538 per hectare and CoC $A_2 + FL$ was ₹1,20,572 per hectare. However, substantial variations were observed in the cost of cultivation across States, mainly driven by differences in yield and farm input costs. The varying shares of ratoon crop and fresh planting in different States also influence cost of cultivation. The CoC in Gujarat, Haryana, Karnataka and Uttar Pradesh was lower than all-India average, while the cost was higher in Maharashtra and Tamil Nadu. The CoC was highest in Tamil Nadu due to relatively high cost of human labour, while Karnataka has the lowest CoC mainly due to lower cost of seed as share of ratoon crop was significantly high in the State, machine labour, insecticides and irrigation.
- 5.8 The CoC for sugarcane during the last three years showed different trends among States. In Gujarat, Maharashtra and Uttar Pradesh, per hectare cost decreased in 2022-23 but increased in 2023-24, while in case of Haryana, Karnataka and Tamil Nadu, cost registered an increasing trend during 2022-23 and 2023-24. The break-up of cost of cultivation estimates of sugarcane from 2021-22 to 2023-24 for Gujarat, Haryana, Karnataka, Maharashtra, Tamil Nadu and Uttar Pradesh are given in Annex Tables 5.4(a) to 5.4(f).



Table 5.1: Average Costs and Gross Returns over Cost of Cultivation of Sugarcane, TE2023-24

(₹/ha)

State	Cost of Cultivation		GVO	Gross Returns over	
	A_2	$A_2 + FL$		A_2 (Col.4 - Col.2)	$A_2 + FL$ (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Gujarat	92080	101341	275832	183752	174491
Haryana	89831	92912	282513	192681	189601
Karnataka	68120	84319	257524	189404	173205
Maharashtra	152000	180864	272160	120160	91296
Tamil Nadu	178705	195891	311895	133189	116003
Uttar Pradesh	77368	95399	218868	141500	123469
All-India	100538	120572	244332	143794	123760

Note: All-India CoC, GVO and gross returns are weighted average of respective CoCs, GVOs and gross returns of projected States

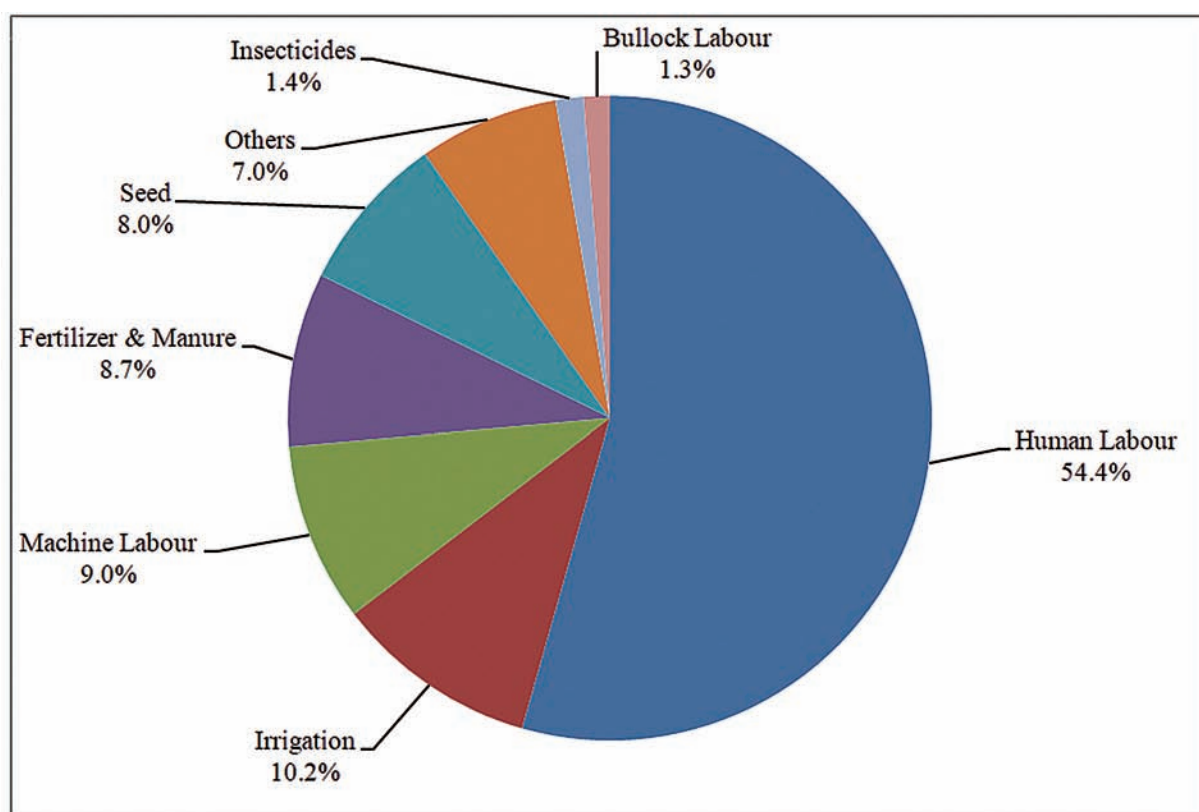
Source: CACP Calculations using CS data

Share of Farm Inputs in Cost of Cultivation

- 5.9 Chart 5.1 shows share of major farm inputs in actual all-India $A_2 + FL$ cost of cultivation during TE2023-24. Human labour accounted for the highest share at 54.4 percent, followed by irrigation cost (10.2%), machine labour (9%), fertiliser and manure (8.7%), seed (8%), insecticides (1.4%) and bullock labour (1.3%). The share of human labour in the cost of cultivation varied from 41.2 percent in Maharashtra to 71.3 percent in Karnataka. Sugarcane cultivation operations, particularly, planting, weeding and interculture, application of fertilizers and pesticides and harvesting, are labour-intensive, therefore, mechanization of sugarcane cultivation would help in reducing costs and improving profitability.



**Chart 5.1: Share of Farm Inputs in Actual Cost of Cultivation
 A_2 +FL of Sugarcane, TE2023-24**



Note 1: Others include Interest on Working Capital, Rent Paid for Leased-in-Land, Land Revenue, Cesses & Taxes, Depreciation on Implements & Farm Buildings and Miscellaneous Expenses

2: All-India actual CoC is weighted average of projected States

Source: CACP Calculations using CS data

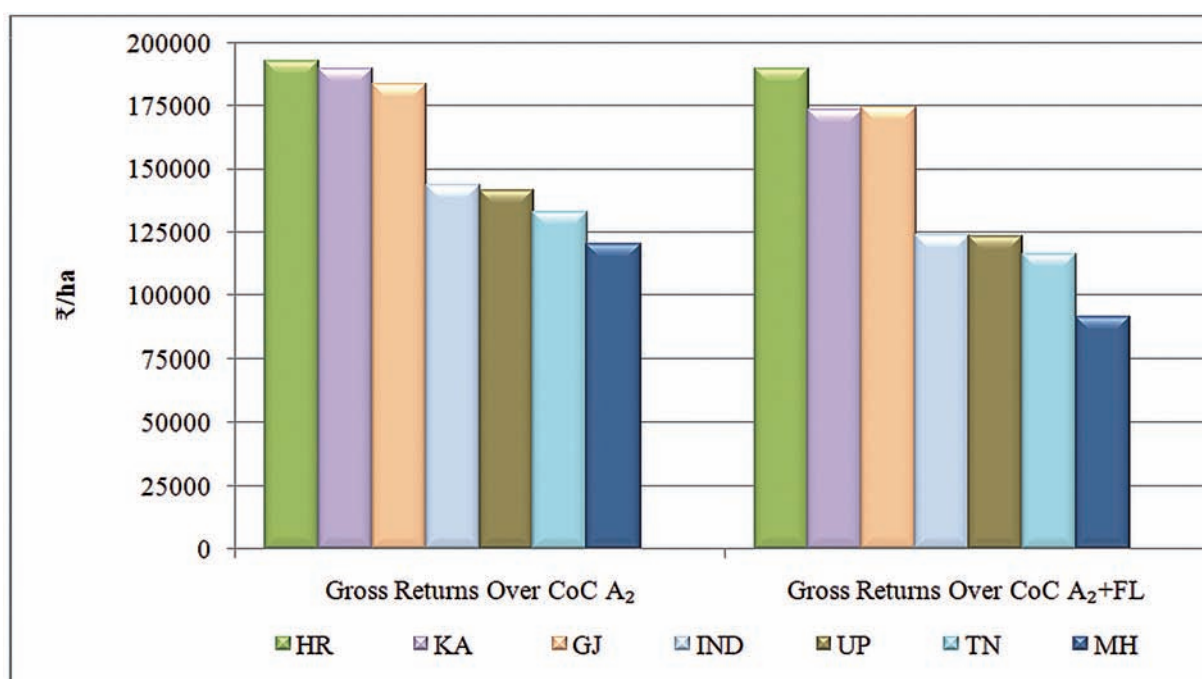
Gross Value of Output and Returns

- 5.10 All-India average GVO for sugarcane during TE2023-24 was ₹2,44,332 per hectare, which varied from ₹2,18,868 in Uttar Pradesh to ₹3,11,895 in Tamil Nadu. The GVO per hectare at all-India level declined in 2022-23, due to decrease in yield but increased in 2023-24, mainly due to increase in cane price. Gujarat, Haryana and Tamil Nadu registered an increase in GVO during TE2023-24 due to higher yield and price in Gujarat and Haryana, and higher price in Tamil Nadu. The GVO in Karnataka, Maharashtra and Uttar Pradesh witnessed a decline in 2022-23 mainly due to fall in yield, but recovered in 2023-24 primarily due to improvement in prices.
- 5.11 The average gross returns over CoC A_2 and A_2 +FL during TE2023-24 for major States and at all-India level have been presented in Chart 5.2. All-India per hectare average gross returns over CoC A_2 and A_2 +FL were ₹1,43,794 and ₹1,23,760, respectively. Haryana exhibited highest per hectare gross returns over both CoC A_2 and A_2 +FL driven by lower cost of cultivation, while Maharashtra reported the lowest returns, primarily due to high input cost.



- 5.12 In percentage terms, all-India gross returns over CoC A_2 were 143 percent, and over A_2 +FL the returns were 102.6 percent. The percentage returns over A_2 and A_2 +FL were highest in Karnataka, whereas, Tamil Nadu registered the lowest returns over A_2 and Maharashtra had the lowest returns over A_2 +FL.
- 5.13 The average per hectare gross returns over CoC A_2 and A_2 +FL showed a fluctuating trend over the last three years. At all-India level, average gross returns declined in 2022-23 due to lower yield and higher cost of production but improved in 2023-24 due to increase in prices in major sugarcane growing States. In Gujarat and Haryana, returns over CoC increased during 2021-22 to 2023-24 due to increase in GVO, while in Uttar Pradesh and Karnataka, returns decreased during 2022-23 due to lower GVO but increased in 2023-24. Tamil Nadu recorded an increasing trend in returns over A_2 and Maharashtra experienced increasing returns over A_2 +FL during the last three years. The year-wise CoC, GVO, gross returns over CoC for major States and at all-India level are given in Annex Table 5.1.

Chart 5.2: Average Gross Returns in Major Sugarcane Growing States, TE2023-24



Note: All-India CoC, GVO and gross returns are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data

Agriculture Wage Rate Trends and Patterns

- 5.14 Table 5.2 analyses growth in average daily wage rates of agriculture labour (in nominal terms) in major sugarcane growing States and at all-India level during the sugar season 2022-23, 2023-24 and 2024-25. At all-India level, wage rate increased by 6.2 percent in 2022-23, 6.3 percent in 2023-24 and eight percent in 2024-25 over the corresponding period of previous year. The increase in average daily wage rate in 2024-25 over 2023-24 was highest in Gujarat (12%), followed by Haryana (11.4%), Karnataka (8.9%) and lowest in Maharashtra (6.9%).

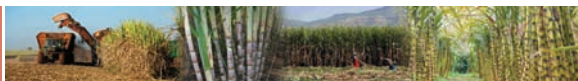


Table 5.2: Growth (%) in Average Daily Wage Rates of Agriculture Labour in Major Sugarcane Growing States

State	2022-23	2023-24	2024-25
Gujarat	8.0	6.0	12.0
Haryana	8.4	5.8	11.4
Karnataka	9.0	6.5	8.9
Maharashtra	5.0	6.2	6.9
Tamil Nadu	4.8	7.0	7.1
Uttar Pradesh	6.6	5.0	7.6
All-India	6.2	6.3	8.0

Note 1: Average daily wage rates of agriculture labour are related to men only

2: Growth is derived from the average taken for the period from October to September

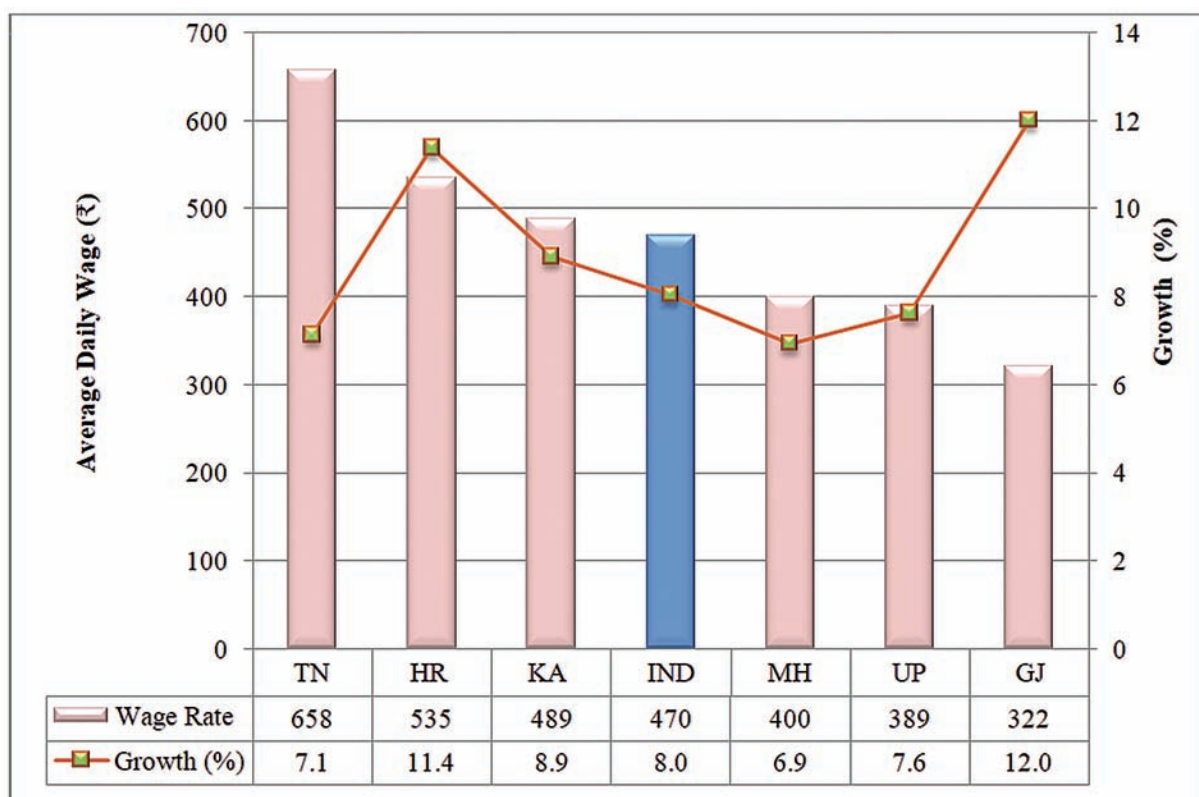
3: All-India daily wage rate is weighted average of daily wage rates of States

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

- 5.15 Chart 5.3 shows the average daily wages of agriculture labour in selected States and at all-India level during 2024-25 and growth in wages over 2023-24. All-India average daily wage of agriculture labour was ₹470 in 2024-25. The wage rate was highest in Tamil Nadu (₹658) and the lowest for Gujarat (₹322). Haryana, Karnataka and Tamil Nadu had higher average daily wages than all-India average, while wages were lower for Gujarat, Maharashtra and Uttar Pradesh. The monthly average daily wage rates for agriculture labour in major sugarcane growing States from 2021 to 2025 are given in Annex Table 5.2.



Chart 5.3: Average Daily Wage Rates in Major Sugarcane Growing States



Note 1: Average daily wage rates of agriculture labour are related to men only

2: Average has been taken for the period from October to September

3: All-India daily wage rate is weighted average of daily wage rates of States mentioned in the Chart

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

Movement in Farm Input Prices

- 5.16 The trends in prices of major farm inputs were analysed to assess the impact of changes in input prices on cost of cultivation. The changes in Wholesale Price Index (WPI) for various farm inputs (base 2011-12=100) during 2024-25 (October to September) over previous year are presented in Chart 5.4. The indices of High Speed Diesel (HSD) (-4.4%), cattle feed (-3.5%) and fodder (-0.5%) declined in 2024-25 over 2023-24, whereas indices of manufacture of fertilizers and nitrogen compounds (0.1%), electricity (0.2%), manufacture of pesticides and other agrochemical products (0.4%), and manufacture of agricultural and forestry machinery¹ (1.5%) increased in 2024-25. The overall weighted average price of these inputs declined by 1.4 percent in 2024-25 over 2023-24. The monthly wholesale price indices of various farm inputs from 2023 to 2025 are given in Annex Table 5.3.

¹Manufacture of agricultural and forestry machinery consists of 'agricultural tractors', 'harvesters', 'agriculture implements', 'threshers' and 'soil preparation & cultivation machinery (other than tractors)'.

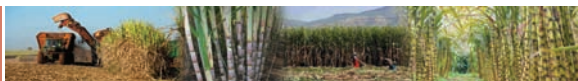
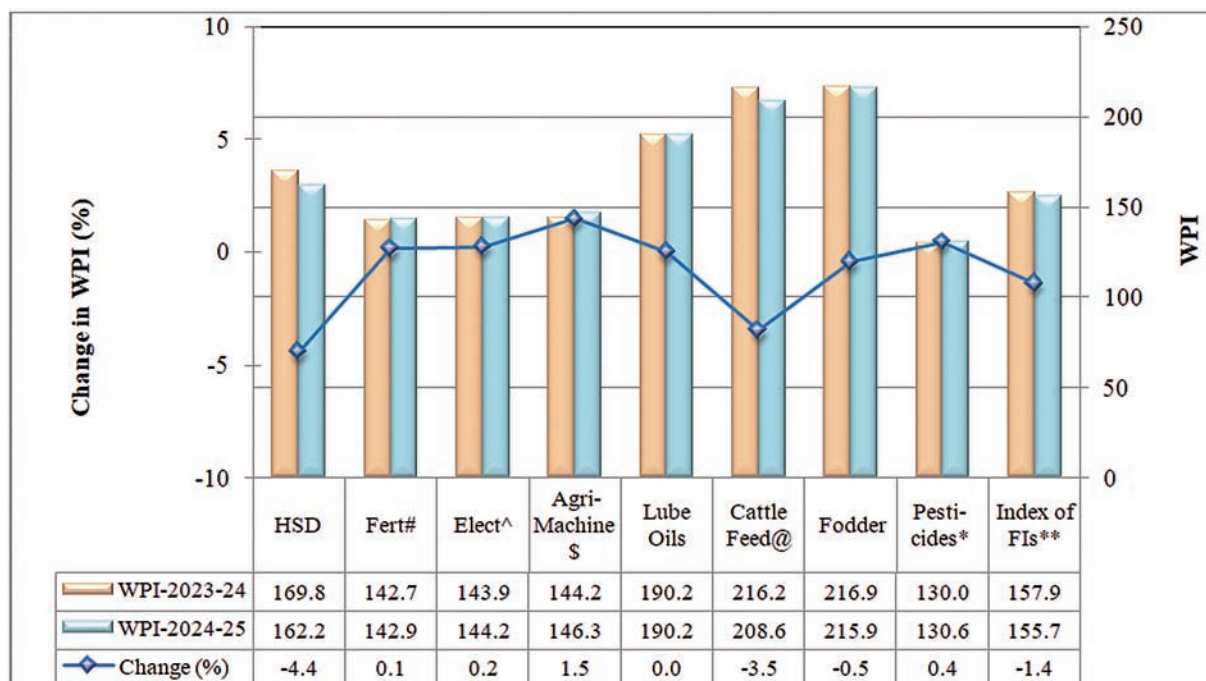


Chart 5.4: Movement in Wholesale Price Index (WPI) of Farm Inputs



Note 1: WPI of a farm input shown in the Chart is average of monthly WPIS from October to September

2: Index of Farm Inputs is weighted average of WPIS of farm inputs mentioned in the Chart

#: Manufacture of Fertilizers and Nitrogen Compounds

^: Electricity

\$: Manufacture of Agricultural and Forestry Machinery

@: Gola & similar Cattle Feed

*: Manufacture of Pesticides and other Agrochemical Products

** : Index of Farm Inputs

Source: Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations

Cost Projections: Sugar Season 2026-27

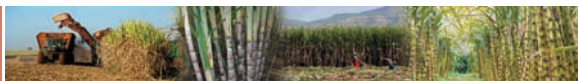
Projection of Farm Input Price Index

- 5.17 Based on actual farm input price indices up to 2024-25 and farm input weights up to 2023-24 for Gujarat, Haryana, Karnataka, Maharashtra, Tamil Nadu and Uttar Pradesh and prices of farm inputs up to 2024-25, the Commission has projected Composite Input Price Indices (CIPIs) up to sugar season 2026-27. Using State-wise actual cost estimates up to 2023-24 and projected CIPIs, CoC A_2 , A_2+FL and C_2 estimates for sugar season 2026-27 were projected for each State. Using these estimated CoC, average ratio of Main Product (MP) to Gross Value of Output (GVO) during TE2023-24 and yield, State-wise CoP A_2 , A_2+FL and C_2 for sugarcane at State-specific sugar recovery rate were projected for sugar season 2026-27.
- 5.18 The projected CoP at State-specific recovery rate for the sugar season 2026-27 has been adjusted at a basic sugar recovery rate of 10.25 percent using State-specific recovery rate of the respective State. Subsequently, all-India weighted average CoP A_2 , A_2+FL and C_2 at State-specific recovery rate and at a basic recovery rate of 10.25 percent, with weights being the respective share of the State in total sugarcane production during TE2024-25, have been worked out for sugar season 2026-27.



Projected Cost of Production

- 5.19 The State-specific recovery rates and all-India average recovery rate based on recovery rates of Gujarat, Haryana, Karnataka, Maharashtra, Tamil Nadu and Uttar Pradesh during TE2024-25, projected CoP A_2 , A_2 +FL and C_2 at State-specific recovery rates and basic recovery rate of 10.25 percent, and production share of the States are presented in Table 5.3. The all-India average sugar recovery rate (including sugar diverted to ethanol production) during TE2024-25 was estimated at 10.98 percent. The recovery rate was highest for Uttar Pradesh (11.19%) and lowest for Tamil Nadu (9.15%).
- 5.20 The all-India per quintal CoP A_2 , A_2 +FL and C_2 of sugarcane at State-specific recovery rates for sugar season 2026-27 are projected at ₹161, ₹194 and ₹276, respectively. The projected CoP A_2 (₹97/ qtl), A_2 +FL (₹120/qtl) and C_2 (₹179/qtl) are lowest in Karnataka and highest in Tamil Nadu at ₹225, ₹247 and ₹305, respectively.
- 5.21 The projected CoPs for major sugarcane growing States at State-specific recovery rate and all-India average for sugar seasons 2025-26 and 2026-27 are given in Annex Table 5.6. The projected CoP A_2 at all-India level at State-specific recovery rate increased by 2.4 percent in sugar season 2026-27 over 2025-26, while A_2 +FL and C_2 increased by 3.1 percent and 2.9 percent, respectively.
- 5.22 The CoP estimate at 10.25 percent basic sugar recovery rate for a State for sugar season 2026-27 has been derived by using sugar recovery rate during TE2024-25, and for sugar season 2025-26, sugar recovery rate during TE2023-24 was used. All-India per quintal CoP A_2 , A_2 +FL and C_2 at 10.25 percent basic recovery rate for sugar season 2026-27 are estimated at ₹151, ₹182 and ₹258, respectively, and CoP is lowest for Karnataka (₹112) and highest for Tamil Nadu (₹276).
- 5.23 Projected CoPs for States and all-India, calculated at 10.25 percent basic sugar recovery rate for the sugar seasons 2025-26 and 2026-27 are presented in Annex Table 5.7. At basic recovery rate of 10.25 percent, all-India CoP A_2 of sugarcane increased by 4.1 percent, A_2 +FL by 5.2 percent, and C_2 by 4.9 percent in sugar season 2026-27 over 2025-26 and the highest increase was recorded in Haryana.
- 5.24 Based on inputs received from the State Governments and using WPI of transport equipment and fuels, transportation cost and insurance charges for sugarcane have been estimated at ₹29 per quintal during the sugar season 2026-27 (details in Annex Table 5.5). All-India modified A_2 , A_2 +FL and C_2 per quintal of sugarcane, inclusive of transportation cost and insurance premium, for sugar season 2026-27 at 10.25 percent basic sugar recovery rate are estimated at ₹180, ₹211 and ₹287, respectively.

**Table 5.3: Projected Cost of Production (CoP) of Sugarcane for Sugar Season 2026-27**

State	Average State-specific Recovery Rate (%) during TE2024-25	CoP at State-specific Recovery Rate (₹/qtl)			CoP at Basic Recovery Rate of 10.25 Percent (₹/qtl)			Production Share (%) during TE 2024-25
		A_2	A_2+FL	C_2	A_2	A_2+FL	C_2	
Gujarat	10.43	131	144	202	128	141	199	3.0
Haryana	10.09	146	151	272	149	154	277	1.6
Karnataka	10.97	97	120	179	90	112	167	10.8
Maharashtra	10.97	185	220	277	173	205	259	24.8
Tamil Nadu	9.15	225	247	305	252	276	341	3.4
Uttar Pradesh	11.19	162	199	300	148	182	275	46.8
All-India	10.98	161	194	276	151	182	258	90.4

Note: All-India CoP of sugarcane is weighted average of CoPs of projected States

Source: CACP Calculations using Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution & CS data

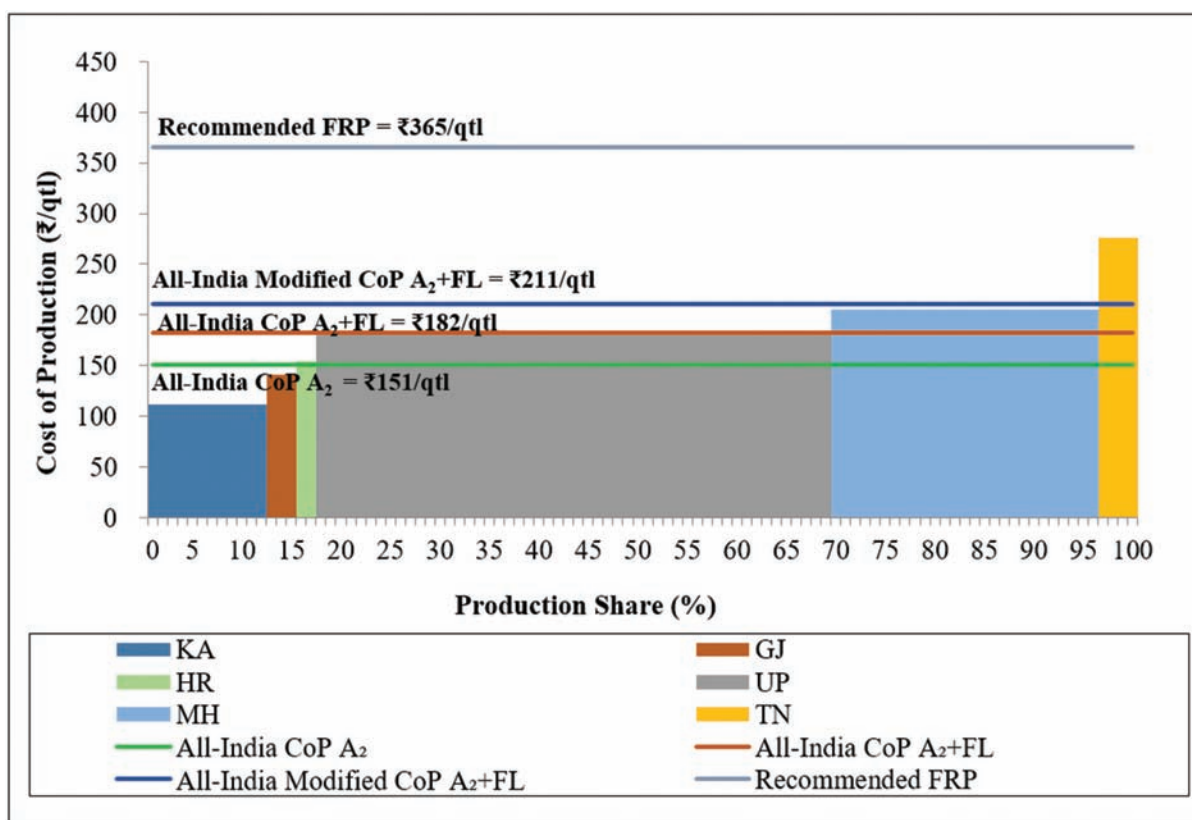
Sugarcane Production and Cost of Production

- 5.25 Chart 5.5 shows supply curve² for A_2+FL CoP at 10.25 percent basic recovery rate for States as well as at all-India level in increasing order of cost with their corresponding relative production shares in the projected States. The supply curve illustrates the percentage of production of major sugarcane growing States that is covered by all-India weighted cost of production. As it is evident from the Chart, more than 62 percent cane production of major producer States has either lower or equal to all-India A_2+FL cost of production. Gujarat, Haryana and Karnataka have lower than all-India cost of production, while in Maharashtra and Tamil Nadu, the cost is higher than all-India average.
- 5.26 The recommended FRP of ₹365 per quintal for the sugar season 2026-27 covers all-India weighted CoP and modified CoP A_2+FL of all major sugarcane producing States at 10.25 percent recovery rate. At all-India level, at 10.25 percent basic recovery rate, FRP margins over all-India CoP A_2+FL and Modified CoP A_2+FL were 100.5 percent and 73 percent, respectively. Margin over CoP A_2+FL was the highest at 225.9 percent in Karnataka, followed by Gujarat (158.9%), Haryana (137%), Uttar Pradesh (100.5%), Maharashtra (78%) and lowest in Tamil Nadu (32.2%).

²Supply curve is graphical representation of CoP, representing the quantum of production of sugarcane at different CoPs in various States.



Chart 5.5: Supply Curve and Projected Cost of Production (A_2 +FL) for Sugarcane at 10.25 Percent Basic Recovery Rate for Sugar Season 2026-27



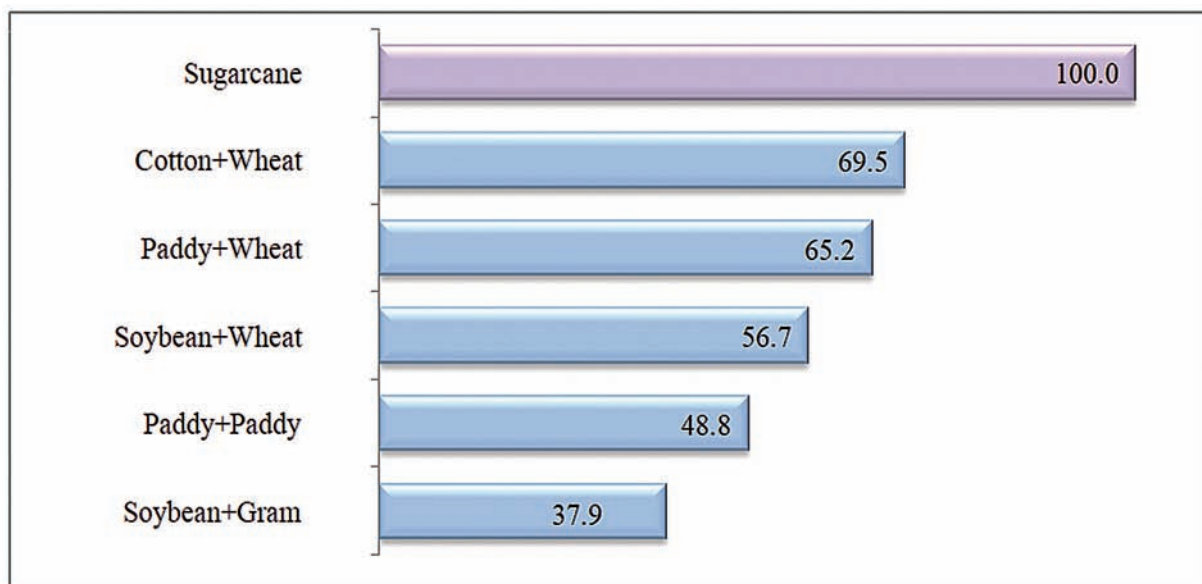
Note 1: Production shares are related to production of sugarcane of projected States mentioned in Chart
 2: All-India CoP of sugarcane are weighted average of CoPs of projected States mentioned in Chart
 Source: CACP Calculations using CS data

Inter-Crop Parity in Returns

- 5.27 Inter-crop parity is an important factor for determination of FRP, therefore, per hectare average gross returns of different combinations of competing mandated crops in major sugarcane growing States for which CS data are available, are compared. The per hectare relative average gross returns of major competing crop combinations during TE2023-24 have been computed at all-India level and are presented in Chart 5.6. The relative average gross returns are measured in percentage terms over $CoC A_2$ +FL for various crop combinations with reference to sugarcane. It is evident from the Chart that sugarcane is the most profitable crop *vis-à-vis* other mandated crop combinations. The relative returns from sugarcane at all-India level are 1.4 times of cotton+wheat, 1.5 times of paddy+wheat, 1.8 times of soybean+wheat, two times of paddy+paddy and 2.6 times of soybean+gram.
- 5.28 The gross returns over $CoC A_2$ and A_2 +FL and relative average gross returns over $CoC A_2$ and A_2 +FL for major competing crop combinations in major States with respect to sugarcane for 2021-22, 2022-23 and 2023-24 are given in Annex Table 5.8. The Table shows that sugarcane is more profitable than all other crop combinations considered in all States in TE2023-24.



Chart 5.6: Relative Average Gross Returns (%) over CoC A_2 +FL with Respect to Sugarcane, TE2023-24



Note: All-India gross returns are weighted average of respective gross returns of projected States

Source: CACP Calculations using CS data

CACP Cost Estimates vis-a-vis State Estimates

- 5.29 A comparison of CACP cost projections and cost of production estimates provided by the States for sugar season 2026-27 has been made and is given in Annex Table 5.9. Gujarat, Haryana, Karnataka, Tamil Nadu and Uttar Pradesh have provided the estimates of cost of cultivation/production for sugarcane. There are variations in State estimates and CACP cost projections, mainly attributed to differences in methodology and cost concepts used by the States and CACP. For example, Haryana has included 10 percent management charges and 10 percent risk management cost and Gujarat has added 10 percent management cost, while estimating cost of production.

Recapitulation

- 5.30 During TE2023-24, average A_2 +FL cost of cultivation of sugarcane at all-India level was ₹1,20,572 per hectare, varied from ₹84,319 per hectare in Karnataka to ₹1,95,891 per hectare in Tamil Nadu. At all-India level, per hectare average GVO was estimated at ₹2,44,332 and varied from ₹2,18,868 in Uttar Pradesh to ₹3,11,895 in Tamil Nadu. The per hectare average gross returns over CoC A_2 +FL at all-India level were ₹1,23,760, highest at ₹1,89,601 in Haryana, and lowest in Maharashtra at ₹91,296.
- 5.31 The average daily wages of agriculture labour at all-India level registered an increase of eight percent in 2024-25 (October-September) over the corresponding period in 2023-24, while weighted index of selected farm input prices decreased by (-)1.4 percent. As human labour accounted for nearly 55 percent of total CoC A_2 +FL during TE2023-24, mechanization in sugarcane cultivation should be promoted to reduce cost of cultivation and improve profitability.



- 5.32 The all-India A_2 +FL CoP of sugarcane for sugar season 2026-27 at State-specific recovery rate is projected at ₹194 per quintal and adjusted A_2 +FL at basic recovery rate of 10.25 percent is estimated at ₹182 per quintal. All-India modified CoP A_2 +FL, inclusive of expenses on transportation and insurance premium, at 10.25 percent basic recovery rate, is estimated at ₹211 per quintal. The A_2 +FL CoP at all-India level increased by 3.1 percent at State specific recovery rate and 5.2 percent at 10.25 percent basic recovery rate in sugar season 2026-27 over 2025-26. Sugarcane was the most profitable crop compared to other competing crop combinations of mandated crops in all sugarcane growing States.
- 5.33 The FRP of ₹365 per quintal recommended for the sugar season 2026-27 would cover A_2 +FL and C_2 cost of production of all major sugarcane producing States. The FRP margin over all-India A_2 +FL CoP at 10.25 percent basic recovery rate is estimated at 100.5 percent, and varies from 225.9 percent in Karnataka to 32.2 percent in Tamil Nadu.



CHAPTER 6



Considerations and Recommendations for Price Policy

- 6.1 The Commission has considered various factors including the cost of production of sugarcane, price of sugar, availability of sugar to the consumer at a fair price, recovery of sugar from sugarcane, realization made from sale of by-products viz. molasses, bagasse and press mud or their imputed value, reasonable margins for the growers of sugarcane on account of risk and profits and general trend of prices of agricultural commodities, for recommending the Fair and Remunerative Price (FRP) for sugarcane for the sugar season 2026-27. In addition, the Commission held extensive consultations with various stakeholders, including the Central and State Governments, sugar industry representatives and industry associations, research institutions, etc. before finalizing the price and non-price recommendations.

Demand-Supply Situation

Domestic

- 6.2 Sugarcane production in the country marginally increased (0.3%) to 454.6 million tonnes in 2024-25 from 453.2 million tonnes in 2023-24 but is significantly lower (-7.3%) than 490.5 million tonnes in 2022-23. Area expansion has been a major driver of sugarcane production, while yield has become stagnant during the last few years. However, a strong rebound in production is estimated in 2025-26 and as per First Advance Estimates sugarcane production is estimated at 475.6 million tonnes, up 4.6 percent from 2024-25. Area expansion (4.5%) is estimated to contribute to higher output in 2025-26 while yield is expected to remain at same level as in 2024-25.
- 6.3 The all-India recovery of sugar from sugarcane witnessed a continuous decline during the last two seasons and fell from 11.2 percent in 2022-23 to 10.5 percent in 2024-25. All major sugarcane growing States, including Uttar Pradesh, Maharashtra and Karnataka, recorded a decline in sugar recovery during 2024-25 season.
- 6.4 Domestic sugar production, which decreased for three consecutive years from 2022-23 to 2024-25, is estimated to increase by 17.2 percent reaching at 30.6 million tonnes in 2025-26. On the contrary, sugar consumption is estimated to increase marginally (1.4%) and is estimated at 28.5 million tonnes in



2025-26. Significantly higher production but marginal rise in consumption is set to increase closing stocks by 12 percent and Stocks- to-Use-Ratio (SUR) from 17.8 percent in 2024-25 to 19.6 percent in 2025-26.

Global

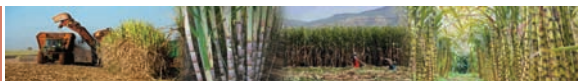
- 6.5 Various agencies expect global sugarcane production to rise in the 2025-26 season, driven by increased output in India, Brazil and Thailand due to favorable weather conditions and expansion in planted area. According to the ISO, FAO, USDA and ABARES, world sugar production is forecast to increase by 3-5 percent in 2025-26, while global consumption is estimated to rise at a significantly lower rate than increase in production. Global sugar trade in 2025-26 is forecast to rise and FAO and ABARES estimate an increase of 3-4 percent in exports, while ISO expects a marginal decline in exports. Both ending stocks and SUR are forecast to be higher in 2025-26 as per the USDA, FAO and ABARES, while ISO expects closing stocks and SUR to be marginally lower than 2024-25 level. Although global sugar market in 2025-26 is projected to remain in deficit, the deficit is likely to be marginal. In contrast, India is expected to have large closing stocks of 5.6 million tonnes of sugar.

Domestic and World Prices

- 6.6 The domestic retail, wholesale and ex-mill prices of sugar showed an increasing trend during the last four seasons and remained much higher than Minimum Selling Price (MSP) of sugar. However, rise in sugar prices was much lower during 2024-25 compared with 2023-24 season. The increase in ex-mill prices was higher than wholesale and retail prices during the last two seasons.
- 6.7 The Wholesale Price Index (WPI) based inflation for sugar, which fell from a high 6.5 percent in 2023-24 to 3.2 percent in 2024-25, marginally increased to 3.9 in 2025-26 (April-September 2025). Molasses and *gur* inflation recorded a continuous decline during the last two years and was negative (-1.2%) for *gur* in 2025-26. The inflation for bagasse, which showed a steep decline from 11.6 percent in 2022-23 to (-)1.1 percent in 2024-25, marginally rose to 0.3 percent in 2025-26. The rate of inflation based on WPI for sugar and *gur* has remained moderate in the recent years.
- 6.8 Global sugar prices have been more volatile than domestic prices and remained significantly higher than domestic prices during 2021 to 2025(Q1). Global prices recorded a generally declining trend from February 2024, while domestic prices continued a rising trend, surpassed international prices in June 2025, and have remained above global prices. The FAO Sugar Price Index averaged 94.1 points in October 2025, down 5.3 points from September 2025 but 35.5 points below its level in October 2024 and the lowest level since December 2020. The decline in international sugar prices was mainly driven by expectations of substantial global surplus and weaker global economic growth.

Trade Performance

- 6.9 India has remained a trade surplus country during the last decade, except in 2016-17, and as a result net exporter of sugar. India's sugar exports, after reaching the record level 11.6 million tonnes in 2021-22, recorded a significant decline during the last three seasons due to lower domestic production and restrictions on exports.



- 6.10 According to ISO estimates, global sugar trade in 2025-26 is anticipated to remain steady with exports marginally up (63.9 million tonnes) from 63.3 million tonnes in the last season, but much lower than the 2023-24 level of 69.3 million tonnes. Import demand is estimated at about 63.8 million tonnes, marginally lower than 2023-24.

Cost of Production and Inter-Crop Parity

- 6.11 During TE2023-24, the all-India average Cost of Cultivation (CoC) A_2 +FL was estimated at ₹1,20,572 per hectare. The CoC was the highest in Tamil Nadu (₹1,95,891/ha) and lowest in Karnataka (₹84,319/ha). Human labour accounted for nearly 55 percent of total costs at the all-India level in TE2023-24. The average daily wages of agriculture labour rose by eight percent in 2024-25 over 2023-24 and index of farm inputs declined by 1.4 percent. At all-India level, sugarcane is more profitable than other competing crop combinations and relative returns from sugarcane were 1.4 times of cotton+wheat, 1.5 times of paddy+wheat, 1.8 times of soybean+wheat, two times of paddy +paddy and 2.6 times of soybean+gram.
- 6.12 For the 2026-27 sugar season, the projected all-India A_2 +FL Cost of Production (CoP) of sugarcane is estimated at ₹194 per quintal based on the State-specific recovery rates and ₹182 per quintal at the basic recovery rate of 10.25 percent, an increase of 5.2 percent over 2025-26 season. The all-India modified A_2 +FL CoP, which includes transportation cost and insurance premium, is projected at ₹211 per quintal at 10.25 percent basic sugar recovery rate.

Non-Price Recommendations

Improve Varietal and Seed Replacement Rate

- 6.13 The introduction of high-yielding varieties like Co 0238 in sub-tropical region and Co 86032 in tropical region has played a pivotal role in improving sugarcane yield and production, but continuous cultivation of these varieties has led to degeneration and increased vulnerability to pests and diseases, causing stagnation/decline in cane and sugar yield. The varietal replacement rate (VRR) and seed replacement rate (SRR) in sugarcane remain extremely low primarily due to non-availability of quality seed/seedlings, and suitable varieties and high costs. Seed treatment, which is crucial in minimizing seed-borne diseases, particularly red rot, is not common among farmers due to various reasons including lack of awareness. Therefore, the Commission recommends that State Governments should formulate an action plan in collaboration with sugar mills and research institutions for developing high cane and sugar yielding varieties, which are tolerant to biotic and abiotic stresses, and ensure availability of seed cane through participatory seed production at farmers' fields under strict monitoring to improve VRR and SRR. Special training and awareness campaigns should be launched to educate farmers about importance of seed treatment and ensure availability of seed treatment devices and promote improved methods of planting like single bud sett planting.

Intercropping for Improving Soil Health and Farmers Income

- 6.14 Intercropping of short duration crops in sugarcane helps in improving soil health, supplementing income and promotes efficient use of available resources. The Indian Council of Agricultural Research (ICAR)



and State Agricultural Universities (SAUs) have developed and demonstrated potential of various sugarcane-based intercropping models. Some State Governments and sugar mills are also promoting sugarcane-based intercropping but adoption among farmers is quite low. Therefore, there is a need to promote intercropping and crop rotation by organizing large-scale field demonstrations and awareness programmes.

Promote Efficient Water Management

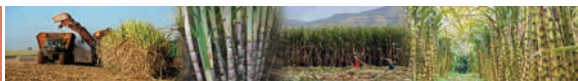
- 6.15 Sugarcane is a water-intensive crop and groundwater is a major source of irrigation in most of sugarcane growing areas. The groundwater extraction is very high in many sugarcane growing areas with a very low coverage of micro-irrigation in such areas. Therefore, there is a need to promote efficient water management technologies and practices such as micro-irrigation, skip-furrow system, trash mulching, etc. to improve water use efficiency. The Commission recommends that a special component or separate allocation for promoting micro-irrigation in sugarcane should be created under *Pradhan Mantri Krishi Sinchayee Yojana- Per Drop More Crop (PMKSY-PDMC)* and State Governments and sugar mills should be encouraged to provide additional subsidy on micro-irrigation as well as actively involved in promoting micro-irrigation in sugarcane.

High-Labour Costs and Mechanisation

- 6.16 Human labour cost accounts for nearly 55 percent of the total cost of cultivation in sugarcane, while share of machine labour is less than 10 percent. Therefore, mechanization in labour-intensive operations like planting, inter-culture and harvesting should be promoted on a massive scale. It is worth mentioning that recent reduction in GST on tractors and farm machinery is a significant move to make farm machinery more affordable and accessible to farmers. However, high cost of machinery & equipment, particularly sugarcane harvesters, and small operational holding are the major constraints in adoption of mechanization. Provisions of subsidy have been made for purchase of high cost machinery like, sugarcane harvester to be promoted through Custom Hiring Centers and Farm Machinery Banks, and other equipments such as sugarcane thrash cutter, settling planter, ratoon manager, power operated sugarcane sett treatment device, etc. under the Sub-Mission on Agricultural Mechanization (SMAM) but allocation for sugarcane remains quite low. Therefore, to promote mechanization in sugarcane cultivation, special provisions or a special component under SMAM should be made for mechanization of sugarcane cultivation. The R&D efforts are required for developing low-cost improved machinery suitable for smallholders, and increasing adoption among small and marginal farmers through collective ownership.

Address Distortions in Sugarcane Pricing

- 6.17 Some States, such as Haryana, Punjab, Uttar Pradesh and Uttarakhand, announce their own State Advised Price (SAP), while some including Tamil Nadu and Bihar announce an incentive/special incentive over and above the Fair and Remunerative Price (FRP). High SAP or additional incentive by State Governments without any linkage to sugar recovery and sugar price can lead to crisis in sugar sector as mills are not able to pay high cane prices, especially when sugar prices are low. A durable solution to this problem lies in moving over to a system where farmers are paid cane price based on



FRP or Revenue Sharing Formula (RSF). Both Rangarajan Committee (2012) and the NITI Aayog Task Force (2020) have recommended adoption of the Revenue Sharing Formula (RSF) to ensure remunerative price to farmers as well as financial viability of sugar mills. The Commission has been recommending discontinuation of SAP by State Governments and adoption of RSF or FRP to encourage and incentivise higher sugar recovery, remove inter-State price distortions and also help in improving financial health of sugar mills. The Commission is of the view that those States, which want to continue with SAP/incentive, should pay the difference between FRP/RSF and SAP directly to farmers through Direct Benefit Transfer (DBT) as adopted by Government of Haryana, Punjab and Tamil Nadu to reduce financial burden on sugar mills and encourage sugarcane cultivation. Considering the structural transformation of the sugar sector through modernization and diversification into ethanol and co-generation, the Commission recommends that a High-Power Expert Committee should be constituted to review the recommendations on RSF and Price Stabilization Fund (PSF) in line with existing industry structure and market conditions.

Review Cane Area Reservation and Minimum Distance Criteria

- 6.18 Cane area reservation and minimum distance criteria are intended to ensure adequate cane supply to mills, prevent unhealthy competition, and committing the sugar mill to procure sugarcane from farmers at FRP/SAP. The Rangarajan Committee had recommended that States should encourage market-based, long-term contractual arrangements and phase out cane reservation and bonding but the Central Government left the recommendation to the State Governments for adoption and implementation. Maharashtra is the only major sugarcane growing State that has abolished cane area reservation provision, while other sugarcane growing States are still continuing with such restrictions. All cane-growing States have the minimum distance criteria of 15-25 km for establishment of a sugar factory under provisions of Sugarcane (Control) Order.
- 6.19 India's installed sugar capacity as well as capacity per factory has more than doubled during the last one and a half decades, while average capacity utilization has declined from over 85 percent to 55 percent, which implies that economies of scale are not fully exploited and thus resulting in higher cost of production. Based on a comprehensive analysis of changing size and structure of Indian sugar industry, the Commission recommends that cane area reservation and minimum distance criteria should be reviewed to avoid unhealthy competition amongst sugar factories and ensure adequate supply of sugarcane.

Link Ethanol Price to FRP of Sugarcane

- 6.20 The Ethanol Blended Petrol (EBP) programme, apart from promoting clean fuel, has played a crucial role in supporting sugarcane farmers, sugar industry and stabilizing India's sugar sector. The administered price mechanism for ethanol produced from sugarcane-based feedstocks and guaranteed procurement by OMCs under EBP programme have helped in stabilizing Indian sugar sector by diversion of surplus sugar and derivatives for ethanol production as well as timely cane payment to farmers. However, it should be noted that during the last five ESYs (2021-22 to 2024-25), grain-based ethanol prices have significantly increased but price of ethanol from sugarcane juice/sugar/sugar syrup and B-Heavy molasses



increased marginally compared with increase in FRP of sugarcane. The Commission, therefore, recommends that the Government should revise the administered prices of sugarcane-based ethanol in view of increase in FRP of sugarcane.

Dual Pricing of Sugar

- 6.21 In view of rising cost of sugar production, low-price realization from sugar and ethanol, and reduced allocation for ethanol from sugar-based feedstocks, sugar industry is facing significant challenges. Since industrial/commercial sector uses about 60-65 percent of total sugar production in the country, the Commission recommends that feasibility of a dual pricing mechanism of sugar for domestic and commercial users should be explored. Various stakeholders, including sugar industry, industry associations and some State Governments, have also suggested for adopting dual price policy for sugar.

Fair and Remunerative Price (FRP) Recommendations

- 6.22 Based on the Cost of Production of sugarcane and other relevant factors associated with sugarcane and sugar economy, the Commission recommends FRP of sugarcane for 2026-27 sugar season to be ₹365 per quintal linked to a basic sugar recovery rate of 10.25 percent. To promote efficiency and incentivise farmers for higher sugar recovery, for every 0.1 percentage point increase in sugar recovery over and above 10.25 percent, the FRP should be increased by ₹3.56 per quintal. The Commission also recommends that for every 0.1 percentage point decline in sugar recovery below 10.25 percent, the FRP should be reduced by ₹3.56 per quintal. The Commission believes that the recommended FRP and non-price measures would incentivise and encourage sugarcane farmers and sugar mills to adopt modern technologies and practices to improve competitiveness and efficiency of Indian sugarcane based economy.

(Vijay Paul Sharma)
Chairman

(Prem Chand)
Member (Official)

(Ratan Lal Daga)
Member (Non-Official)

(Cyriac George)
Member Secretary



Annex Tables

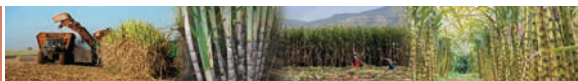
**Annex Table 1.1: Area under Cultivation of Sugarcane in Major States**

(‘000 ha)

State/Year	BR	GJ	HR	KA	MP	MH	PB	TN	UP	All India
2011-12	218	202	95	430	69	1022	80	346	2162	5038
2012-13	250	176	101	425	60	933	83	347	2212	4999
2013-14	258	174	102	420	73	937	89	313	2228	4993
2014-15	254	208	97	480	111	1030	94	263	2141	5067
2015-16	244	157	93	450	103	987	90	252	2169	4927
2016-17	240	169	102	397	92	633	88	218	2160	4436
2017-18	234	182	114	370	98	902	96	172	2234	4737
2018-19	226	155	109	471	108	1163	95	166	2224	5061
2019-20	224	161	96	429	125	822	91	131	2208	4603
2020-21	221	219	99	443	95	1143	89	128	2180	4851
2021-22	211	223	108	637	93	1262	87	148	2177	5175
2022-23	210	181	108	659	92	1359	90	159	2795	5885
2023-24	202	202	88	543	115	1437	90	152	2653	5740
2024-25	193	189	72	540	107	1167	89	131	2720	5450
2025-26*	202	186	94	648	36	1372	95	94	2720	5693

Note: *First Advance Estimates

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare



Annex Table 1.2: Production of Sugarcane in Major States

('000 tonnes)

State/Year	BR	GJ	HR	KA	MP	MH	PB	TN	UP	All India
2011-12	11289	12750	6959	38808	2677	86733	5653	38576	128819	361036
2012-13	12741	12690	7437	35732	2642	69648	5919	33919	132428	341200
2013-14	12882	12550	7499	37905	3174	76901	6675	32454	134689	352142
2014-15	14034	14330	7169	43776	4567	84699	7039	28093	133061	362333
2015-16	12649	11120	6692	37834	5281	73680	6607	25494	145385	348448
2016-17	13036	11950	8223	27378	4730	52262	7152	18988	140169	306069
2017-18	13825	12072	9633	31135	5430	82984	8024	17154	177033	379905
2018-19	20116	11326	8505	42408	5282	89768	7774	17140	179715	405416
2019-20	13579	11570	7730	38181	7434	69312	7302	14119	179539	370500
2020-21	12110	16955	8532	42528	5441	105138	7487	13285	178339	405399
2021-22	12026	17459	8823	61152	5379	116076	7131	16166	179167	439425
2022-23	12743	13461	8861	59346	6444	123968	7506	17658	224246	490533
2023-24	12084	14480	7374	41811	7590	112088	7510	15927	215811	453158
2024-25	12150	13714	5830	48060	6109	109970	7279	13351	220801	454611
2025-26*	12721	13492	7611	55080	2054	129271	7874	9616	220801	475614

Note: *First Advance Estimates

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare



Annex Table 1.3: Yield of Sugarcane in Major States

(t/ha)

State/Year	BR	GJ	HR	KA	MP	MH	PB	TN	UP	All India
2011-12	51.7	63.1	73.3	90.3	38.7	84.9	70.7	111.4	59.6	71.7
2012-13	50.9	72.1	73.6	84.1	44.4	74.6	71.3	97.7	59.9	68.3
2013-14	49.9	72.1	73.5	90.3	43.4	82.1	75.0	103.6	60.5	70.5
2014-15	55.2	68.9	73.9	91.2	41.1	82.2	74.9	106.8	62.2	71.5
2015-16	51.8	70.8	72.0	84.1	51.3	74.7	73.4	101.1	67.0	70.7
2016-17	54.4	70.7	80.6	69.0	51.4	82.5	81.3	87.0	64.9	69.0
2017-18	59.1	66.3	84.5	84.1	55.4	92.0	83.6	99.8	79.2	80.2
2018-19	89.2	73.2	78.2	90.0	48.9	77.2	81.8	103.0	80.8	80.1
2019-20	60.6	71.9	80.3	89.0	59.5	84.3	80.2	107.6	81.3	80.5
2020-21	54.8	77.3	86.2	96.0	57.3	92.0	83.8	104.0	81.8	83.6
2021-22	56.9	78.3	81.9	96.0	57.8	92.0	82.2	109.2	82.3	84.9
2022-23	60.6	74.3	82.2	90.0	70.0	91.2	83.1	111.1	80.2	83.3
2023-24	59.8	71.6	83.6	77.0	66.0	78.0	83.3	105.0	81.3	79.0
2024-25	63.1	72.7	81.4	89.0	57.1	94.2	82.1	102.0	81.2	83.4
2025-26*	63.1	72.7	81.2	85.0	57.1	94.2	82.9	102.0	81.2	83.5

Note: *First Advance Estimates

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare



Annex Table 1.4: Production of Sugar in Major States, 2011-12 to 2024-25

(Lakh Tonnes)

States/ Year	BH	HR	PB	UK	UP	AP	TG	GJ	MH	KA	TN	Others	All India
2011-12	4.5	4.9	3.9	3.3	69.6	11.4		10.0	90.0	38.7	23.8	3.3	263.4
2012-13	5.1	5.1	4.4	3.4	75.0	9.8		11.3	79.9	34.4	19.3	4.1	251.8
2013-14	6.0	5.4	4.7	3.0	66.1	6.8	3.3	11.8	77.2	41.6	14.2	5.4	245.5
2014-15	5.3	5.7	5.4	3.3	71.4	5.7	3.2	11.5	105.4	49.9	12.6	5.5	284.6
2015-16	5.0	5.4	6.7	2.7	68.5	5.5	2.8	11.6	84.2	40.5	13.7	4.8	251.2
2016-17	5.3	6.7	6.6	3.5	87.7	3.9	1.2	8.8	41.9	21.4	10.5	4.8	202.3
2017-18	7.2	7.3	8.2	4.5	120.5	4.6	2.7	11.1	107.1	36.9	5.5	6.5	322.0
2018-19	8.4	6.8	7.9	3.9	118.6	4.9	2.6	13.7	107.2	44.3	7.2	5.8	331.3
2019-20	7.2	7.4	5.4	4.6	128.7	2.9	1.4	9.3	61.8	33.8	7.3	4.3	274.0
2020-21	4.6	7.7	5.5	4.1	110.6	2.0	1.2	10.5	106.4	42.9	8.8	5.7	310.0
2021-22	4.6	7.2	6.0	4.5	102.0	2.1	2.3	12.1	137.4	62.1	12.7	6.3	359.0
2022-23	6.3	7.5	6.6	4.9	104.8	1.9	2.6	9.9	105.3	59.8	14.8	5.6	330.0
2023-24	6.9	6.1	6.0	3.1	104.0	1.6	1.8	9.2	111.0	52.9	10.5	6.9	320.0
2024-25	6.2	5.2	5.7	3.7	92.7	0.9	1.7	8.9	81.0	42.4	6.5	6.2	261.1

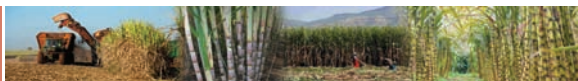
Source: Directorate of Sugar & Vegetable Oils, Department of Food and Public Distribution



Annex Table 1.5: Cane Reservation Area and Minimum Distance Criteria in Major Sugarcane Producing States

State	Whether Cane Reservation Area exists	Existing Minimum Distance Criteria norm (Kms) between two sugar mills
Bihar	Yes	15
Haryana	Yes	25 for 2500 TCD, 40 for 5000 TCD, 60 for 10000 TCD
Karnataka	Yes	15
Madhya Pradesh	Yes	15
Maharashtra	No	25
Punjab	Yes	25
Tamil Nadu	Yes	15
Uttar Pradesh	Yes	15
Uttarakhand	Yes	15

Source: Directorate of Sugar & Vegetable Oils, Department of Food and Public Distribution & State Replies



Annex Table 2.1: Sugar Balance Sheet as per the National Federation of Cooperative Sugar Factories Limited

(Lakh Tonnes)

Sl. No.	Particulars	2022-23	2023-24	2024-25	2025-26 [#]
1	Opening Stocks [@]	71.2	57.2	80.2	50.0
2	Gross Sugar Production	375.0	343.0	296.8	350.0
3	Diversion in to the Ethanol	45.0	24.0	35.0	35.0
4	Net Sugar Production	330.0	319.0	261.8	315.0
5	Total Net Availability	401.2	376.2	342.0	365.0
6	Domestic consumption	280.0	295.0	284.0	290.0
7	Exports during the Season	64.0	1.0	8.0	15.0
8	Closing Stocks [*]	57.2	80.2	50.0	60.0
9	Stocks-to-Use Ratio (%)	16.6	27.1	17.1	19.7

Note: 1. [#] Anticipated

2. [@] As on 01st October

3. ^{*} As on 30th September

4. Stocks-to-Use Ratio shows closing stocks as percentage of (Consumption+Exports)

Source: National Federation of Cooperative Sugar Factories Ltd.



Annex Table 2.2: Sugar Balance Sheet as per the Indian Sugar & Bio-energy Manufacturers Association

(Lakh Tonnes)

Sl. No.	Particulars	2022-23	2023-24	2024-25 [#]	2025-26 ^{##}
1	Opening Stocks ^{\$}	70.0	55.7	79.0 ^{\$\$}	50.0
2	Gross Sugar Production [@]	366.2	341.1	296.0	343.5 ^{@@}
3	Diversion for Ethanol ^{##}	38.0	21.5	35.0	34.0
4	Net Production	328.2	319.6	261.0	309.0
5	Imports [*]	0.0	0.0	0.0	0.0
6	Total Availability	398.2	375.3	340.0	359.5
7	Internal Consumption	278.5	290.0	281.0	285.0
8	Exports [*]	64.0	0.5	9.0	15.0
9	Total Offtake	342.5	290.5	290.0	300.0
10	Closing Stocks ^{**}	55.7	84.8	50.0	59.5
11	Stocks-to-Use Ratio (%)	16.2	29.2	17.2	19.8

Note: 1. [#] Provisional

2. ^{##} Estimated

3. ^{\$} As on 01st October

4. ^{\$\$} Opening Stocks of 2024-25 reconciled with Government Data

5. [@] Without diversion for Ethanol

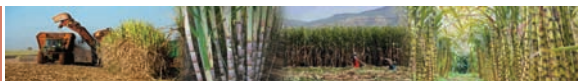
6. ^{@@} As per pre estimates

7. ^{*} Imports and Exports are under O.G.L. and as reported by sugar mills to GoI

8. ^{**} As on 30th September i.e. at the end of each Sugar Season

9. Stocks-to-Use Ratio shows closing stocks as percentage of (Consumption+Exports)

Source: Indian Sugar & Bio-energy Manufacturers Association (ISMA).



Annex Table 2.3: All-India Monthly Average Retail Prices of Sugar

(₹/qtl)

Month	2020-21	2021-22	2022-23	2023-24	2024-25
Oct	4019	4230	4251	4396	4490
Nov	4003	4244	4261	4462	4495
Dec	3990	4195	4220	4468	4471
Jan	3977	4143	4183	4453	4465
Feb	3934	4114	4172	4442	4512
Mar	3929	4107	4161	4428	4553
Apr	3947	4110	4192	4433	4571
May	3985	4144	4238	4457	4584
Jun	3970	4167	4272	4479	4590
Jul	3950	4164	4299	4487	4594
Aug	4002	4191	4318	4482	4620
Sep	4144	4244	4343	4472	4642
Average	3988	4171	4243	4455	4549

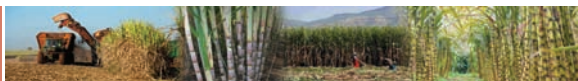
Source: Department of Consumer Affairs, Ministry of Consumer Affairs, Food & Public Distribution

**Annex Table 2.4: All-India Monthly Average Wholesale Prices of Sugar**

(₹/qtl)

Month	2020-21	2021-22	2022-23	2023-24	2024-25
Oct	3694	3905	3915	4080	4171
Nov	3681	3916	3925	4132	4172
Dec	3667	3854	3904	4137	4141
Jan	3618	3812	3869	4118	4135
Feb	3577	3787	3848	4099	4191
Mar	3574	3772	3843	4082	4232
Apr	3605	3786	3886	4093	4250
May	3657	3821	3926	4122	4266
Jun	3646	3831	3947	4145	4265
Jul	3621	3815	3974	4151	4267
Aug	3670	3844	3998	4163	4296
Sep	3829	3903	4025	4157	4311
Average	3653	3837	3922	4123	4225

Source: Department of Consumer Affairs, Ministry of Consumer Affairs, Food & Public Distribution



Annex Table

Annex Table 2.5: Ex-Mill Prices of Sugar in Major Sugar Producing States

(₹/qtl)

State	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Andhra Pradesh	2460	3127	3658	3162	3191	3310	3260	3260*	3260*	3260*	3260*
Gujarat	2404	3064	3615	3196	3119	3314	3252	3460	3491	3675	3834
Karnataka	2318	3035	3516	3068	3080	3270	3250	3342	3356	3780	3912
Maharashtra	2340	3045	3515	3045	3067	3256	3155	3355	3378	3549	3700
Punjab	2575	3195	3688	3274	3190	3363	3247	3247*	3247*	3247*	3247*
Tamil Nadu	2454	3183	3623	3293	3233	3405	3290	3462	3518	3548	3704
Uttar Pradesh	2578	3207	3649	3246	3183	3361	3166	3525	3608	3831	3930

Note: * Figures not available so 2020-21 price is considered

Source: Directorate of Sugar and Vegetable Oils, Ministry of Consumer Affairs, Food and Public Distribution



Annex Table 2.6: FAO Price Index of Sugar & Wholesale Price Indices of Sugar, Molasses, Bagasse and Gur

Month	FAO Price Index	Wholesale Price Index			
	<i>Sugar</i>	<i>Sugar</i>	<i>Molasses</i>	<i>Bagasse</i>	<i>Gur</i>
Apr-20	63.9	116.7	124.8	149.4	127.0
May-20	68.6	115.3	122.4	146.6	134.4
Jun-20	75.7	117.8	119.2	145.1	134.6
Jul-20	76.8	119.2	117.4	144.9	139.5
Aug-20	82.0	118.6	118.4	143.3	141.2
Sep-20	79.8	118.3	114.1	142.5	143.6
Oct-20	85.6	117.3	113.1	142.2	140.5
Nov-20	88.5	117.7	110.9	128.3	132.5
Dec-20	88.1	117.7	113.1	125.3	125.5
Jan-21	93.7	116.3	111.9	120.7	126.1
Feb-21	99.7	116.1	114.0	122.9	124.9
Mar-21	95.7	116.0	117.0	125.8	124.4
Apr-21	99.5	117.5	121.3	125.5	127.4
May-21	106.2	118.2	124.3	128.6	131.5
Jun-21	107.2	117.4	127.4	126.9	132.2
Jul-21	109.0	116.4	127.8	126.9	134.1
Aug-21	119.9	119.6	127.9	126.9	135.3
Sep-21	121.2	125.6	131.0	127.4	137.0
Oct-21	119.1	126.8	131.5	129.5	143.7
Nov-21	120.2	124.4	131.1	128.4	136.8
Dec-21	116.4	123.7	132.1	133.7	135.5
Jan-22	112.7	123.4	133.2	134.6	130.0
Feb-22	110.5	123.2	136.1	135.5	138.8
Mar-22	117.9	123.1	136.3	138.2	131.9
Apr-22	121.5	123.5	139.0	143.1	134.3
May-22	120.4	124.5	140.7	145.9	135.1
Jun-22	117.3	123.9	141.7	144.6	137.8
Jul-22	112.8	123.5	144.6	146.1	142.9
Aug-22	110.5	124.3	144.7	145.8	137.8
Sep-22	109.7	125.0	144.7	146.4	150.9
Oct-22	109.0	125.1	140.9	144.8	154.8
Nov-22	114.4	126.2	137.5	146.7	149.1

(Contd...)



Dec-22	117.2	126.3	137.8	144.4	135.3
Jan-23	116.8	125.5	138.8	142.8	134.4
Feb-23	125.2	125.5	139.3	145.2	135.8
Mar-23	127.0	124.7	139.8	147.8	138.1
Apr-23	149.4	127.7	140.4	147.0	141.8
May-23	157.2	128.2	143.9	149.3	148.5
Jun-23	152.2	129.2	146.9	148.3	152.0
Jul-23	146.3	129.3	147.0	148.1	155.3
Aug-23	148.2	130.5	148.6	149.8	160.6
Sep-23	162.7	132.6	149.8	147.1	161.3
Oct-23	159.2	135.1	149.5	146.2	158.7
Nov-23	161.4	138.9	149.0	151.1	150.6
Dec-23	134.2	136.6	152.0	154.8	147.0
Jan-24	136.4	136.2	154.8	153.9	150.0
Feb-24	140.8	135.1	154.8	150.0	150.2
Mar-24	133.4	134.8	156.8	148.3	150.3
Apr-24	126.6	135.7	157.3	153.2	151.1
May-24	117.1	137.8	156.7	151.3	157.2
Jun-24	119.4	137.3	157.3	148.2	162.8
Jul-24	119.5	136.3	157.1	149.0	165.4
Aug-24	113.9	137.6	156.5	149.3	165.0
Sep-24	126.3	136.1	157.9	150.2	165.5
Oct-24	129.6	136.9	158.4	146.2	168.3
Nov-24	126.4	136.0	158.7	147.4	163.4
Dec-24	119.3	134.0	157.6	144.4	153.4
Jan-25	111.2	136.8	158.2	143.6	152.7
Feb-25	118.5	140.1	157.6	144.5	150.0
Mar-25	116.9	142.1	158.4	147.7	151.6
Apr-25	112.3	142.6	159.5	149.2	155.0
May-25	109.4	142.6	162.0	150.0	157.7
Jun-25	103.6	141.5	162.3	149.9	159.7
Jul-25	103.3	141.3	162.4	151.7	157.7
Aug-25	103.6	142.7	162.6	152.7	161.3
Sep-25	99.4	142.1	163.1	153.5	164.2
Oct-25	94.1	142.6	163.9	152.4	167.8

Note: FAO Sugar Price Index Base: 2014-16, WPI Base: 2011-12

Source: 1) Food & Agriculture Organization

2) Office of the Economic Advisor, Ministry of Commerce & Industry

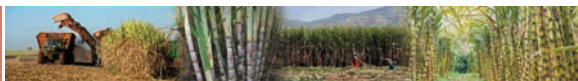


Annex Table 2.7: State-Wise SAP

(₹/qtl)

Year	Uttar Pradesh			Punjab			Haryana			Uttarakhand		
	Early Varieties	Mid Varieties	Late Varieties	Early Varieties	Mid Varieties	Late Varieties	Early Varieties	Mid Varieties	Late Varieties	Early Varieties	Mid Varieties	Late Varieties
2012-13	290	280	275	250	240	235	276	271	266	295	285	310
2013-14	290	280	275	290	280	275	301	295	290	295	285	295
2014-15	290	280	275	295	285	280	310	305	300	290	280	310
2015-16	290	280	275	295	285	280	310	305	300	290	280	295
2016-17	315	305	300	300	290	285	320	315	310	317	307	310
2017-18	325	315	310	310	300	295	330	325	320	326	316	295
2018-19	325	315	310	310	300	295	340	335	335	327	317	310
2019-20	325	315	310	310	300	295	340	335	335	327	317	-
2020-21	325	315	310	310	300	295	340	335	335	327	317	-
2021-22	350	340	-	360	360	360	362	355	355	355	345	-
2022-23	350	340	335	380	370	365	372	365	365	355	345	345
2023-24	370	360	355	391	381	381	386	379	379	375	365	365
2024-25	370	360	355	401	391	391	400	393	393	375	365	365
2025-26	400	390	390	NA	NA	NA	415	408	408	NA	NA	NA

Source: Directorate of Sugar, Ministry of Consumer Affairs, Food & Public Distribution



Annex Table 2.8: Sugar Recovery in Major States of India

State/Year	AP	BR	GJ	HR	KR	MH	PB	TN	TG	UP	UK	All India	FRP (₹ per quintal)	Basic Recovery Level
1999-00	10.09	9.20	10.61	9.27	10.65	11.39	9.10	9.20	NA	9.34		10.20		
2000-01	10.36	9.11	10.42	9.80	10.75	11.63	9.70	9.64		9.71	NA	10.48	-	-
2001-02	10.01	8.82	10.79	9.95	10.72	11.60	9.45	9.61		9.53	9.42	10.27	-	-
2002-03	10.15	9.05	10.58	10.13	10.80	11.68	9.72	9.87		9.54	9.47	10.38	-	-
2003-04	10.32	9.33	10.93	10.47	10.21	10.93	9.72	9.92		9.82	9.75	10.22	69	8.50
2004-05	10.65	9.58	10.76	10.16	10.11	11.39	9.79	9.64		9.79	9.63	10.17	73	8.50
2005-06	10.05	9.48	10.82	9.78	10.83	11.66	9.19	9.24		9.49	9.42	10.22	74.5	9.00
2006-07	9.69	8.67	10.68	9.74	10.69	11.39	9.54	9.31		9.49	9.54	10.16	79.5	9.00
2007-08	10.10	9.20	10.90	9.90	10.10	11.80	9.30	9.30		9.30	9.80	10.30	80	9.00
2008-09	9.88	9.30	9.50	9.05	10.30	11.52	9.33	9.56		8.91	9.20	10.05	81	9.00
2009-10	9.28	9.49	10.52	9.37	10.67	11.51	8.59	8.94		9.13	9.19	10.20	130	9.50
2010-11	9.77	9.30	9.99	9.02	10.92	11.30	8.80	9.10		9.15	9.34	10.17	139	9.50
2011-12	9.80	9.10	10.61	9.07	11.18	11.67	9.12	9.35		9.08	9.10	10.26	145	9.50
2012-13	9.65	8.91	10.77	9.76	10.42	11.45	9.09	8.88		9.17	9.19	10.01	170	9.50
2013-14	9.77	9.40	11.11	9.40	10.98	11.24	9.40	9.00	10.53	9.44	9.23	10.25	210	9.50
2014-15	9.72	9.10	10.48	9.89	10.90	11.67	9.37	8.10	10.47	9.49	9.15	10.44	220	9.50
2015-16	9.31	9.69	10.37	9.39	10.59	11.19	10.01	8.47	10.83	10.51	9.52	10.47	230	9.50
2016-17	9.34	9.08	10.56	10.19	10.28	11.19	9.34	9.08	10.33	10.26	8.20	10.25	230	9.50
2017-18	9.43	9.54	10.51	10.46	10.57	11.09	9.74	8.50	10.84	10.73	10.18	10.64	255	9.50
2018-19	9.33	10.35	10.65	10.31	10.25	10.07	10.10	8.90	10.21	10.91	10.91	10.37	275	10.00
2019-20	9.41	11.13	10.75	10.55	10.16	11.54	9.76	8.58	10.27	11.74	11.21	11.24	275	10.00
2020-21	8.95	10.03	10.23	10.24	9.74	10.49	9.03	8.96	10.31	10.76	10.87	10.37	285	10.00
2021-22	9.38	11.39	10.60	9.85	10.91	11.20	9.71	9.26	10.78	11.46	10.38	11.00	290	10.00
2022-23	9.40	11.60	10.70	10.10	11.80	11.30	10.60	9.46	10.57	11.40	10.12	11.20	305	10.25
2023-24	9.00	11.70	10.30	10.50	10.60	11.10	10.10	8.80	10.60	11.50	10.40	11.00	315	10.25
2024-25	8.01	10.14	10.30	9.68	10.50	10.50	9.50	9.20	10.70	10.68	10.90	10.50	340	10.25

Note: Sugar recovery Rate from 2019-20 to 2024-25 includes sugar diverted to ethanol.

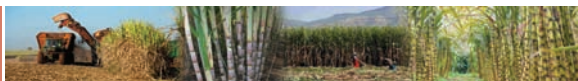
Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution



**Annex Table 2.9(a): State-Wise Existing Ethanol Producing Capacity of Distilleries in 2022-23
(As on 30th September)**

State/UT	Molasses/Dual		Grain		Total	
	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)
Maharashtra	133	293	29	43	162	336
Uttar Pradesh	68	276	7	20	75	296
Karnataka	41	196	7	25	48	220
Punjab	4	15	19	82	23	97
Haryana	4	9	18	68	22	77
Bihar	8	17	9	52	17	69
Andhra Pradesh	8	11	16	40	24	50
Madhya Pradesh	5	10	13	39	18	49
West Bengal	-	-	4	20	4	20
Telangana	3	7	3	12	6	19
Gujarat	12	15	1	3	13	19
Tamil Nadu	12	18	-	-	12	18
Jharkhand	-	-	2	14	2	14
Rajasthan	-	-	9	13	9	13
Himachal Pradesh	-	-	3	9	3	9
Uttarakhand	3	7	-	-	3	7
Chhattisgarh	-	-	3	5	3	5
Odisha	-	-	3	4	3	4
Assam	-	-	2	3	2	3
Sikkim	-	-	1	2	1	2
Total	301	873	149	455	450	1328

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution



**Annex Table 2.9(b): State-Wise Existing Ethanol Producing Capacity of Distilleries in 2023-24
(As on 30th September)**

State/UT	Molasses/Dual		Grain		Total	
	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)
Maharashtra	133	336	29	50	162	386
Uttar Pradesh	65	263	9	27	74	290
Karnataka	41	225	6	21	47	246
Punjab	4	15	23	115	27	130
Haryana	4	9	18	69	22	78
Bihar	8	18	14	70	22	88
Andhra Pradesh	8	11	12	34	20	45
Madhya Pradesh	5	10	20	94	25	104
West Bengal	-	-	7	50	7	50
Telangana	4	8	8	37	12	45
Gujarat	13	16	2	12	15	28
Tamil Nadu	13	20	-	-	13	20
Jharkhand	-	-	5	32	5	32
Rajasthan	-	-	10	21	10	21
Himachal Pradesh	-	-	2	9	2	9
Uttarakhand	3	8	1	1	4	9
Chhattisgarh	1	2	6	24	7	26
Odisha	-	-	5	20	5	20
Assam	-	-	5	19	5	19
Sikkim	-	-	1	2	1	2
Total	302	941	183	707	485	1648

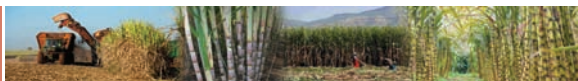
Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution



**Annex Table 2.9(c): State-Wise Existing Ethanol Producing Capacity of Distilleries in 2024-25
(As on 30th September)**

State/UT	Molasses/Dual		Grain		Total	
	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)	No. of projects	Installed Capacity (in cr ltrs p.a.)
Maharashtra	136	327	32	71	168	398
Uttar Pradesh	63	267	16	67	79	333
Karnataka	42	259	8	52	50	311
Punjab	4	15	23	112	27	127
Haryana	4	10	20	99	24	109
Bihar	8	19	15	77	23	96
Andhra Pradesh	8	11	9	51	17	62
Madhya Pradesh	4	8	20	97	24	105
West Bengal	0	0	8	59	8	59
Telangana	4	8	8	42	12	50
Gujarat	13	16	3	23	16	40
Tamil Nadu	13	20	2	12	15	32
Jharkhand	0	0	5	32	5	32
Rajasthan	0	0	11	28	11	28
Himachal Pradesh	0	0	3	16	3	16
Uttarakhand	3	12	3	30	6	42
Chhattisgarh	1	2	8	32	9	34
Odisha	0	0	6	23	6	23
Assam	0	0	5	31	5	31
Sikkim	0	0	1	2	1	2
Total	303	973	206	957	509	1930

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution



Annex Table 2.10: State-Wise Weighted Prices of Sugar Primary By-Products

(₹/tonne)

States	2020-21	2021-22	2022-23	2023-24	2024-25
Molasses B -Heavy					
MH	6,565.2	4,468.9	3,808.3	4,348.3	4,197.3
UP	-	3,584.0	4,211.6	4,518.4	5,556.1
KR	3,347.1	2,523.4	3,136.0	2,637.5	2,867.6
BR	135.0	-	-	331.5	-
PB	-	208.0	236.7	7.3	265.0
Molasses C -Heavy					
MH	2,686.7	2,996.1	2,507.5	3,077.3	3,256.4
UP	2,396.9	1,728.6	2,795.1	2,886.7	3,242.0
KR	1,084.4	1,533.9	1,803.8	1,639.1	1,620.3
TN	283.2	276.4	356.5	309.4	250.2
GJ	258.2	339.0	290.8	344.5	414.7
BR	44.7	73.0	110.1	123.9	137.3
HR	168.3	173.8	191.1	199.7	232.3
PB	108.9	130.6	154.8	6.3	213.7
Bagasse					
MH	716.5	998.7	835.8	923.2	851.6
UP	711.9	472.3	825.2	910.7	975.3
KR	289.2	360.9	379.7	345.1	437.3
TN	68.1	90.3	117.1	91.8	81.0
GJ	34.3	77.2	84.9	88.6	73.4
BR	26.4	21.2	40.0	53.3	-
HR	44.1	46.9	69.2	48.1	51.5
PB	33.0	33.5	45.7	1.5	54.9
Pressmud					
MH	163.7	182.2	153.5	222.2	116.9
UP	57.7	61.8	145.8	144.5	159.4
KR	58.7	82.3	96.9	114.7	139.9
TN	4.5	5.6	7.2	5.3	4.0
GJ	0.6	2.7	2.4	2.3	4.6
HR	1.7	2.3	7.8	10.5	13.5
PB	3.6	2.4	7.1	0.2	6.0

Note: Weighted prices of sugar primary by-products have been calculated by taking sugar production in respective States as weights

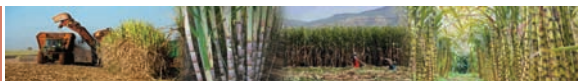
Source: 1. State replies for State-wise by-products prices
 2. Directorate of Sugar & Vegetable Oils, Department of Food and Public Distribution, Ministry of Consumer Affairs, Food & Public Distribution for sugar production



Annex Table 2.11: Status of Sugar Mills and Recovery Rate in Major Sugar Producing States during 2024-25 Sugar Season

States	Number of Mills	Number of Operational Mills	Sugar Production (exluding diversion to ethanol) (lakh tonnes)	Recovery Rate (%)
Maharashtra	261	200	81.0	10.50
Uttar Pradesh	159	122	92.7	10.68
Karnataka	94	79	42.4	10.50
Tamil Nadu	46	30	6.5	9.20
Gujarat	29	15	8.9	10.30
Haryana	16	14	5.2	9.68
Punjab	24	15	5.7	9.50
Bihar	28	10	6.2	10.14
Madhya Pradesh	34	22	5.1	10.70
Uttarakhand	10	8	3.7	10.90
Others	73	19	3.7	-
All India	774	534	261.1	10.50

Source: Directorate of Sugar & Vegetable Oils, Ministry of Consumer Affairs, Food & Public Distribution

**Annex Table 3.1: Share of Ratooning in Sugarcane Area in Karnataka**

Year	Share (%)	Year	Share (%)
2013-14	80.62	2019-20	96.47
2014-15	57.37	2020-21	73.97
2015-16	95.48	2021-22	77.44
2016-17	97.43	2022-23	86.23
2017-18	64.73	2023-24	68.56
2018-19	92.07		

Source: Cost of Cultivation Scheme data from Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare



Annex Table 3.2: Top-3 Varieties of Sugarcane in Major States

State	TE2016-17			TE2023-24		
	Top-1	Top-2	Top-3	Top-1	Top-2	Top-3
Andhra Pradesh	NA	NA	NA	2003V46 (48.0)	87A298 (25.1)	2009A107 (7.1)
Bihar	BO-110 (16.7)	BO-91 (15.8)	Co 0238 (10.7)	Co 0238 (64.5)	CoP 9301 (16.5)	Co 0118 (4.5)
Chhattisgarh	Co 86032 (38.5)	Co 8036 (11.9)	Co 62175 (10.7)	Co 86032 (41.3)	CO-085 (17.9)	Co 9805 (12.3)
Gujarat	CoM 265 (21.7)	Co 86032 (18.2)	Co 86002 (15.4)	Co 86032 (28.9)	Phule 265 (15.2)	VSI 08005 (11.8)
Haryana	Co 0238 (29.2)	CoH-119 (21.5)	Co-89003 (12.9)	Co 0238 (50.4)	Co 0118 (13.5)	CoH 160 (11.6)
Karnataka	Co 86032 (61.4)	Co 62175 (15.0)	CoVCF 015 (10.0)	NA	NA	NA
Maharashtra	Co 86032 (53.4)	CoM-265 (31.1)	Co-92005 (6.2)	Co 86032 (45.4)	CoM 265 (26.9)	VSI 08005 (11.3)
Odisha	Co 86032 (9.4)	Co 86V96 (9.4)	CoA 89085 (9.3)	CoA 92081 (18.2)	CoV-09-0356 (13.2)	Co 62175 (8.9)
Punjab	Co 0238 (59.6)	CoJ85 (12.6)	Co-89003 (11.5)	Co 0238 (70.8)	Co 0118 (16.1)	CoJ 85 (2.9)
Tamil Nadu	Co 86032 (62.3)	CoV 09356 (11.2)	CoC 24 (8.0)	Co 86032 (48.7)	CoV 09356 (28.9)	Co 11015 (9.7)
Telangana	Co 86032 (57.9)	93V297 (18.8)	87A298 (8.2)	93V297 (31.9)	Co86032 (28.2)	2003V46 (18.7)
Uttar Pradesh	Co 0238 (22.6)	Co.S.767 (19.2)	Cos.91269 (5.0)	Co 0238 (71.2)	Co 0118 (11.7)	CoLK 94184 (6.9)
Uttarakhand	Cos.767 (32.6)	Cos.8436 (13.6)	Cos.97264 (10.0)	Co 0238 (82.8)	Co 0118 (8.8)	Co 0118 (3.3)
West Bengal	Co 527 (28.0)	BO-91 (25.0)	CoB 94164 (20.0)	Co 527 (26.9)	BO-91 (23.6)	CoB 94164 (17.8)

Note: Figures in parentheses are area shares

Source: Respective States Department of Sugarcane/Sugar/Agriculture



Annex Table 3.3: Status of Groundwater Extraction, Micro-irrigation and Sugarcane Yield in Major Districts

State	District	Stage of Groundwater Extraction (%)	Area under Sugarcane (%)		Micro-irrigation Coverage (%)	Yield (t/ha)
			w.r.t. All-India Cane Area	w.r.t. District's NSA*		
UP	Kheri	62.0	5.6	70.9	0.3	86.4
KA	Belagavi	71.9	4.3	55.4	14.3	92.8
MH	Solapur	77.6	4.1	25.9	8.4	94.9
UP	Bijnor	70.4	4.2	77.3	0.6	89.4
UP	Muzaffarnagar	77.4	2.9	83.6	0.7	94.4
MH	Pune	64.5	2.6	37.8	7.8	107.0
UP	Sitapur	57.4	3.4	50.1	0.5	79.3
MH	Ahmednagar	79.0	2.9	15.3	7.6	89.7
MH	Kolhapur	42.5	3.0	51.2	2.9	86.0
UP	Meerut	76.9	2.5	18.6	2.2	91.6
MH	Satara	62.6	2.0	22.6	5.5	107.2
MH	Sangli	53.5	2.1	24.5	8.9	90.3
UP	Saharanpur	96.8	2.0	57.0	0.8	83.9
BR	Pashchim Champaran	28.8	2.4	56.0	0.2	61.6
KA	Bagalkote	84.4	2.0	49.3	18.6	74.7
UP	Bareilly	72.5	1.9	35.3	0.6	78.5
UP	Pilibhit	62.8	1.7	48.0	0.7	81.2
UP	Shahjahanpur	64.5	1.7	30.2	0.6	80.0
UP	Hardoi	61.2	1.5	21.1	0.6	81.5
UP	Shamli	101.3	1.2	75.3	1.2	102.4
UP	Amroha	95.9	1.4	63.0	1.0	83.2
KA	Vijayapura	43.2	1.3	15.8	24.0	86.4
UP	Gonda	61.3	1.5	32.6	0.5	78.6
UK	Haridwar	65.4	1.3	66.9	1.1	90.8
UP	Kushi Nagar	52.0	1.4	40.5	1.3	76.2
GJ	Surat	42.4	1.4	39.0	5.6	78.3
UP	Balrampur	57.3	1.6	0.8	1.6	66.2
UP	Moradabad	84.9	1.3	46.0	1.0	79.1
UP	Bulandshahr	95.0	1.1	27.8	0.4	89.7
MH	Beed	59.4	1.2	10.4	9.0	72.3
UP	Bahraich	59.2	1.2	22.6	0.9	77.8
MH	Latur	53.8	1.1	8.3	9.8	82.9
MH	Osmanabad	59.7	1.4	10.6	10.6	62.7
UP	Sambhal	87.9	1.0	26.1	1.0	83.6

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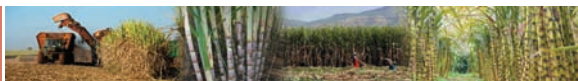


Annex Table 3.3: Status of Groundwater Extraction, Micro-irrigation and Sugarcane Yield in Major Districts

State	District	Stage of Groundwater Extraction (%)	Area under Sugarcane (%)		Micro-irrigation Coverage (%)	Yield (t/ha)
			w.r.t. All-India Cane Area	w.r.t. District's NSA*		
KA	Mandya	63.8	0.6	19.1	34.9	114.5
UP	Baghpat	95.9	1.2	78.3	1.9	88.4
UP	Ayodhya	62.3	0.9	28.5	1.0	78.0
GJ	Bharuch	26.6	1.0	18.9	5.4	67.7
MH	Parbhani	46.0	0.8	7.8	9.2	85.0
UP	Basti	65.4	0.9	28.4	1.6	67.9
UP	Hapur	95.7	0.7	48.3	2.1	84.7
KA	Kalaburagi	43.5	0.7	7.0	80.2	79.1
UP	Rampur	71.5	0.6	18.9	1.2	80.8
MH	Jalna	52.2	0.7	8.3	15.3	69.3
MH	Aurangabad	64.7	0.7	5.5	21.4	73.9
TN	Kallakurichi	83.4	0.4	16.6	8.8	110.5
TN	Erode	81.6	0.3	13.4	52.6	120.5
MH	Nanded	32.1	0.6	5.5	7.9	61.5
MP	Narsinghpur	65.5	0.8	16.6	1.0	48.8
GJ	Tapi	32.7	0.4	21.3	15.7	81.7
UP	Ghaziabad	124.0	0.4	59.6	3.0	89.6
UP	Ambedkar Nagar	62.8	0.4	17.1	1.8	83.6
PB	Hoshiarpur	116.2	0.4	12.4	1.0	79.1
PB	Gurdaspur	136.4	0.4	10.5	0.4	86.9
MH	Nashik	57.7	0.4	1.5	10.2	83.8
TN	Cuddalore	65.0	0.3	7.1	19.3	114.0
TN	Tiruvannamalai	82.9	0.3	6.9	23.5	98.6
TN	Villupuram	87.3	0.2	8.2	51.8	106.0
UP	Budaun	83.0	0.3	6.4	0.4	80.9
KA	Mysuru	51.6	0.2	4.4	65.1	122.5
TN	Namakkal	112.4	0.2	7.5	48.0	129.2
MH	Hingoli	39.3	0.3	2.5	9.3	73.2
KA	Bidar	60.4	0.3	6.7	117.2	86.4
MH	Nandurbar	41.8	0.4	9.0	11.4	52.3
BR	Purbi Champaran	43.3	0.3	7.1	0.3	59.9

Note: * Net Sown Area

Source: 1. Dynamic Ground Water Resources of India Report, 2024
 2. UPAg Portal, Ministry of Agriculture and Farmers Welfare
 3. Portal of Pradhan Mantri Krishi Sinchayee Yojana



Annex Table

Annex Table 4.1: Top-10 Export Destinations of Indian Sugar (Sugar Season)

2020-2021			2021-22			2022-23			2023-24			2024-25*		
Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share
Indonesia	5473	23.0	Indonesia	5593	13.6	Sudan	6634	18.3	Libya	2789	13.0	Sudan	2255	14.1
Sudan	2719	11.4	Sudan	5037	12.2	Bangladesh	3418	9.5	Sudan	2535	11.8	Libya	1732	10.8
UAE	1850	7.8	Bangladesh	4319	10.5	Sri Lanka	2700	7.5	Somalia	2237	10.4	Somalia	1685	10.5
Somalia	1659	7.0	Saudi Arabia	2661	6.5	Djibouti	2479	6.9	Djibouti	2027	9.4	Sri Lanka	1398	8.7
Sri Lanka	1424	6.0	Somalia	2545	6.2	Somalia	2476	6.8	Sri Lanka	1774	8.3	Djibouti	1297	8.1
Bangladesh	1339	5.6	UAE	2111	5.1	UAE	1708	4.7	Tanzania	1664	7.7	Tanzania	921	5.7
Afghanistan	1275	5.4	Iraq	1959	4.8	Indonesia	1444	4.0	Kenya	1517	7.1	UAE	898	5.6
Djibouti	1237	5.2	Djibouti	1867	4.5	Iraq	1302	3.6	Pakistan	907	4.2	Bangladesh	835	5.2
Saudi Arabia	930	3.9	Pakistan	1689	4.1	Kenya	1056	2.9	Yemen Republic	739	3.4	Kenya	695	4.3
Pakistan	794	3.3	Malaysia	1672	4.1	Libya	954	2.6	Bangladesh	629	2.9	Lebanon	650	4.1
Others	5063	21.4	Others	11747	28.4	Others	11994	33.2	Others	4687	21.8	Others	3673	22.9
Total	23764	100	Total	41199	100	Total	36164	100	Total	21504	100	Total	16040	100

Note: 1. Commodity - 1701 Cane/Beet Sugar chemically pure sucrose in solid form

2. * for October 2024 to August 2025

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Annex Table 4.2: Top-10 Import Origins of Indian Sugar (Sugar Season)

2020-2021			2021-22			2022-23			2023-24			2024-25*		
Country	Values (₹ lakh)	% share	Country	Values (₹ lakh)	% share	Country	Values (₹ lakh)	% share	Country	Values (₹ lakh)	% share	Country	Values (₹ lakh)	% share
Brazil	318056	94.7	Brazil	131657	93.8	Brazil	641142	97.2	Brazil	1674894	99.4	Brazil	928511	98.0
South Africa	6665	2.0	Germany	4203	3.0	Sudan	12218	1.9	UAE	4095	0.2	UAE	8319	0.9
Germany	4982	1.5	USA	1525	1.1	Germany	2611	0.4	Germany	2439	0.2	Germany	4181	0.4
UAE	1935	0.6	UAE	1062	0.8	USA	1232	0.2	USA	1429	0.1	Poland	1924	0.2
USA	1198	0.4	New Zealand	707	0.5	New Zealand	717	0.1	Belgium	803	0.1	USA	1588	0.2
New Zealand	1066	0.3	Italy	425	0.3	Belgium	585	0.1	New Zealand	547	0.0	Belgium	1052	0.2
Italy	580	0.2	Belgium	391	0.3	France	342	0.1	Italy	110	0.0	France	514	0.1
Belgium	548	0.2	Netherland	103	0.1	Italy	145	0.0	Netherland	109	0.0	New Zealand	328	0.0
Japan	110	0.0	France	83	0.1	Netherland	91	0.0	Japan	27	0.0	Kenya	276	0.0
Korea Rp	104	0.0	Spain	42	0.0	Japan	28	0.0	Mauritius	9	0.0	Italy	208	0.0
Others	313	0.1	Others	42	0.0	Others	23	0.0	Others	13	0.0	Others	182	0.0
Total	335556	100	Total	140240	100	Total	659134	100	Total	1684475	100	Total	947084	100

Note: 1. Commodity - 1701 Cane / Beet Sugar chemically pure sucrose in solid form

2. * for October 2024 to August 2025

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Annex Table 4.3: World Sugar Projections

WORLD	Unit	Average 2022-24 (E)	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Sugarbeet												
Production	Mt	270.4	271.6	272.1	273.5	273.8	274.2	274.4	274.8	275.2	275.7	276.2
Area	Mha	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Yield	t/ha	62.60	62.96	63.25	63.45	63.62	63.82	64.01	64.21	64.42	64.64	64.89
Biofuel use	Mt	10.5	10.0	9.9	9.8	9.6	9.2	8.8	8.5	8.2	7.9	7.5
Sugarcane												
Production	Mt	1836	1897	1913	1940	1967	1992	2015	2038	2059	2078	2100
Area	Mha	24.9	25.3	25.4	25.6	25.9	26.1	26.3	26.5	26.7	26.8	27.0
Yield	t/ha	73.62	74.97	75.26	75.66	75.96	76.29	76.62	76.94	77.25	77.57	77.89
Biofuel use	Mt	375.2	426.6	452.7	467.6	482.4	494.2	504.8	514.6	523.7	531.3	538.9
Sugar												
Production	Mt tq	178.5	185.9	185.4	188.0	190.5	193.3	195.8	198.3	200.5	202.7	205.0
Consumption	Mt tq	176.8	181.5	183.7	186.1	188.3	190.5	192.9	195.3	197.7	199.9	202.3
Closing	Mt tq	84.4	84.7	84.2	83.9	84.0	84.6	85.3	86.2	86.9	87.5	88.1
Price, raw sugar	USD/t	464.5	383.8	400.0	405.1	411.5	417.1	422.0	426.0	429.9	434.0	437.5
Price, white sugar	USD/t	581.0	489.3	507.0	513.3	524.4	533.0	539.1	544.5	551.4	560.0	566.6

Note: 1. E: Estimated; Mt: Million Metric tonnes; Mha: Million hectares; t/ha: tonnes/hectare, tq: Tel quel basis or "as it comes" for sugar

2. Raw sugar world price, ICE contract No11 nearby (October/September).

3. Refined sugar price, White Sugar Futures Contract No. 407, Euronext market, Liffe, London, Europe (October/September).

Source: OECD-FAO Agricultural Outlook 2025-2034

**Annex Table 4.4(a): Trade policy (Export) of Sugar since 2011**

Month	Trade Policy	Description
March 2011	Export under OGL	Exports of 15 lakh tonnes in 2010-11 and 20 lakh tonnes in 2011-12 allowed under Open General License (OGL).
May 2012	Export under OGL ceased	Exports of 15 lakh tonnes in 2010-11 and 20 lakh tonnes in 2011-12 under OGL were ceased.
February 2014	New Scheme	The scheme envisaged an export target of 40 lakh tonnes during 2013-14 and 2014-15 sugar seasons subject to the condition that the incentives available under the scheme shall be utilized by the sugar mills for making payment to the farmer.
February 2015	Increase in export incentive	The export incentive amount was increased to ₹ 4,000 per tonne of raw sugar (on flat basis) for 2014-15 sugar season, subject to ceiling of 14 lakh tonnes.
June 2016	Discontinuation of export incentive	Anticipating fall in sugar production during 2016-17, the export incentive was discontinued and Government imposed an export duty of 20 percent on sugar exports under OGL to protect the domestic market.
March 2018	Withdrawal of export duty	Domestic sugar production rebounded and Government withdrew the export duty on sugar. Moreover, mill-wise Minimum Indicative Export Quota (MIEQ) of 20 lakh tonnes was fixed to promote exports and stabilize the falling domestic sugar prices. Additionally, for the duty free import of raw sugar intended for export, Government allowed Duty Free Import Authorization (DFIA) Scheme.
September 2018	Export target fixed under MIEQ	To promote exports, mill-wise export targets were fixed allocating MIEQ of 50 lakh tonnes of sugar, for sugar season 2018-19.
October 2018	New Scheme launched	In order to facilitate exports, during the sugar season 2018-19, a scheme for extending assistance towards internal transport, freight, handling and other charges was also notified for sugar mills.
September 2019	New Scheme launched	Government of India notified a scheme for providing assistance at the rate of ₹10,448 per tonne to sugar mills for expenses on marketing costs including handling, upgrading and other processing costs, cost of international and internal transport and freight charges on export of 60 lakh tonnes under Maximum Admissible Export Quantity (MAEQ).
September 2020	Extension of Scheme duration	Government also allowed those sugar mills, which had partially exported their MAEQ quota of sugar season 2019-20 till 30 th September 2020, to export the balance quantity of their quota by 31 st December 2020.
December 2020	Export incentives under MAEQ	Export incentives at the rate of ₹ 6,000 per tonne (for marketing expenses, internal transport and ocean freight) were notified for MAEQ of 60 lakh tonnes for the sugar season 2020-21. Sugar mills, which exported domestically manufactured sugar (of their own or through an exporter) under OGL and not under “advance license”, were provided the lump sum assistance under this scheme.
May 2021	Reduction in export incentives under	The previous incentive was reduced to ₹ 4,000 per tonne of sugar. Moreover, the reduced assistance was limited to only internal trans-

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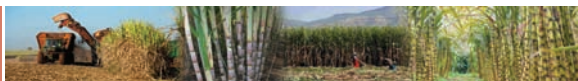
	MAEQ with extended timeline	port and ocean freight. Further, in September 2021, the timeline for availing MAEQ benefits for sugar season 2020-21 was extended from 30 th September 2021 to 30 th November 2021.
October 2021	TRQ for EU	Total 5,841 tonnes of raw/refined sugar was authorized to export to EU under the TRQ scheme from 1 st October 2021 to 30 th September 2022.
December 2020 & November 2021	TRQ for USA	DGFT allocated 8,727 MTRV (Metric Tonne Raw Value) of sugar (in the form of raw cane sugar / refined and specialty sugar/sugar-containing products) for the export to USA for fiscal year 2021 till 31 st December 2021.
May 2022 & October 2022	TRQ for USA	DGFT allocated a total quantity of 10,475 metric tonnes of sugar for the export to USA till 31 st December 2022 under the TRQ for fiscal year 2022.
May 2022	Free to Restricted	DGFT modified the export policy of sugar (raw, refined and white sugar) from “Free” to “Restricted” w.e.f 1 st June 2022.
October 2022	Export restriction extended	The export restriction on sugar was initially put in place for the period 1 st June 2022 to 31 st October 2022 but the same was extended till 31 st October 2023.
November 2022	Export quota announced	During 2022-23 season, under the restricted export regime, Government announced sugar export quota for exports of sugar up to 60 lakh tonnes w.e.f 1 st November 2022 till 31 st May 2023. For this purpose, mill-wise export quotas were fixed at 18.23 percent of the average production share of a sugar mill in the last three years. Additionally, mills had also been given the flexibility to surrender the quota partially/fully or swap their export quota with domestic quota of another sugar mill within 60 days of the said order.
April 2023	TRQ for USA	Allocation of additional quantity of 2360 tonnes for export of raw cane sugar to USA from India under TRQ up to 30 th September 2023.
August 2023	TRQ for EU	Total 5,841 tonnes of raw/refined sugar has again been authorized to be exported to EU from India under the TRQ scheme from October 2023 to September 2024.
October 2023	Export restriction extended	The export restrictions on sugar have been further extended beyond 31 st October 2023 till further order.
January 2024	TRQ for USA	Allocated 8606 MTRV of raw cane sugar to USA from India under TRQ Scheme from 1 st October, 2023 to 30 th September, 2024.
September 2024	TRQ for USA	Allocated 8606 MTRV of raw cane sugar to USA from India under TRQ Scheme from 1 st October, 2024 to 30 th September, 2025.
September 2024	TRQ for EU	Allocated 5841 MT of sugar to be exported to EU from India under TRQ Scheme for the year 2024-25 (October 2024 to September 2025).
June 2025	Exports of Pharma Grade Sugar	Allowed exports of Pharma Grade Sugar, up to a total 25,000 MTs per financial year to bonafide pharma exporters against a Restricted Export Authorization on 17th June 2025.
August 2025	TRQ for EU	Allocated 5841 MT of sugar to be exported to EU from India under TRQ Scheme for the year 2025-26 (October 2025 to September 2026) on 1st August 2025.

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November 2025	Export quota announced	On 14 th November 2025, Government allocated export quota of 15 LMT of sugar for the sugar season 2025-26 <i>vide</i> notification No.1(4)/2025-SP. The export quota has been allocated on pro-rata basis amongst operational sugar mills by taking into their average production of sugar during last three sugar seasons.
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Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry
2. Central Board of Indirect Taxes and Customs, Department of Revenue

**Annex Table 4.4(b): Trade policy (Import) of Sugar since 2009**

Month	Trade Policy	Description
April 2009	Import under AAS	To boost the domestic availability of sugar, as a result of fall in sugar production in 2008-09 and 2009-10 seasons, import of raw sugar was allowed into the country at zero import duty under Advance Authorization Scheme (AAS) from 17 th April 2009 to 30 th June 2012.
July 2012	Import duty	The import duty was increased from zero to 10 percent.
July 2013	Import duty	The import duty was increased from 10 to 15 percent.
August 2014	Import duty	The import duty was increased from 15 to 25 percent.
April 2015	Import duty	The import duty was increased from 25 to 40 percent.
April 2015	Import under OGL	Government allowed import of 5 lakh tonnes of raw sugar at zero duty under OGL for eastern, southern and western zones with quantity and port restrictions to mitigate the regional production gaps and to maintain domestic prices at reasonable levels.
July 2016	Import duty under AAS	Sugar exported under AAS was exempted from import duty.
July 2017	Import duty	The import duty was increased from 40 to 50 percent.
February 2018	Import duty	The import duty was increased from 50 to 100 percent and at present the rate is continued to be at the same level.

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry
 2. Central Board of Indirect Taxes and Customs, Department of Revenue



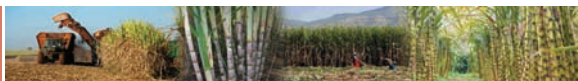
Annex Table 4.5: Top-5 Export Destinations of Indian Molasses (Sugar Season)

2020-2021			2021-22			2022-23			2023-24			2024-25*		
Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share	Country	Values (₹ crore)	% share
Thailand	478	28.3	Netherland	586	30.8	Netherland	587	32.8	Bangladesh	304	62.7	Bangladesh	179	99.3
Korea Rp	290	17.2	Philippines	370	19.4	Philippines	247	13.8	Korea Rp	78	16.2	Bhutan	1	0.6
Netherland	267	15.9	Vietnam	192	10.1	Bangladesh	161	9.0	Philippines	62	12.8	UK	0	0.1
Philippines	174	10.3	Spain	139	7.3	UK	154	8.6	UK	15	3.1	Malaysia	0	0.0
Vietnam	94	5.6	Korea Rp	114	6.0	Italy	128	7.1	Turkey	8	1.8	UAE	0	0.0
Others	384	22.7	Others	501	26.4	Others	513	28.7	Others	17	3.4	Others	0	0.0
Total	1687	100	Total	1902	100	Total	1790	100	Total	484	100	Total	180	100

Note: 1. Commodity - 1703 Molasses resulting from the extraction or refining of sugar

2. * for October 2024 to August 2025

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Annex Table 5.1: Actual Costs and Gross Returns over Actual Cost of Cultivation (₹/ha) of Sugarcane in Major Sugarcane Growing States: 2021-22 to 2023-24

State/Year	Cost of Cultivation		GVO	Gross Returns over			
	A ₂	A ₂ +FL		A ₂		A ₂ +FL	
				(Col.4 - Col.2)	Percent (Col.5 / Col.2) *100	(Col.4 - Col.3)	Percent (Col.7 / Col.3)*100
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Gujarat							
2021-22	90664	99151	253543	162879	179.7	154392	155.7
2022-23	87225	96372	264094	176869	202.8	167722	174.0
2023-24	98352	108500	309860	211508	215.1	201360	185.6
TE2023-24	92080	101341	275832	183752	199.6	174491	172.2
Haryana							
2021-22	86531	89497	263705	177174	204.8	174208	194.7
2022-23	90068	93144	279379	189311	210.2	186235	199.9
2023-24	92895	96095	304454	211559	227.7	208359	216.8
TE2023-24	89831	92912	282513	192681	214.5	189601	204.1
Karnataka							
2021-22	64904	78681	252931	188027	289.7	174250	221.5
2022-23	68042	84651	251470	183428	269.6	166819	197.1
2023-24	71413	89625	268171	196758	275.5	178546	199.2
TE2023-24	68120	84319	257524	189404	278.0	173205	205.4
Maharashtra							
2021-22	150029	180228	269802	119773	79.8	89574	49.7
2022-23	145128	173561	263678	118549	81.7	90117	51.9
2023-24	160843	188804	283002	122159	75.9	94198	49.9
TE2023-24	152000	180864	272160	120160	79.1	91296	50.5
Tamil Nadu							
2021-22	175133	189742	307226	132093	75.4	117484	61.9
2022-23	177758	194743	310269	132511	74.5	115526	59.3
2023-24	183225	203188	318189	134964	73.7	115001	56.6
TE2023-24	178705	195891	311895	133189	74.5	116003	59.2

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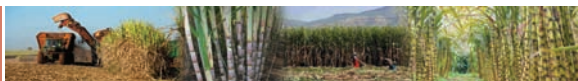


Annex Table 5.1: Actual Costs and Gross Returns over Actual Cost of Cultivation (₹ /ha) of Sugarcane in Major Sugarcane Growing States: 2021-22 to 2023-24

State/Year	Cost of Cultivation		GVO	Gross Returns over			
	A ₂	A ₂ +FL		A ₂		A ₂ +FL	
				(Col.4 - Col.2)	Percent (Col.5 / Col.2) *100	(Col.4 - Col.3)	Percent (Col.7 / Col.3)*100
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Uttar Pradesh							
2021-22	80651	98968	236112	155462	192.8	137145	138.6
2022-23	75697	93448	209630	133933	176.9	116183	124.3
2023-24	75755	93782	210861	135105	178.3	117079	124.8
TE2023-24	77368	95399	218868	141500	182.9	123469	129.4
All-India							
2021-22	101367	121377	251614	150247	148.2	130238	107.3
2022-23	96452	116199	235183	138731	143.8	118983	102.4
2023-24	103794	124139	246198	142404	137.2	122059	98.3
TE2023-24	100538	120572	244332	143794	143.0	123760	102.6

Note: All-India CoC, GVO and gross returns are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.2: Monthly Average Daily Wage Rates (₹/day) for Agriculture Labour (Men) in Selected Sugarcane Growing States during 2021 to 2025

Year/Month	GJ	HR	KA	MH	TN	UP	IND
2021							
October	242	405	374	331	544	309	377
November	242	406	374	328	543	310	376
December	244	410	378	329	543	315	378
2022							
January	245	410	379	331	544	319	380
February	247	412	387	332	550	320	383
March	248	413	389	333	550	323	384
April	252	421	389	334	550	327	386
May	253	422	388	340	551	328	388
June	256	422	391	341	554	328	390
July	260	427	391	341	549	330	390
August	262	435	396	343	550	335	393
September	262	437	402	343	550	335	394
October	263	442	407	344	551	337	396
November	264	442	416	345	553	337	398
December	266	449	417	347	555	338	400
2023							
January	268	452	419	349	558	341	403
February	269	452	419	350	560	342	403
March	271	452	421	351	561	345	405
April	271	452	423	351	572	346	408
May	272	460	423	352	583	347	412
June	274	461	424	355	590	349	415
July	277	461	427	357	600	350	419
August	277	461	430	362	603	352	422
September	278	461	431	364	605	354	423

(Contd...)



October	279	461	431	366	610	355	426
November	282	461	435	368	610	356	427
December	282	462	438	370	611	356	429
2024							
January	283	466	442	371	611	357	430
February	283	472	447	371	611	358	431
March	282	474	451	372	613	359	433
April	283	481	451	372	613	363	434
May	285	487	453	374	613	365	435
June	291	491	457	376	613	365	437
July	297	496	458	377	618	368	440
August	299	502	458	386	621	369	444
September	302	507	463	389	626	372	448
October	302	512	465	391	630	374	451
November	308	512	465	392	632	376	453
December	312	512	466	395	634	378	455
2025							
January	313	512	467	396	637	381	457
February	313	517	471	397	638	382	458
March	313	520	477	399	639	382	460
April	313	528	482	402	640	384	462
May	315	536	488	403	643	386	465
June	320	541	491	406	645	386	468
July	339	572	513	403	716	409	495
August	354	574	536	407	719	418	504
September	357	577	545	412	724	418	508

Note: 1: Daily Wage rate - Average of agricultural operations, such as, ploughing/tilling; sowing (including planting/transplanting/weeding); harvesting/winnowing/threshing; watering & irrigation and plant protection (applying pesticides, treating seeds, etc.)

2: All-India daily wage rate is weighted average of daily wage rates of States mentioned in the Table

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

Annex Table 5.3: Monthly Wholesale Price Index (Base: 2011-12) of Major Farm Inputs during 2023 to 2025

Year/ Month	High Speed Diesel	Fertilizers and Nitrogen Compounds	Electricity	Manufacture of Agricultural and Forestry Machinery	Lube Oils	Cattle Feed	Fodder	Pesticides and other Agrochemical Products	Index of Farm Inputs
2023									
October	175.1	142.2	142.5	142.5	190.2	218.6	242.5	133.0	160.5
November	174.3	143.0	153.3	143.2	190.2	218.9	230.9	132.8	163.1
December	172.0	142.5	158.4	143.9	190.2	219.1	225.6	132.3	163.7
2024									
January	171.8	141.4	157.2	144.1	190.2	218.3	222.7	131.2	162.9
February	172.1	142.0	157.0	144.1	190.2	215.9	220.5	130.0	162.8
March	170.3	142.6	144.6	143.9	190.2	216.3	219.3	130.5	158.4
April	168.5	143.3	141.1	145.3	190.2	214.7	207.2	127.1	156.2
May	167.7	143.3	135.7	145.2	190.2	214.5	204.4	127.8	154.1
June	166.0	143.3	132.0	145.0	190.2	214.2	204.9	128.0	152.5
July	167.4	143.4	132.9	143.5	190.2	214.1	205.6	128.9	153.2
August	166.7	143.2	134.8	144.3	190.2	214.8	208.8	129.8	153.8
September	165.1	142.7	136.7	145.2	190.2	214.9	210.2	128.8	153.9
October	164.2	142.9	147.4	145.3	190.2	214.0	213.2	129.3	157.1
November	164.4	143.5	149.4	145.6	190.2	212.1	218.9	129.3	158.2
December	164.6	143.0	157.5	145.8	190.2	211.1	219.6	128.7	160.7
2025									
January	165.6	142.9	155.8	146.6	190.2	209.0	226.4	129.2	160.8
February	166.6	142.8	154.4	147.0	190.2	207.7	227.9	129.2	160.8
March	165.4	142.3	151.3	146.7	190.2	207.1	221.5	130.0	159.0
April	160.0	142.7	139.8	146.7	190.2	208.0	212.9	131.7	153.5
May	157.3	143.3	137.9	146.8	190.2	206.2	207.8	132.6	151.9
June	157.5	143.0	134.5	146.8	190.2	206.3	206.8	131.6	150.8
July	160.2	142.9	133.7	146.4	190.2	206.7	206.8	131.2	151.3
August	160.8	142.9	133.3	146.2	190.2	207.7	213.8	131.6	151.8
September	160.0	143.0	134.9	145.6	190.2	207.3	215.2	132.4	152.1

Note : Index of Farm Inputs is weighted average of WPIs of farm inputs mentioned in the Table

Source : Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations



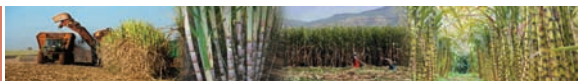


Annex Table 5.4(a): Sugarcane : Break-up of cost of Cultivation of Gujarat

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	98629	95837	107778	-2.8	12.5
Human Labour					
<i>Casual</i>	33443	34792	34439	4.0	-1.0
<i>Attached</i>	115	0	0	-	-
<i>Family</i>	8487	9147	10148	7.8	10.9
Total	42045	43939	44587	4.5	1.5
Bullock Labour					
<i>Hired</i>	3	0	151	-	-
<i>Owned</i>	0	0	0	-	-
Total	3	0	151	-	-
Machine Labour					
<i>Hired</i>	8340	7342	8679	-12.0	18.2
<i>Owned</i>	4475	5037	2050	12.6	-59.3
Total	12815	12379	10729	-3.4	-13.3
Seed	8824	4548	14347	-48.5	215.5
Fertilisers and Manure					
<i>Fertilisers</i>	11960	11624	12747	-2.8	9.7
<i>Manure</i>	156	180	162	15.2	-10.0
Total	12116	11804	12908	-2.6	9.4
Insecticides	711	767	810	7.8	5.7
Irrigation charges	16600	17082	18280	2.9	7.0
Interest on working capital	5302	5099	5743	-3.8	12.6
Miscellaneous	213	219	223	2.7	1.8
Crop Insurance	0	0	0	-	-
Fixed Cost	48592	50720	58168	4.4	14.7
Rental value of owned land	42266	44025	51130	4.2	16.1
Rent paid for leased-in land	0	0	154	-	-
Land revenue, cesses & taxes	10	10	11	8.2	6.0
Depreciation on implements & Farm buildings	513	525	558	2.4	6.3
Interest on fixed capital	5805	6161	6315	6.1	2.5
Yield(Qtl/ha)	868.67	889.79	911.69	2.4	2.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.4(b): Sugarcane : Break-up of cost of Cultivation of Haryana

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	86964	92011	94986	5.8	3.2
Human Labour					
<i>Casual</i>	38526	41780	43599	8.4	4.4
<i>Attached</i>	609	653	672	7.2	3.0
<i>Family</i>	2966	3076	3199	3.7	4.0
Total	42101	45509	47471	8.1	4.3
Bullock Labour					
<i>Hired</i>	446	471	465	5.4	-1.3
<i>Owned</i>	0	0	0	-	-
Total	446	471	465	5.4	-1.3
Machine Labour					
<i>Hired</i>	4700	4742	4581	0.9	-3.4
<i>Owned</i>	7518	7799	7933	3.7	1.7
Total	12218	12541	12513	2.6	-0.2
Seed	11107	11734	12240	5.6	4.3
Fertilisers and Manure					
<i>Fertilisers</i>	6870	7023	7360	2.2	4.8
<i>Manure</i>	226	243	251	7.5	3.6
Total	7096	7265	7611	2.4	4.8
Insecticides	3847	3957	4019	2.8	1.6
Irrigation charges	5056	5157	5118	2.0	-0.7
Interest on working capital	4941	5231	5399	5.9	3.2
Miscellaneous	151	145	150	-3.9	3.0
Crop Insurance	0	0	0	-	-
Fixed Cost	88859	96483	94716	8.6	-1.8
Rental value of owned land	77420	86274	84907	11.4	-1.6
Rent paid for leased-in land	1428	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-
Depreciation on implements & Farm buildings	1104	1133	1109	2.6	-2.2
Interest on fixed capital	8906	9075	8700	1.9	-4.1
Yield(Qtl/ha)	721.23	728.80	758.40	1.0	4.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

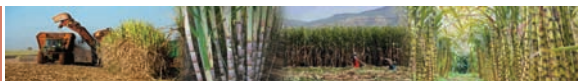


Annex Table 5.4(c): Sugarcane : Break-up of cost of Cultivation of Karnataka

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	78459	84436	89406	7.6	5.9
Human Labour					
<i>Casual</i>	41079	44491	46101	8.3	3.6
<i>Attached</i>	24	43	48	81.2	12.7
<i>Family</i>	13777	16608	18212	20.5	9.7
Total	54880	61142	64361	11.4	5.3
Bullock Labour					
<i>Hired</i>	390	326	392	-16.6	20.4
<i>Owned</i>	777	583	531	-24.9	-9.0
Total	1167	909	923	-22.1	1.6
Machine Labour					
<i>Hired</i>	3725	3534	3788	-5.1	7.2
<i>Owned</i>	396	127	131	-67.9	2.9
Total	4122	3662	3919	-11.2	7.0
Seed	2871	1200	1294	-58.2	7.9
Fertilisers and Manure					
<i>Fertilisers</i>	8072	9287	10243	15.1	10.3
<i>Manure</i>	45	59	61	31.0	3.4
Total	8117	9347	10304	15.1	10.2
Insecticides	756	731	766	-3.4	4.8
Irrigation charges	2408	3143	3341	30.5	6.3
Interest on working capital	3805	3990	4188	4.9	5.0
Miscellaneous	333	313	309	-5.9	-1.4
Crop Insurance	0	0	0	-	-
Fixed Cost	53143	54198	58809	2.0	8.5
Rental value of owned land	50803	51984	56376	2.3	8.4
Rent paid for leased-in land	0	0	0	-	-
Land revenue, cesses & taxes	13	14	13	1.4	-7.9
Depreciation on implements & Farm buildings	209	201	207	-3.7	2.8
Interest on fixed capital	2118	1999	2214	-5.6	10.8
Yield (Qtl/ha)	939.84	911.14	893.78	-3.1	-1.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.4(d): Sugarcane : Break-up of cost of Cultivation of Maharashtra

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	178446	171655	186764	-3.8	8.8
Human Labour					
<i>Casual</i>	42330	38466	45684	-9.1	18.8
<i>Attached</i>	3147	4981	2521	58.2	-49.4
<i>Family</i>	30199	28432	27960	-5.9	-1.7
Total	75677	71879	76165	-5.0	6.0
Bullock Labour					
<i>Hired</i>	4091	2820	2203	-31.1	-21.9
<i>Owned</i>	1791	2547	2651	42.2	4.1
Total	5883	5367	4854	-8.8	-9.6
Machine Labour					
<i>Hired</i>	30770	30923	36794	0.5	19.0
<i>Owned</i>	2130	2422	2879	13.7	18.9
Total	32899	33344	39674	1.4	19.0
Seed	9865	8273	7370	-16.1	-10.9
Fertilisers and Manure					
<i>Fertilisers</i>	20211	20074	23223	-0.7	15.7
<i>Manure</i>	3049	2771	2815	-9.1	1.6
Total	23260	22845	26038	-1.8	14.0
Insecticides	1896	1884	1816	-0.6	-3.6
Irrigation charges	19101	18479	20340	-3.3	10.1
Interest on working capital	8720	8425	9341	-3.4	10.9
Miscellaneous	1143	1157	1163	1.2	0.6
Crop Insurance	3	2	2	-34.3	11.8
Fixed Cost	64152	63904	65178	-0.4	2.0
Rental value of owned land	44976	43955	47176	-2.3	7.3
Rent paid for leased-in land	0	0	0	-	-
Land revenue, cesses & taxes	41	42	40	2.5	-4.2
Depreciation on implements & Farm buildings	1741	1863	1999	7.0	7.3
Interest on fixed capital	17394	18044	15962	3.7	-11.5
Yield(Qtl/ha)	1060.67	999.03	995.48	-5.8	-0.4

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

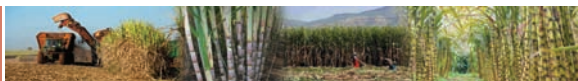


Annex Table 5.4(e): Sugarcane : Break-up of cost of Cultivation of Tamil Nadu

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	188011	192940	201173	2.6	4.3
Human Labour					
<i>Casual</i>	106774	108393	111394	1.5	2.8
<i>Attached</i>	670	788	827	17.6	5.0
<i>Family</i>	14609	16985	19963	16.3	17.5
Total	122053	126166	132184	3.4	4.8
Bullock Labour					
<i>Hired</i>	650	719	689	10.6	-4.2
<i>Owned</i>	23	45	49	96.7	9.4
Total	673	765	739	13.5	-3.4
Machine Labour					
<i>Hired</i>	8372	8343	8192	-0.4	-1.8
<i>Owned</i>	1610	630	1023	-60.9	62.4
Total	9983	8973	9215	-10.1	2.7
Seed	10866	8487	8924	-21.9	5.2
Fertilisers and Manure					
<i>Fertilisers</i>	13966	16853	19815	20.7	17.6
<i>Manure</i>	4920	4749	4449	-3.5	-6.3
Total	18886	21603	24264	14.4	12.3
Insecticides	1034	961	1052	-7.1	9.5
Irrigation charges	13943	15349	13834	10.1	-9.9
Interest on working capital	10200	10350	10659	1.5	3.0
Miscellaneous	372	287	301	-22.8	4.8
Crop Insurance	0	0	0	-	-
Fixed Cost	61960	66788	59583	7.8	-10.8
Rental value of owned land	39155	46998	38958	20.0	-17.1
Rent paid for leased-in land	70	54	55	-22.9	1.8
Land revenue, cesses & taxes	12	12	12	-2.3	1.4
Depreciation on implements & Farm buildings	1649	1737	1948	5.4	12.1
Interest on fixed capital	21075	17987	18610	-14.6	3.5
Yield(Qtl/ha)	1075.06	1010.16	1018.69	-6.0	0.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.4(f): Sugarcane : Break-up of cost of Cultivation of Uttar Pradesh

(₹/ha)

Cost Items	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	96656	90401	90742	-6.5	0.4
Human Labour					
<i>Casual</i>	40159	36784	34463	-8.4	-6.3
<i>Attached</i>	0	0	156	-	-
<i>Family</i>	18317	17751	18027	-3.1	1.6
Total	58476	54534	52646	-6.7	-3.5
Bullock Labour					
<i>Hired</i>	41	133	0	222.9	-
<i>Owned</i>	449	465	646	3.5	39.0
Total	490	597	646	21.9	8.1
Machine Labour					
<i>Hired</i>	2331	2649	2784	13.7	5.1
<i>Owned</i>	1410	1389	1565	-1.5	12.6
Total	3741	4038	4349	7.9	7.7
Seed	11345	9723	11090	-14.3	14.1
Fertilisers and Manure					
<i>Fertilisers</i>	4654	4237	5154	-9.0	21.6
<i>Manure</i>	489	413	447	-15.6	8.3
Total	5143	4650	5601	-9.6	20.5
Insecticides	1750	1544	1493	-11.7	-3.3
Irrigation charges	11033	10959	10558	-0.7	-3.7
Interest on working capital	4608	4274	4277	-7.3	0.1
Miscellaneous	71	82	81	15.9	-1.2
Crop Insurance	0	0	0	-	-
Fixed Cost	67481	69953	61361	3.7	-12.3
Rental value of owned land	56214	58121	48994	3.4	-15.7
Rent paid for leased-in land	0	507	398	-	-21.4
Land revenue, cesses & taxes	10	9	11	-6.0	22.3
Depreciation on implements & Farm buildings	2301	2531	2630	10.0	3.9
Interest on fixed capital	8956	8785	9328	-1.9	6.2
Yield(Qtl/ha)	659.06	576.94	563.35	-12.5	-2.4

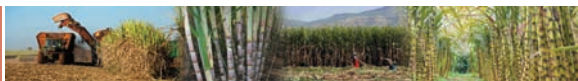
Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

**Annex Table 5.5: Wholesale Price Index of Transport Equipment and Fuels**

Year	Annual WPI Index	Growth(%)
2011-12	105	-
2012-13	110	5.38
2013-14	117	6.20
2014-15	101	-13.27
2015-16	91	-10.00
2016-17	97	6.47
2017-18	103	6.28
2018-19	107	3.10
2019-20	103	-3.76
2020-21	112	8.87
2021-22	145	30.25
2022-23	151	3.86
2023-24	148	-2.27
2024-25	145	-1.74
Average Growth		3.03

- Note* 1: Monthly WPI of Transport Equipment and Fuels is weighted average of monthly WPIs of Tractor Tyre, Agricultural tractors, Light, medium & heavy commercial vehicles, Body (for commercial motor vehicles), Chassis of different vehicle types, Manufacture of parts and accessories for motor vehicles, High Speed Diesel, Lube Oils, Medium & heavy commercial vehicle tyre
- 2: Annual WPI of Transport Equipment and Fuels is average of monthly WPI of Transport Equipment and Fuels from October to September for 2011-12 to 2024-25

Source : Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations



Annex Table 5.6: Projected Cost of Production (CoP) of Sugarcane at State-Specific Recovery Rate for Sugar Season 2025-26 and 2026-27

(₹/qtl)

State	2025-26			2026-27			Change (%) in 2026-27 over 2025-26		
	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2
Gujarat	130	140	194	131	144	202	0.7	2.4	4.1
Haryana	134	138	254	146	151	272	9.5	9.7	7.4
Karnataka	94	116	172	97	120	179	2.6	3.0	4.1
Maharashtra	185	218	276	185	220	277	-0.1	0.7	0.6
Tamil Nadu	210	230	289	225	247	305	7.1	7.4	5.3
Uttar Pradesh	157	193	293	162	199	300	2.7	3.3	2.4
All-India	158	189	268	161	194	276	2.4	3.1	2.9

Note: All-India is weighted average of States at State-specific recovery rates

Source: CACP Calculations using CS data



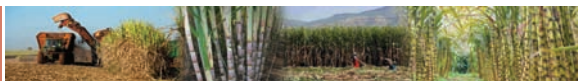
Annex Table 5.7: Cost of Production (CoP) of Sugarcane for Sugar Season 2025-26 and 2026-27 at 10.25 Percent Recovery Rate

(₹/qtl)

State	2025-26			2026-27			Change (%) in 2026-27 over 2025-26		
	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2
Gujarat	126	137	189	128	141	199	1.6	3.3	4.9
Haryana	135	139	257	149	154	277	10.0	10.2	7.8
Karnataka	87	107	159	90	112	167	3.9	4.3	5.4
Maharashtra	169	200	253	173	205	259	1.9	2.7	2.6
Tamil Nadu	237	259	326	252	276	341	6.4	6.7	4.5
Uttar Pradesh	141	173	263	148	182	275	5.1	5.4	4.8
All-India	145	173	246	151	182	258	4.1	5.2	4.9
All-India Modified Cost	173	201	274	180	211	287	4.0	5.0	4.7

Note: All-India is weighted average of States at 10.25 percent basic recovery Rate

Source: CACP calculations using CS data



Annex Table 5.8: Gross Returns and Relative Average Gross Returns with respect to Sugarcane in Selected States

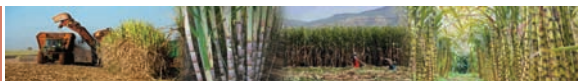
State/Crop/Crop Combination/Year	Gross Returns Over CoC (₹/ha)		Relative Average Gross Returns Over CoC (%)	
	A_2	A_2+FL	A_2	A_2+FL
Gujarat				
Sugarcane				
2021-22	162879	154392	100.0	100.0
2022-23	176869	167722	100.0	100.0
2023-24	211508	201360	100.0	100.0
TE2023-24	183752	174491	100.0	100.0
Paddy+Wheat				
2021-22	80717	65867	49.6	42.7
2022-23	94279	76627	53.3	45.7
2023-24	106544	89029	50.4	44.2
TE2023-24	93847	77174	51.1	44.2
Cotton+Wheat				
2021-22	114553	94177	70.3	61.0
2022-23	146017	123943	82.6	73.9
2023-24	135443	112158	64.0	55.7
TE2023-24	132004	110093	71.8	63.1
Paddy+Paddy				
2021-22	62182	46538	38.2	30.1
2022-23	75009	54590	42.4	32.5
2023-24	88580	68680	41.9	34.1
TE2023-24	75257	56603	41.0	32.4
Soybean+Gram				
2021-22	28420	15092	17.4	9.8
2022-23	45117	34368	25.5	20.5
2023-24	62392	51017	29.5	25.3
TE2023-24	45310	33492	24.7	19.2

(Contd...)



Soybean+Wheat				
2021-22	44768	30969	27.5	20.1
2022-23	61170	49156	34.6	29.3
2023-24	76019	63300	35.9	31.4
TE2023-24	60652	47809	33.0	27.4
Haryana				
Sugarcane				
2021-22	177174	174208	100.0	100.0
2022-23	189311	186235	100.0	100.0
2023-24	211559	208359	100.0	100.0
TE2023-24	192681	189601	100.0	100.0
Paddy+Wheat				
2021-22	159039	143967	89.8	82.6
2022-23	172743	157343	91.2	84.5
2023-24	188745	172483	89.2	82.8
TE2023-24	173509	157931	90.0	83.3
Cotton+Wheat				
2021-22	122644	99792	69.2	57.3
2022-23	130223	107394	68.8	57.7
2023-24	143065	121461	67.6	58.3
TE2023-24	131978	109549	68.5	57.8
Paddy+Paddy				
2021-22	172312	155129	97.3	89.0
2022-23	189022	171653	99.8	92.2
2023-24	216668	197697	102.4	94.9
TE2023-24	192667	174826	100.0	92.2
Karnataka				
2021-22	188027	174250	100.0	100.0
2022-23	183428	166819	100.0	100.0
2023-24	196758	178546	100.0	100.0
TE2023-24	189404	173205	100.0	100.0

(Contd...)



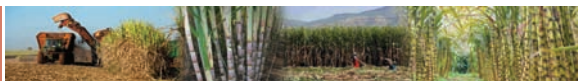
Paddy+Paddy				
2021-22	76095	51574	40.5	29.6
2022-23	115724	81368	63.1	48.8
2023-24	122160	90579	62.1	50.7
TE2023-24	104660	74507	55.3	43.0
Soybean+Gram				
2021-22	52436	42694	27.9	24.5
2022-23	57152	45858	31.2	27.5
2023-24	52224	40716	26.5	22.8
TE2023-24	53937	43089	28.5	24.9
Maharashtra				
Sugarcane				
2021-22	119773	89574	100.0	100.0
2022-23	118549	90117	100.0	100.0
2023-24	122159	94198	100.0	100.0
TE2023-24	120160	91296	100.0	100.0
Paddy+Wheat				
2021-22	11072	-12713	9.2	-14.2
2022-23	15977	-10933	13.5	-12.1
2023-24	44334	11198	36.3	11.9
TE2023-24	23795	-4149	19.8	-4.5
Cotton+Wheat				
2021-22	45696	23743	38.2	26.5
2022-23	69913	44469	59.0	49.3
2023-24	72572	44911	59.4	47.7
TE2023-24	62727	37708	52.2	41.3
Paddy+Paddy				
2021-22	-23832	-52126	-19.9	-58.2
2022-23	-22162	-52990	-18.7	-58.8
2023-24	13714	-29083	11.2	-30.9
TE2023-24	-10760	-44733	-9.0	-49.0

(Contd...)



Soybean+Gram				
2021-22	60230	49341	50.3	55.1
2022-23	57131	44143	48.2	49.0
2023-24	76479	63593	62.6	67.5
TE2023-24	64613	52359	53.8	57.4
Soybean+Wheat				
2021-22	55698	41473	46.5	46.3
2022-23	55334	38248	46.7	42.4
2023-24	69772	52139	57.1	55.4
TE2023-24	60268	43953	50.2	48.1
Tamil Nadu				
Sugarcane				
2021-22	132093	117484	100.0	100.0
2022-23	132511	115526	100.0	100.0
2023-24	134964	115001	100.0	100.0
TE2023-24	133189	116003	100.0	100.0
Paddy+Paddy				
2021-22	55840	41584	42.3	35.4
2022-23	80655	67401	60.9	58.3
2023-24	91585	76601	67.9	66.6
TE2023-24	76027	61862	57.1	53.3
Uttar Pradesh				
Sugarcane				
2021-22	155462	137145	100.0	100.0
2022-23	133933	116183	100.0	100.0
2023-24	135105	117079	100.0	100.0
TE2023-24	141500	123469	100.0	100.0
Paddy+Wheat				
2021-22	69910	48386	45.0	35.3
2022-23	73484	51579	54.9	44.4
2023-24	107370	85235	79.5	72.8
TE2023-24	83588	61733	59.1	50.0

(Contd...)



Paddy+Paddy				
2021-22	34271	10809	22.0	7.9
2022-23	38771	14540	28.9	12.5
2023-24	76521	52667	56.6	45.0
TE2023-24	49855	26005	35.2	21.1
All-India				
Sugarcane				
2021-22	150247	130238	100.0	100.0
2022-23	138731	118983	100.0	100.0
2023-24	142404	122059	100.0	100.0
TE2023-24	143794	123760	100.0	100.0
Paddy+Wheat				
2021-22	90338	68189	60.1	52.4
2022-23	101151	78496	72.9	66.0
2023-24	117963	95331	82.8	78.1
TE2023-24	103151	80672	71.7	65.2
Cotton+Wheat				
2021-22	96431	73791	64.2	56.7
2022-23	116411	92559	83.9	77.8
2023-24	116917	91634	82.1	75.1
TE2023-24	109920	85995	76.4	69.5
Paddy+Paddy				
2021-22	70955	44332	47.2	34.0
2022-23	86201	59392	62.1	49.9
2023-24	103402	77574	72.6	63.6
TE2023-24	86853	60433	60.4	48.8
Soybean+Gram				
2021-22	58241	44796	38.8	34.4
2022-23	58188	43353	41.9	36.4
2023-24	67887	52673	47.7	43.2
TE2023-24	61439	46941	42.7	37.9

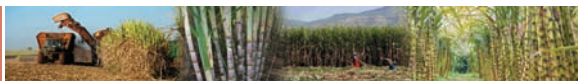
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Soybean+Wheat				
2021-22	78967	64983	52.6	49.9
2022-23	82624	67668	59.6	56.9
2023-24	93834	77964	65.9	63.9
TE2023-24	85142	70205	59.2	56.7

Note: All-India gross returns of a crop are weighted average of respective gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.9: Comparison of CACP and State Projected Cost of Production (C_2) at State Specific Recovery Rate for Sugar Season 2026-27

State	CACP Projections on the basis of CS data		State Projections	
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)
Gujarat	890.05	202	854.52	318
Haryana	736.14	272	815.00	362
Karnataka	914.92	179	900.00	353
Maharashtra	1018.39	277	Inputs not Received	
Tamil Nadu	1034.64	305	1045.00	370
Uttar Pradesh	599.78	300	800.00	334

Source: CACP Calculations using CS data, and State Governments

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November 2025